Comprehensive Quant Interview Guide

XeLaTeX Edition

This document is a compact but rigorous interview companion. Each section contains the definitions and formulas that are most often used in quant interviews, followed by worked examples with complete solutions. The emphasis is on tools that recur in trading, research, and mathematical finance interviews: linear algebra, analysis, combinatorics, and probability.

Contents

1 Linear Algebra

Quant interviews use linear algebra for covariance matrices, regressions, factor models, optimization, and stability questions. The focus is usually on structure: what a matrix or operator can and cannot do.

1.1 Systems, Rank, Determinants, and Inverses

Definitions and formulas.

- A *permutation* of $\{1, \ldots, n\}$ is a bijection σ. Its sign is $\operatorname{sgn}(\sigma) = (-1)^{\operatorname{inv}(\sigma)}$, where $\operatorname{inv}(\sigma)$ is the number of inversions. Every permutation decomposes into disjoint cycles and also into a product of transpositions. These signs are exactly the coefficients that appear in determinant expansions.
- A complex number has Cartesian and polar forms

  $$z = x + iy = r(\cos\theta + i\sin\theta) = re^{i\theta}, \qquad |z| = \sqrt{x^2 + y^2}.$$

  De Moivre's formula gives $(re^{i\theta})^n = r^n e^{in\theta}$, and the nth roots of unity are $e^{2\pi ik/n}$ for $k = 0, \ldots, n-1$. These facts are useful whenever eigenvalues or characteristic roots are complex.
- A linear system is written $Ax = b$. It is consistent if and only if

  $$\operatorname{rank}(A) = \operatorname{rank}([A \mid b]).$$

  If this common rank equals the number of unknowns, the solution is unique. Otherwise,

  $$x = x_p + x_h,$$

  where x_p is one particular solution and x_h runs over the nullspace of A. Rank counts the number of independent constraints, so free variables appear exactly when the rank is too small.
- Elementary row operations preserve the solution set of a system, and Gaussian elimination reduces a system to echelon form. In practice this is the fastest way to detect pivots, free variables, and inconsistency.
- For an invertible square matrix,

  $$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A), \qquad \det(AB) = \det(A)\det(B).$$

  Also, A is invertible if and only if $\det(A) \neq 0$. Invertibility means every right-hand side b has a unique solution to $Ax = b$.
- The determinant is multilinear in rows (or columns), alternates under row swaps, and measures signed volume. A zero determinant signals linear dependence of rows or columns.
- A useful rank inequality is

  $$\operatorname{rank}(AB) \le \min\{\operatorname{rank}(A), \operatorname{rank}(B)\}.$$

  Composition cannot create more independent directions than either map already had.
- Iterative solvers appear in numerical interviews. Writing $A = D + L + U$ gives the Jacobi iteration

  $$x^{(k+1)} = D^{-1}(b - (L + U)x^{(k)}),$$

and Gauss–Seidel iteration

$$x^{(k+1)} = (D+L)^{-1}(b - Ux^{(k)}).$$

A sufficient condition for convergence is strict diagonal dominance. These methods trade exact elimination for repeated cheap updates.

Example 1.1 (Consistency and general solution)**.** Solve the system

$$\begin{aligned} x + y + z &= 1, \\ 2x + 2y + 2z &= 2, \\ x - y &= 0. \end{aligned}$$

Determine whether the solution is unique.

Solution. The second equation is just twice the first, so it contributes no new information. The third equation gives $x = y$. Substituting into the first equation yields

$$2x + z = 1.$$

Let $z = t$. Then

$$x = y = \frac{1-t}{2}, \qquad z = t.$$

Therefore the solution set is

$$\left\{ \left(\frac{1-t}{2}, \frac{1-t}{2}, t \right) : t \in \mathbb{R} \right\}.$$

This matches the rank test: $\operatorname{rank}(A) = \operatorname{rank}([A \mid b]) = 2 < 3$, so the system is consistent but has infinitely many solutions. □

Example 1.2 (Independence of shifted determinants, and where it must fail)**.** Let E be the 3×3 identity. (a) Prove that $\det(X)$, $\det(X+E)$, $\det(X-E)$, viewed as functions on the space of complex 3×3 matrices, are linearly independent. (b) Prove there is an $m \in \mathbb{N}$ for which $\det(X - mE), \det(X-(m-1)E), \ldots, \det(X+mE)$ (a run of $2m+1$ such functions) is linearly *dependent*.

Solution. **Key reduction.** For a 3×3 matrix X, expanding $\det(X - kE)$ as a polynomial in the scalar k gives exactly (up to sign) the characteristic polynomial of X:

$$\det(X - kE) = -k^3 + \operatorname{tr}(X)\, k^2 - p_2(X)\, k + \det(X),$$

where $p_2(X)$ is the sum of the principal 2×2 minors of X. So each function $f_k(X) := \det(X - kE)$ is a fixed linear combination of the four functions $\{1,\ \operatorname{tr}(X),\ p_2(X),\ \det(X)\}$ — namely $f_k = -k^3 \cdot 1 + k^2 \cdot \operatorname{tr}(X) - k \cdot p_2(X) + 1 \cdot \det(X)$ — with coefficient vector $v_k = (-k^3, k^2, -k, 1) \in \mathbb{C}^4$. Since $1, \operatorname{tr}, p_2, \det$ have different degrees in the entries of X, they are linearly independent functions; hence the functions $\{f_k\}$ are linearly (in)dependent exactly when their coefficient vectors $\{v_k\}$ are linearly (in)dependent in $\mathbb{C}^4$.

(a) Here $k \in \{-1, 0, 1\}$: $v_{-1} = (1,1,1,1)$, $v_0 = (0,0,0,1)$, $v_1 = (-1,1,-1,1)$. Suppose $\alpha v_{-1} + \beta v_0 + \gamma v_1 = 0$. The third coordinate gives $\alpha - \gamma = 0$, i.e. $\alpha = \gamma$; the second coordinate gives $\alpha + \gamma = 0$, i.e. $\alpha = -\gamma$. Together $\alpha = \gamma = 0$, and then the fourth coordinate gives $\beta = 0$. So v_{-1}, v_0, v_1 are linearly independent in $\mathbb{C}^4$, hence so are f_{-1}, f_0, f_1:

$$\boxed{\det(X),\ \det(X+E),\ \det(X-E) \text{ are linearly independent}}.$$

(b) Since $\mathbb{C}^4$ is 4-dimensional, any 5 vectors in it are automatically linearly dependent. Taking $m = 2$ gives $2m + 1 = 5$ functions $f_{-2}, f_{-1}, f_0, f_1, f_2$, whose coefficient vectors $v_{-2}, \ldots, v_2 \in$

$\mathbb{C}^4$ (five vectors in a four-dimensional space) must be dependent, hence so are the functions themselves:

$$\boxed{m = 2 \text{ works: } \det(X - 2E), \ldots, \det(X + 2E) \text{ are linearly dependent}}$$

(indeed any $m \geq 2$ works, by the same pigeonhole argument). □

Example 1.3 (A rank-drop inequality for A and A^2)**.** Let A be an $n \times n$ matrix. Prove $n - \operatorname{rank} A \geq \operatorname{rank} A - \operatorname{rank} A^2$.

Solution. Restrict A to a map $\operatorname{im}(A) \to \mathbb{R}^n$. Its image is $A(\operatorname{im} A) = \operatorname{im}(A^2)$, and its kernel is $\ker(A) \cap \operatorname{im}(A)$. By rank–nullity applied to this restriction,

$$\operatorname{rank}(A^2) = \dim \operatorname{im}(A^2) = \dim \operatorname{im}(A) - \dim\big(\ker(A) \cap \operatorname{im}(A)\big)$$

$$= \operatorname{rank}(A) - \dim\big(\ker(A) \cap \operatorname{im}(A)\big),$$

so

$$\operatorname{rank}(A) - \operatorname{rank}(A^2) = \dim\big(\ker(A) \cap \operatorname{im}(A)\big).$$

Since $\ker(A) \cap \operatorname{im}(A)$ is a subspace of $\ker(A)$, its dimension is at most $\dim \ker(A) = n - \operatorname{rank}(A)$ (ordinary rank–nullity for A). Combining,

$$\boxed{n - \operatorname{rank}(A) \geq \operatorname{rank}(A) - \operatorname{rank}(A^2)}.$$

□

Example 1.4 (Every square matrix is a sum of two invertible matrices)**.** Prove that every real $n \times n$ matrix A is the sum of two invertible matrices.

Solution. A has at most n (real) eigenvalues, so among the infinitely many real numbers there is some c that is neither 0 nor an eigenvalue of A (finitely many values are excluded). For such c, set $B = cI$ and $C = A - cI$:

- $B = cI$ is invertible, since $c \neq 0$.
- $C = A - cI$ is invertible, since c is not an eigenvalue of A (so 0 is not an eigenvalue of $A - cI$, i.e. $A - cI$ has trivial kernel).

And $B + C = cI + (A - cI) = A$. So

$$\boxed{A = B + C \text{ with both } B, C \text{ invertible}}.$$

□

Example 1.5 (A rank-one perturbation determinant that is always 1)**.** Let J be a $2n \times 2n$ real skew-symmetric matrix, $\beta > 0$ a scalar, and $u \in \mathbb{R}^{2n}$ a nonzero vector. Find $\det(I + \beta u u^T J)$.

Solution. The matrix $\beta u u^T J$ has rank ≤ 1: it equals $u\,(\beta J^T u)^T = u(-\beta J u)^T$, i.e. it is ab^T with $a = u$ and $b = -\beta J u$ (using $J^T = -J$). By the matrix determinant lemma, $\det(I + ab^T) = 1 + b^T a$, so

$$\det(I + \beta u u^T J) = 1 + b^T a = 1 + (-\beta J u)^T u = 1 - \beta\, u^T J^T u = 1 + \beta\, u^T J u.$$

But $u^T J u$ is a scalar, so it equals its own transpose: $u^T J u = (u^T J u)^T = u^T J^T u = -u^T J u$, forcing $u^T J u = 0$ (this holds for *any* skew-symmetric matrix and any vector). So

$$\boxed{\det(I + \beta u u^T J) = 1}$$

regardless of β, u, or the specific skew-symmetric J (verified numerically for random J, u). □

Example 1.6 (Does $ABA = A$ force $BAB = B$?)**.** Let A, B be square matrices of the same size with $ABA = A$. Must $BAB = B$?

Solution. No. Take

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $AB = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, so $ABA = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = A$ — the hypothesis holds.

But $BA = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$, so $BAB = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = B$.

So

$$\boxed{\text{no — } ABA = A \text{ does not imply } BAB = B}$$

(the hypothesis $ABA = A$ says B is a *generalized inverse* of A in the weak sense; $BAB = B$ additionally is the stronger "reflexive" generalized-inverse condition, and the two are genuinely different). ☐

Example 1.7 (Determinant of the symmetric Pascal matrix)**.** Let $A = (a_{ij})_{i,j=1}^n$ with $a_{ij} = \binom{i+j-2}{i-1}$. Find $\det A$.

Solution. This is the classical *symmetric Pascal matrix*. Let $L = (\ell_{ij})$ be the lower-triangular Pascal matrix, $\ell_{ij} = \begin{pmatrix} i-1 \\ j-1 \end{pmatrix}$ for $i \geq j$ (and 0 otherwise), which has all 1's on its diagonal ($\ell_{ii} = \binom{i-1}{i-1} = 1$), so $\det L = 1$.

Claim: $A = LL^T$**.** The (i, j) entry of LL^T is

$$\sum_{m=0}^{\min(i,j)-1} \binom{i-1}{m}\binom{j-1}{m} = \sum_m \binom{i-1}{m}\binom{j-1}{j-1-m} = \binom{i-1+j-1}{j-1}$$

by Vandermonde's identity, and $\binom{i+j-2}{j-1} = \binom{i+j-2}{i-1} = a_{ij}$ (using $\binom{n}{k} = \binom{n}{n-k}$). So indeed $A = LL^T$.

Determinant. Since L is triangular with all diagonal entries 1, $\det L = 1$, so

$$\det A = \det(LL^T) = \det L \cdot \det L^T = 1 \cdot 1 = \boxed{1}$$

for every n (checked directly for $n = 1, 2, 3$: e.g. $n = 3$ gives $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{pmatrix}$, $\det A = 1$). ☐

Example 1.8 (Swapping the order in $\det(I - AB)$)**.** Let A, B be $n \times n$ real matrices. Prove $\det(I - AB) = \det(I - BA)$.

Solution. Consider the $2n \times 2n$ block matrix $M = \begin{pmatrix} I & A \\ B & I \end{pmatrix}$, and compute $\det M$ two ways using the block-determinant (Schur complement) formula $\det \begin{pmatrix} P & Q \\ R & S \end{pmatrix} = \det(P)\det(S - RP^{-1}Q) = \det(S)\det(P - QS^{-1}R)$ (valid whenever P, respectively S, is invertible).

Eliminating using the top-left block ($P = I$): $\det M = \det(I) \cdot \det(I - BI^{-1}A) = \det(I - BA)$.

Eliminating using the bottom-right block ($S = I$): $\det M = \det(I) \cdot \det(I - AI^{-1}B) = \det(I - AB)$.

Both computations give $\det M$, so

$$\boxed{\det(I - AB) = \det(I - BA)}$$

for any square matrices A, B of the same size. ☐

Example 1.9 (A linear relation forcing commutativity)**.** Let X, Y be square matrices of the same size with $XY = \lambda X + \mu Y$ for some nonzero scalars λ, μ. Prove X and Y commute.

Solution. Expand the product $(X - \mu I)(Y - \lambda I)$:

$$(X - \mu I)(Y - \lambda I) = XY - \lambda X - \mu Y + \lambda\mu I = (\lambda X + \mu Y) - \lambda X - \mu Y + \lambda\mu I = \lambda\mu I$$

using the hypothesis. Since $\lambda\mu \neq 0$, this shows $X - \mu I$ has $\frac{1}{\lambda\mu}(Y - \lambda I)$ as a right inverse; for square matrices a one-sided inverse is automatically two-sided, so also

$$(Y - \lambda I)(X - \mu I) = \lambda\mu I.$$

Expanding the left side: $YX - \mu Y - \lambda X + \lambda\mu I = \lambda\mu I$, i.e. $YX = \lambda X + \mu Y$. Comparing with the hypothesis $XY = \lambda X + \mu Y$,

$$\boxed{XY = \lambda X + \mu Y = YX}.$$

□

Example 1.10 (Few zero entries in the inverse of an all-positive matrix)**.** Let A be an invertible real $n \times n$ matrix ($n \geq 2$) all of whose entries are positive. Prove that A^{-1} has at most $n^2 - 2n$ zero entries.

Solution. Let $B = A^{-1}$, so $AB = I$. Fix a column j of B. Since B is invertible, column j is nonzero: $B_{kj} \neq 0$ for at least one k.

Pick any row $i \neq j$ (possible since $n \geq 2$). The (i, j) entry of $AB = I$ is 0 (off-diagonal), i.e.

$$\sum_{k=1}^{n} A_{ik} B_{kj} = 0,$$

where every coefficient A_{ik} is *strictly positive*. If all B_{kj} were ≥ 0, then — since some $B_{kj} > 0$ — the sum would be a positive combination with at least one strictly positive term, hence > 0, contradicting the sum being 0. Likewise, all $B_{kj} \leq 0$ is impossible. So column j must contain both a strictly positive entry and a strictly negative entry — in particular, *at least* 2 *nonzero entries*.

Since this holds for every column $j = 1, \ldots, n$, matrix B has at least $2n$ nonzero entries in total. Hence the number of zero entries is at most

$$\boxed{n^2 - 2n}.$$

□

1.2 Vector Spaces and Linear Maps

Definitions and formulas.

- A set V is a vector space if it is closed under addition and scalar multiplication. A *basis* is a linearly independent spanning set. The number of basis vectors is $\dim(V)$. A basis is the coordinate system intrinsic to the space.
- For subspaces $U, V \subseteq \mathbb{R}^n$,

  $$\dim(U + V) = \dim(U) + \dim(V) - \dim(U \cap V).$$

  If $U \cap V = \{0\}$, then $U \oplus V$ is a direct sum and $\dim(U \oplus V) = \dim(U) + \dim(V)$. The overlap must be subtracted because it would otherwise be counted twice.

- If $L : V \to W$ is linear, then

$$\ker(L) = \{v \in V : L(v) = 0\}, \qquad \operatorname{im}(L) = \{L(v) : v \in V\},$$

and the rank–nullity theorem says

$$\dim(\ker L) + \dim(\operatorname{im} L) = \dim(V).$$

The kernel measures directions lost by the map, while the image measures directions that survive.

- If C is the change-of-basis matrix from the new basis to the old one, then the matrix of an operator changes by similarity:

$$A_{\text{new}} = C^{-1} A_{\text{old}} C.$$

Similar matrices represent the same linear transformation in different coordinates.

- In regression and least squares,

$$\min_x \|Ax - b\|_2^2 \quad \Longrightarrow \quad A^T A x = A^T b.$$

The normal equations say that the residual $b - Ax$ must be orthogonal to the column space of A.

Example 1.11 (Dimension of a constrained symmetric matrix space)**.** Let subspaces $V, U \subseteq \mathbb{R}^4$ be defined by

$$V = \operatorname{span}\{v_1, v_2, v_3\}, \qquad U = \{y \in \mathbb{R}^4 : Ty = 0\},$$

where

$$v_1 = (3, -2, 2, 1)^T, \qquad v_2 = (-1, 2, -2, 3)^T, \qquad v_3 = (3, -3, -1, 2)^T,$$

and

$$T = (2, -1, -3, -2).$$

Define

$$F = \{A \in M_4(\mathbb{R}) : \text{every column of } A \text{ lies in } V, \text{ every row of } A \text{ lies in } U\}.$$

Find

$$W = \{A \in M_4(\mathbb{R}) : A = A^T,\ A \in F\}.$$

Write an integer as the answer, or -1 if the set is not a linear space.

Solution. The set W is a linear subspace because symmetry and the row/column constraints are all linear conditions.

First compute the dimension of $V \cap U$.

- The vectors v_1, v_2, v_3 are linearly independent, so $\dim(V) = 3$.
- Since $U = \ker(T)$ with $T \neq 0$, we have $\dim(U) = 3$.
- Now

$$Tv_2 = 2(-1) - 2 + 6 - 6 = -4 \neq 0,$$

so $v_2 \notin U$. Therefore V is not contained in U, hence $V + U = \mathbb{R}^4$.

Therefore,

$$\dim(V \cap U) = \dim(V) + \dim(U) - \dim(V + U) = 3 + 3 - 4 = 2.$$

Set $I = V \cap U$, so $\dim(I) = 2$.

Now let $A \in W$. Since the columns of A lie in V and $A = A^T$, each row is the transpose of a column. Because each row lies in U, each column also lies in U. Thus every column of A lies in $I = V \cap U$. So the image of A is contained in the two-dimensional space I.

Because A is symmetric, in an orthonormal basis adapted to the decomposition

$$\mathbb{R}^4 = I \oplus I^\perp,$$

the matrix of A has block form

$$\begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix},$$

where B is a symmetric 2×2 real matrix. The space of symmetric 2×2 matrices has dimension

$$\frac{2(2+1)}{2} = 3.$$

Hence

$$\boxed{\dim(W) = 3}.$$

□

Example 1.12 (Least-squares solution via normal equations)**.** Find the least-squares solution of

$$Ax \approx b, \qquad A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

Solution. The least-squares solution satisfies the normal equations

$$A^T A x = A^T b.$$

Compute

$$A^T A = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}, \qquad A^T b = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix},$$

so

$$x_1 = 1, \qquad x_2 = \frac{1}{2}.$$

Hence the least-squares solution is

$$\boxed{x^* = \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix}}.$$

Indeed,

$$Ax^* = \begin{pmatrix} 3/2 \\ 1/2 \\ 1 \end{pmatrix}, \qquad r = b - Ax^* = \begin{pmatrix} -1/2 \\ -1/2 \\ 1 \end{pmatrix},$$

and one checks that r is orthogonal to both columns of A, as it should be. □

Example 1.13 (Prescribing an eigenvector and excluding a vector from the image)**.** Let $v, w \in \mathbb{R}^n$ be linearly independent. Does there always exist an $n \times n$ matrix A such that v is an eigenvector of A with eigenvalue 5, while $w \notin \text{im}(A)$?

Solution. Yes, always. Since v, w are linearly independent, extend $\{v, w\}$ to a basis $\{v, w, e_3, \dots, e_n\}$ of $\mathbb{R}^n$. A linear map is determined by its values on a basis, so define A by

$$Av = 5v, \qquad Aw = 0, \qquad Ae_3 = \cdots = Ae_n = 0.$$

By construction v is an eigenvector of A with eigenvalue 5. Moreover

$$\text{im}(A) = \text{span}\{5v, 0, \dots, 0\} = \text{span}\{v\},$$

and $w \notin \text{span}\{v\}$ precisely because v, w are linearly independent. Hence

$$\boxed{w \notin \text{im}(A)}$$

for this choice of A. Concretely: writing any $x \in \mathbb{R}^n$ in basis coordinates $x = \alpha v + \beta w + \sum_i \gamma_i e_i$, one has $Ax = 5\alpha v$. □

Example 1.14 (Equal rank from an invertibility hypothesis)**.** Let A, B be $n \times n$ matrices with $A^2 = A$, $B^2 = B$, and suppose $I - (A + B)$ is invertible. Prove that $\operatorname{rank} A = \operatorname{rank} B$.

Solution. Let $C = I - A - B$, invertible by hypothesis. Since A is idempotent, every $v \in \text{im}(A)$ satisfies $Av = v$ (write $v = Au$; then $Av = A^2u = Au = v$), and likewise $Bv = v$ for $v \in \text{im}(B)$.

Take $v \in \text{im}(A)$. Then

$$Cv = v - Av - Bv = v - v - Bv = -Bv \in \text{im}(B),$$

so C maps $\text{im}(A)$ into $\text{im}(B)$. As C is invertible it is injective, so its restriction to $\text{im}(A)$ is an injective map into $\text{im}(B)$, giving

$$\operatorname{rank} A = \dim \text{im}(A) \le \dim \text{im}(B) = \operatorname{rank} B.$$

Symmetrically, for $v \in \text{im}(B)$,

$$Cv = v - Av - Bv = v - Av - v = -Av \in \text{im}(A),$$

so the same argument gives $\operatorname{rank} B \le \operatorname{rank} A$. Combining,

$$\boxed{\operatorname{rank} A = \operatorname{rank} B}.$$

□

1.3 Eigenvalues, Jordan Form, and Singular Value Decomposition

Definitions and formulas.

- A scalar λ is an eigenvalue of A if there exists $v \neq 0$ such that $Av = \lambda v$. The eigenspace is $E_\lambda = \ker(A - \lambda I)$. Eigenvectors are directions whose orientation is preserved by the transformation.
- The characteristic polynomial is $p_A(\lambda) = \det(\lambda I - A)$. The Hamilton–Cayley theorem states that $p_A(A) = 0$. This lets you reduce high powers of a matrix to lower ones.
- For an $n \times n$ matrix, the sum of eigenvalues counted with algebraic multiplicity is $\text{tr}(A)$ and the product is $\det(A)$. These are quick consistency checks when you compute a spectrum.
- A matrix is diagonalizable if and only if for every eigenvalue, algebraic multiplicity equals geometric multiplicity. Equivalently, the space has a full basis of eigenvectors.

- In particular, if the characteristic polynomial splits and has distinct roots, the matrix is automatically diagonalizable. Distinct eigenvalues automatically give linearly independent eigenvectors.
- A nilpotent operator satisfies $N^k = 0$ for some k. Its Jordan form is a direct sum of Jordan blocks with zero diagonal. Nilpotent parts measure failure of diagonalizability.
- If N is nilpotent on an n-dimensional space, then its only eigenvalue is 0 and its characteristic polynomial is λ^n. Repeated application eventually kills every vector.
- The singular value decomposition is

$$A = U\Sigma V^T,$$

where U, V are orthogonal and Σ is diagonal with nonnegative entries $\sigma_i = \sqrt{\lambda_i(A^TA)}$. The SVD separates input directions, stretching factors, and output directions.

- The singular values measure the principal stretching factors of the linear map $x \mapsto Ax$. The rank of A equals the number of nonzero singular values. For rectangular matrices they play the role that absolute eigenvalues play for symmetric matrices.
- If A is real symmetric, then the spectral theorem gives

$$A = Q\Lambda Q^T,$$

where Q is orthogonal and Λ is diagonal with real eigenvalues. Symmetric matrices therefore act like independent scalar multipliers along orthogonal directions.

- The Frobenius norm is

$$\|A\|_F = \sqrt{\operatorname{tr}(A^TA)} = \left(\sum_{i,j} a_{ij}^2\right)^{1/2} = \left(\sum_i \sigma_i^2\right)^{1/2}.$$

It is the Euclidean norm of the matrix entries viewed as one long vector, and it is the natural norm for entrywise least-squares error.

- The best rank-k approximation in Frobenius norm is obtained by truncating the SVD:

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T.$$

Keeping the largest singular values preserves the most important directions of variation.

Example 1.15 (Using Hamilton–Cayley to compute powers)**.** Let

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

Find a formula for A^n.

Solution. Write

$$A = 2I + N, \qquad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0.$$

Because $N^2 = 0$, the binomial expansion truncates:

$$A^n = (2I + N)^n = \sum_{k=0}^{n} \binom{n}{k} 2^{n-k} N^k = 2^n I + n2^{n-1} N.$$

Hence

$$A^n = \begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix}.$$

This is exactly what Hamilton–Cayley predicts, since the characteristic polynomial is $(\lambda - 2)^2$ and therefore $(A - 2I)^2 = 0$. □

Example 1.16 (Singular values of a rank-one matrix)**.** Let

$$A = \begin{pmatrix} 3 & 0 \\ 4 & 0 \end{pmatrix}.$$

Find the singular values of A and write down one SVD.

Solution. Compute

$$A^T A = \begin{pmatrix} 25 & 0 \\ 0 & 0 \end{pmatrix}.$$

Its eigenvalues are 25 and 0, so the singular values are

$$\sigma_1 = 5, \qquad \sigma_2 = 0.$$

Take

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Then

$$u_1 = \frac{Av_1}{\|Av_1\|} = \frac{1}{5}\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}.$$

Choose an orthonormal vector perpendicular to u_1, for example

$$u_2 = \begin{pmatrix} -4/5 \\ 3/5 \end{pmatrix}.$$

Therefore one valid SVD is

$$U = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix}, \qquad V = I.$$

So

$$\boxed{A = U\Sigma V^T}$$

with singular values 5 and 0. □

Example 1.17 (Centralizer dimension for cyclic vs. non-cyclic matrices)**.** For an $n \times n$ real matrix A and $v \in \mathbb{R}^n$, let $U(A) = \{X \in \mathrm{Mat}_n(\mathbb{R}) : AX = XA\}$ and $W(A, v) = \mathrm{span}\{v, Av, A^2v, \dots\}$ (the A-cyclic subspace generated by v). (a) If $\dim W(A, v) = n$ for *every* $v \neq 0$, what is the largest possible $\dim U(A)$? (b) If $\dim W(A, v) < n$ for *every* v, what is the smallest possible $\dim U(A)$?

Solution. **Background fact.** For any A, $\dim U(A) \geq n$, with equality iff A is *non-derogatory* (its minimal polynomial has degree n, i.e. equals the characteristic polynomial). More precisely, if the Jordan blocks for a single eigenvalue have sizes $n_1 \geq n_2 \geq \cdots \geq n_r$ (a partition of that eigenvalue's multiplicity), that eigenvalue contributes $\sum_i (2i-1)n_i$ to $\dim U(A)$; summing over eigenvalues and minimizing $\sum (2i-1)n_i$ for fixed $\sum n_i$ shows single blocks ($r = 1$) are optimal, giving the bound $\dim U(A) \geq n$.

(a) If v_0 is any vector with $\dim W(A, v_0) = n$, then $v_0, Av_0, \dots, A^{n-1}v_0$ are linearly independent, so the minimal polynomial of A has degree $\geq n$, hence exactly n: A is non-derogatory. By the background fact, $\dim U(A) = n$ exactly. So under the hypothesis of (a) (which certainly implies *some* cyclic vector exists), the dimension is forced to be

$$\boxed{n}.$$

(Such A exist for even n: e.g. for $n = 2$, $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has no real eigenvalues, so $\{v, Av\}$ is dependent only if v is a real eigenvector — impossible — hence every nonzero v is cyclic.)

(b) Now no vector is cyclic, so A is derogatory: some eigenvalue has $r \geq 2$ Jordan blocks. To minimize $\dim U(A) = \sum(2i-1)n_i$ (summed over all eigenvalues) subject to this, isolate the derogatory eigenvalue's contribution $n_1 + 3n_2$ with $n_1 \geq n_2 \geq 1$: writing the eigenvalue's multiplicity as $m = n_1 + n_2$, this equals $m + 2n_2 \geq m + 2$, minimized at $n_2 = 1$ (a single "stray" Jordan block of size 1 split off from an otherwise full block of size $n-1$). Every other eigenvalue, kept non-derogatory, contributes exactly its own multiplicity. Summing, the total is minimized at

$$\boxed{n+2},$$

attained by a single eigenvalue with Jordan blocks of sizes $(n-1, 1)$. □

Example 1.18 (When is a matrix forced to be invertible by $A^T = p(A)$?)**.** Let A be a square real matrix with $A^T = p(A)$ for some polynomial p with nonzero constant term. Prove A is invertible. Is it true that for *every* linear operator $\varphi : \mathbb{R}^n \to \mathbb{R}^n$ there is a polynomial p and a basis in which the matrix of φ satisfies $A^T = p(A)$ (dropping the nonzero-constant-term requirement)?

Solution. **Invertibility.** Write $p(x) = c_0 + c_1 x + \cdots + c_k x^k$ with $c_0 \neq 0$. Suppose toward a contradiction that A is singular, so $Av = 0$ for some $v \neq 0$. Then $A^j v = 0$ for all $j \geq 1$, so $p(A)v = c_0 v$, i.e. $A^T v = c_0 v$. Now use the general identity $v^T A v = v^T A^T v$ (true for any square matrix and vector, since $v^T A v$ is a scalar equal to its own transpose $v^T A^T v$). The left side is $v^T(Av) = v^T \cdot 0 = 0$; the right side is $v^T(A^T v) = v^T(c_0 v) = c_0\|v\|^2$. So $c_0\|v\|^2 = 0$, and since $v \neq 0$ this forces $c_0 = 0$ — contradicting $c_0 \neq 0$. Hence A is invertible.

Every operator? No. If $A^T = p(A)$ for any polynomial p, then A commutes with A^T (it commutes with any polynomial in itself), so A is a *normal* matrix in that basis. Take φ with matrix $A_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ in the standard basis (a nonzero nilpotent operator, $A_0^2 = 0$). Any matrix B representing φ in another basis is similar to A_0, hence also satisfies $B^2 = 0$ and $B \neq 0$; by Cayley–Hamilton such B has $\operatorname{tr} B = 0, \det B = 0$. Since $B^2 = 0$, every polynomial reduces to a linear one: $p(B) = c_0 I + c_1 B$ for $c_0 = p(0), c_1 = p'(0)$. Write $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$ with $b_{11} + b_{22} = 0$, $b_{11}b_{22} = b_{12}b_{21}$. Matching $B^T = c_0 I + c_1 B$ entrywise gives $b_{21} = c_1 b_{12}$ and $b_{12} = c_1 b_{21}$, so $b_{12}(1 - c_1^2) = 0$; since $B \neq 0$ forces $(b_{12}, b_{21}) \neq (0,0)$ (otherwise trace/det = 0 would force $B = 0$ too), we need $c_1 = \pm 1$.

- $c_1 = 1$: the diagonal equations give $c_0 = 0$ and $b_{21} = b_{12}$; combined with $b_{11} = -b_{22}$ and $b_{11}b_{22} = b_{12}b_{21} = b_{12}^2$, this forces $-b_{22}^2 = b_{12}^2$, impossible over $\mathbb{R}$ unless $b_{12} = b_{22} = 0$, which forces $B = 0$.
- $c_1 = -1$: similarly forces $b_{11} = b_{22} = 0$ and then $b_{12}^2 = -b_{11}b_{22} = 0$, again $B = 0$.

Both cases force $B = 0$, contradicting $B \neq 0$. So *no* basis makes φ's matrix satisfy $A^T = p(A)$ for any polynomial p:

$$\boxed{\text{no, not for every operator}}$$

— this nilpotent φ is a counterexample. □

Example 1.19 (An operator built by patching rows, and how many eigenvalues it can have)**.** For $n \times n$ matrices A, B, define $A \star B$ by $(A \star B)_{ij} = (AB)_{ij}$ if i is odd, and $(A \star B)_{ij} = b_{ij}$ if i is even. This makes $\Phi_A : B \mapsto A \star B$ a linear operator on the n^2-dimensional space $\mathrm{Mat}_n(\mathbb{R})$. (a) Can Φ_A have eigenvalue 2 for some A? (b) What is the largest possible number of distinct eigenvalues of Φ_A, for fixed n?

Solution. **Reformulating Φ_A.** Row i of $A \star B$ is $(\mathrm{row}_i A)B$ if i is odd, and just row i of B if i is even. Let M be the matrix obtained from A by replacing each even-indexed row with the corresponding standard basis row e_i^T. Then row i of MB equals $(\mathrm{row}_i M)B$; for odd i this is $(\mathrm{row}_i A)B$ (since row i of M still equals row i of A there), and for even i it is $e_i^T B = \mathrm{row}_i B$. So

$$A \star B = MB \qquad \text{for all } B,$$

i.e. Φ_A is exactly *left multiplication by* M. Left multiplication by a fixed matrix M acts identically and independently on each of the n columns of B (column j of MB is M times column j of B), so the eigenvalues of Φ_A (as an operator on the n^2-dimensional space of matrices) are exactly the eigenvalues of M, each occurring with n times the multiplicity it has in M. In particular the *distinct* eigenvalues of Φ_A are exactly the distinct eigenvalues of M.

(a) Since odd rows of M are free (they equal A's rows, and A is arbitrary), choose A's odd rows so that M is diagonal with a 2 somewhere on the diagonal — e.g. row 1 of A equal to $(2, 0, \dots, 0)$, and every other odd row of A equal to the matching standard basis row. Then $M = \mathrm{diag}(2, 1, 1, \dots, 1)$ (the even rows are forced to identity rows regardless), which has eigenvalue 2. So

$$\boxed{\text{yes, eigenvalue 2 is achievable}}.$$

(b) For each even index i, row i of M is e_i^T, which says exactly that $e_i^T M = e_i^T$ — i.e. e_i^T is a *left* eigenvector of M with eigenvalue 1. The vectors e_i^T for the $\lfloor n/2 \rfloor$ even indices are linearly independent, so eigenvalue 1 has geometric (hence algebraic) multiplicity at least $\lfloor n/2 \rfloor$ in M.

If $\lfloor n/2 \rfloor \le 1$ (i.e. $n \le 3$), this is not a real restriction (a single guaranteed copy of eigenvalue 1 is compatible with all n eigenvalues being distinct), and choosing M diagonal with n distinct entries — forced to 1 only where required — achieves n distinct eigenvalues.

If $\lfloor n/2 \rfloor \ge 2$ (i.e. $n \ge 4$), the n eigenvalues (with multiplicity) include at least $\lfloor n/2 \rfloor$ copies of 1, leaving at most $n - \lfloor n/2 \rfloor = \lceil n/2 \rceil$ "slots" that could be distinct from 1 and from each other — so at most $\lceil n/2 \rceil + 1$ distinct eigenvalues in total. This is achieved: take M upper triangular with the (forced) even diagonal entries equal to 1 and the free odd diagonal entries set to $\lceil n/2 \rceil$ pairwise distinct values, none equal to 1 (e.g. $2, 3, 4, \dots$), and all free odd-row off-diagonal entries 0 — a valid choice since odd rows are unconstrained. The eigenvalues of this triangular M are exactly its diagonal entries: 1 (with multiplicity $\lfloor n/2 \rfloor$) together with $\lceil n/2 \rceil$ distinct values $\ne 1$, giving exactly $\lceil n/2 \rceil + 1$ distinct eigenvalues.

So the maximum number of distinct eigenvalues of Φ_A is

$$\boxed{n \text{ for } n \le 3, \qquad \left\lceil \frac{n}{2} \right\rceil + 1 \text{ for } n \ge 4}.$$

□

Example 1.20 (Eigenvalues of an operator whose cube is a projection)**.** A linear operator $A : \mathbb{R}^n \to \mathbb{R}^n$ satisfies: A^3 is a projection operator. What eigenvalues can A have? Must A be diagonalizable in some basis of $\mathbb{R}^n$?

Solution. "A^3 is a projection" means $(A^3)^2 = A^3$, i.e. $A^6 = A^3$, i.e. $A^3(A^3 - I) = 0$. So the minimal polynomial of A divides

$$x^3(x^3 - 1) = x^3(x - 1)(x^2 + x + 1).$$

Over $\mathbb{C}$, the roots of this polynomial are 0 (from x^3), 1, and the two primitive cube roots of unity $\omega, \overline{\omega} = \omega^2$ (roots of $x^2 + x + 1$). So the eigenvalues of A can only be

$$\boxed{0,\ 1,\ \omega,\ \omega^2} \qquad \left(\omega = e^{2\pi i/3}\right),$$

and since A is a *real* matrix, if ω occurs as an eigenvalue so must its conjugate ω^2.

Not necessarily diagonalizable. Two explicit examples show A need not have a diagonal matrix in any real basis:

- $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$: here $A^3 = 0$, and 0 is trivially a projection (onto $\{0\}$), so the hypothesis holds. But A is a nontrivial nilpotent Jordan block — not diagonalizable even over $\mathbb{C}$.
- A = rotation by 120° in $\mathbb{R}^2$: $A^3 = I$ (a full turn), and I is trivially a projection. Here A^3 satisfies the hypothesis, but A's eigenvalues are the genuinely complex ω, ω^2, so A cannot be diagonalized over $\mathbb{R}$ at all — no real basis makes its matrix diagonal.

So

$$\boxed{\text{no, } A \text{ need not be diagonalizable in any basis of } \mathbb{R}^n}.$$

□

Example 1.21 (Matrices invariant under a 90° rotation of their entries)**.** Call an $n \times n$ matrix *rotational* if rotating its grid of entries by 90° about the center leaves it unchanged. (a) For any reals $\lambda_1, \dots, \lambda_k$, show there is an n and a rotational $n \times n$ matrix having $\lambda_1, \dots, \lambda_k$ among its eigenvalues. (b) Show that every real eigenvector v of a rotational matrix, for a nonzero real eigenvalue, is symmetric: $v_i = v_{n-i+1}$.

Solution. **Reformulating "rotational."** A 90° rotation sends the entry at (i, j) to where the entry at $(n + 1 - j, i)$ used to be, so A is rotational iff $A_{i,j} = A_{n+1-j,i}$ for all i, j. Let J be the reversal (anti-diagonal) permutation matrix, $(Jv)_i = v_{n+1-i}$; one checks $J^T = J$, $J^2 = I$. The rotational condition is exactly

$$AJ = A^T$$

(entrywise, $(AJ)_{i,j} = A_{i,n+1-j}$, and setting this equal to $(A^T)_{i,j} = A_{j,i}$ and relabeling recovers $A_{i,j} = A_{n+1-j,i}$).

A **commutes with** J. Transposing $AJ = A^T$ gives $JA^T = A$ (using $J^T = J$), i.e. $A^T = JA$. Combined with $AJ = A^T$, this gives $AJ = JA$: every rotational matrix commutes with J.

Block structure. Since J is a symmetric orthogonal involution, $\mathbb{R}^n = W_+ \oplus W_-$ orthogonally, where $W_\pm = \{v : Jv = \pm v\}$ ($\dim W_+ = \lceil n/2 \rceil$, $\dim W_- = \lfloor n/2 \rfloor$). Since A commutes with J, it preserves W_+ and W_-, so in an orthonormal basis adapted to this splitting, $A = \mathrm{diag}(B_+, B_-)$ and $J = \mathrm{diag}(I, -I)$. Substituting into $A^T = JA$ (transpose of a block-diagonal matrix, in an orthonormal adapted basis, is block-diagonal in the transposed blocks):

$$\mathrm{diag}(B_+^T, B_-^T) = \mathrm{diag}(I, -I)\,\mathrm{diag}(B_+, B_-) = \mathrm{diag}(B_+, -B_-),$$

so $B_+^T = B_+$ (*symmetric*) and $B_-^T = -B_-$ (*antisymmetric*). Conversely any symmetric B_+ and antisymmetric B_- assembled this way, then rotated back to the standard basis, give a valid rotational matrix (verified directly by construction).

(a) Take $n = 2k$, so $\dim W_+ = \dim W_- = k$. Let $B_+ = \mathrm{diag}(\lambda_1, \dots, \lambda_k)$ (symmetric — it's diagonal) and $B_- = 0$ ($k \times k$, trivially antisymmetric). The resulting rotational matrix has eigenvalues $\lambda_1, \dots, \lambda_k$ (from B_+) together with k copies of 0 (from B_-):

$$\boxed{\text{such } n \text{ and } A \text{ always exist}}.$$

(b) Let $Av = \lambda v$, v real, $\lambda \neq 0$ real. Since $A^T = JA$, $A(Jv) = (AJ)v = A^T v = (JA)v = J(Av) = \lambda(Jv)$: so $w := Jv$ is also a λ-eigenvector (or 0; but $w = 0 \Rightarrow v = Jw = 0$, excluded). Let $z = v - w = v - Jv$; being a linear combination of two λ-eigenvectors, $Az = \lambda z$ if $z \neq 0$. But directly, $Jz = Jv - J^2 v = Jv - v = -z$ always (no eigenvector assumption needed). If $z \neq 0$, apply the general fact (from $A^T Av$-type identities, or directly: $v^T A^T v = (Av)^T v = \lambda \|v\|^2$, and also $v^T A^T v = v^T (AJ)v = v^T Aw = v^T(\lambda w) = \lambda(v^T Jv)$, giving $v^T Jv = \|v\|^2$ since $\lambda \neq 0$) applied to z in place of v: $z^T Jz = \|z\|^2$. But $Jz = -z$ gives $z^T Jz = -\|z\|^2$. So $\|z\|^2 = -\|z\|^2$, forcing $z = 0$. Hence $v = Jv$, i.e.

$$\boxed{v_i = v_{n-i+1} \text{ for all } i}.$$

□

Example 1.22 (Eigenvalues of a rotation-type differential operator)**.** On the space $\mathbb{R}[x, y]$ of real polynomials in x, y, consider the operator $D = y\dfrac{\partial}{\partial x} + x\dfrac{\partial}{\partial y}$. Show every integer is an eigenvalue of D. Find all its eigenvalues, and determine whether D is diagonalizable.

Solution. **A change of variables.** Let $u = x + y$, $v = x - y$. By the chain rule, $\partial_x = \partial_u + \partial_v$ and $\partial_y = \partial_u - \partial_v$ (since $\partial u/\partial x = \partial u/\partial y = 1$ and $\partial v/\partial x = 1, \partial v/\partial y = -1$). Substituting,

$$D = y(\partial_u + \partial_v) + x(\partial_u - \partial_v) = (x + y)\partial_u - (x - y)\partial_v = u\,\partial_u - v\,\partial_v.$$

Diagonal action on monomials. In the coordinates (u, v) — a linear, hence invertible, change of variables, so $\{u^a v^b : a, b \geq 0\}$ is still a basis of $\mathbb{R}[x, y]$ — the operator acts diagonally:

$$D(u^a v^b) = a\,u^a v^b - b\,u^a v^b = (a - b)\,u^a v^b.$$

So every monomial $u^a v^b$ is an eigenvector of D with eigenvalue $a - b$.

(a) Every integer occurs. For any $m \in \mathbb{Z}$, taking $a = \max(m, 0)$, $b = \max(-m, 0)$ gives $a - b = m$, so $u^a v^b$ is an eigenvector with eigenvalue m.

(b) These are all the eigenvalues, and D is diagonalizable. Since $\{u^a v^b\}_{a,b \geq 0}$ is a basis on which D acts diagonally, D is diagonalizable by definition (this eigenbasis exhibits it). Moreover any eigenvector of D, expanded in this basis as a finite sum $\sum c_{a,b} u^a v^b$, satisfies $D(\text{it}) = \sum c_{a,b}(a - b)u^a v^b$; for this to equal $\lambda \sum c_{a,b} u^a v^b$, every term with $c_{a,b} \neq 0$ must have $a - b = \lambda$, so no eigenvalue outside $\{a - b : a, b \geq 0\} = \mathbb{Z}$ can occur. So

$$\boxed{\text{the eigenvalues of } D \text{ are exactly all of } \mathbb{Z}, \text{ and } D \text{ is diagonalizable}}$$

(each eigenvalue $m \in \mathbb{Z}$ has an infinite-dimensional eigenspace, spanned by $\{u^{m+b} v^b : b \geq 0\}$ for $m \geq 0$, or $\{u^a v^{a-m} : a \geq 0\}$ for $m < 0$). □

Example 1.23 (Eigenvalues of an antisymmetric block matrix built from A)**.** A square matrix A of size $n \times n$ has distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ (possibly complex). Find all eigenvalues, including complex ones, of $\begin{pmatrix} 0 & -A \\ A & 0 \end{pmatrix}$.

Solution. Let $M = \begin{pmatrix} 0 & -A \\ A & 0 \end{pmatrix}$, a $2n \times 2n$ matrix, and let v be an eigenvector of A with $Av = \lambda v$.

Constructing eigenvectors of M directly. Try $M\binom{v}{cv}$ for a scalar c to be determined:

$$M\begin{pmatrix} v \\ cv \end{pmatrix} = \begin{pmatrix} -A(cv) \\ Av \end{pmatrix} = \begin{pmatrix} -c\lambda v \\ \lambda v \end{pmatrix}.$$

For this to equal $\mu\binom{v}{cv} = \binom{\mu v}{\mu cv}$, we need $-c\lambda = \mu$ and $\lambda = \mu c$. Substituting the first into the second: $\lambda = (-c\lambda)c = -c^2\lambda$, so (for $\lambda \neq 0$) $c^2 = -1$, i.e. $c = \pm i$. Each choice gives a genuine eigenvector: $c = -i$ gives $\mu = -c\lambda = i\lambda$, and $c = i$ gives $\mu = -i\lambda$. So both $\pm i\lambda$ are eigenvalues of M, with explicit eigenvectors $\binom{v}{\mp iv}$.

The case $\lambda = 0$. If some $\lambda_k = 0$ with eigenvector v, then $M\binom{av}{bv} = \binom{-A(bv)}{A(av)} = \binom{0}{0}$ for *any* scalars a, b (since $Av = 0$), so $\mu = 0$ is an eigenvalue of M with (at least) a 2-dimensional eigenspace $\{(av, bv)\}$ — consistent with the general pattern, since $+i \cdot 0 = -i \cdot 0 = 0$ coincide.

Accounting for all $2n$ eigenvalues. Since A has n distinct eigenvalues $\lambda_1, \dots, \lambda_n$, each contributes the pair $\pm i\lambda_k$ via the construction above, giving $2n$ eigenvalues (as a multiset, with repetitions exactly when $\lambda_j = -\lambda_k$ for some $j \neq k$, or when a $\lambda_k = 0$). This already accounts for all $2n$ eigenvalues of the $2n \times 2n$ matrix M (matching, e.g., $M^2 = \begin{pmatrix} -A^2 & 0 \\ 0 & -A^2 \end{pmatrix}$, whose eigenvalues are $-\lambda_k^2$ each with multiplicity 2 — exactly $(\pm i\lambda_k)^2$). So

$$\boxed{\text{the eigenvalues of } M \text{ are exactly } \pm i\lambda_1,\ \pm i\lambda_2,\ \dots,\ \pm i\lambda_n}$$

(verified numerically against direct eigenvalue computation for random real 4×4 matrices A). ☐

Example 1.24 (A value an integer matrix's eigenvalues can never take)**.** Prove that no matrix with integer entries can have $\lambda = \dfrac{-3 + i\sqrt{5}}{4}$ as an eigenvalue.

Solution. **Eigenvalues of integer matrices are algebraic integers.** If A is an $n \times n$ integer matrix, its characteristic polynomial $\det(xI - A)$ is monic with integer coefficients (each coefficient is an integer polynomial expression in the entries of A). So every eigenvalue of A is a root of some monic polynomial with integer coefficients — i.e. an *algebraic integer*.

λ is not an algebraic integer. From $4\lambda + 3 = i\sqrt{5}$, squaring gives $16\lambda^2 + 24\lambda + 9 = -5$, i.e.

$$8\lambda^2 + 12\lambda + 7 = 0.$$

This quadratic has no rational root (its roots are non-real) and $\gcd(8, 12, 7) = 1$, so it is λ's minimal polynomial over $\mathbb{Q}$ up to scalar — dividing by the leading coefficient 8 to make it monic gives $\lambda^2 + \frac{3}{2}\lambda + \frac{7}{8} = 0$, which does *not* have integer coefficients. Since the minimal polynomial of an algebraic integer must be monic with integer coefficients whenever such a polynomial exists (a standard fact: this monic integer minimal polynomial is unique when it exists), and here it manifestly fails to be integral, λ is not an algebraic integer.

Since every eigenvalue of an integer matrix must be an algebraic integer, and λ is not one,

$$\boxed{\lambda = \tfrac{1}{4}(-3 + i\sqrt{5}) \text{ is never an eigenvalue of an integer matrix}}.$$

☐

Example 1.25 (A pairing strategy forces eigenvalue 1)**.** Dima and Vanya alternately write real numbers into the entries of an (empty) $2n \times 2n$ matrix, one entry per turn, Dima moving first, until every entry is filled. Vanya wants the final matrix to have eigenvalue 1; Dima wants to prevent this. Does either player have a winning strategy?

Solution. Vanya (the second player) wins, via a pairing strategy.

The pairing. In each of the $2n$ rows, pair up its $2n$ entries into n pairs of two cells each (e.g. consecutive columns $(1, 2), (3, 4), \dots$). This partitions all $4n^2$ cells into $2n^2$ pairs.

The strategy. Whenever Dima fills some cell, Vanya immediately fills the *other* cell of that same pair, choosing its value so that the pair's two entries sum to exactly $\frac{1}{n}$. This is always possible (Vanya can write any real number), and it is always available: since Vanya closes every pair the instant Dima opens it, Dima's next move necessarily opens a fresh, untouched pair.

Why this forces eigenvalue 1. By the end of the game, every one of the n pairs within each row sums to $\frac{1}{n}$, so each row's $2n$ entries sum to $n \cdot \frac{1}{n} = 1$. That means $A\mathbf{1} = \mathbf{1}$ for the all-ones vector $\mathbf{1}$ — i.e. $\mathbf{1}$ is an eigenvector of A with eigenvalue 1, regardless of any value Dima ever chooses. So

$$\boxed{\text{Vanya (the second player) has a winning strategy}}.$$

□

Example 1.26 (Asymptotics of a circular gold-dust averaging process)**.** n prospectors sit around a table, each with a pile of gold dust. Every minute, simultaneously, each prospector's pile is replaced by the average of their two neighbors' current piles (i.e. half of the left neighbor's pile plus half of the right neighbor's, as those piles stood at the start of the minute). Describe the long-run behavior of the piles (a) for $n = 3$; (b) for general n.

Solution. Let $p^{(t)} \in \mathbb{R}^n$ be the vector of pile sizes after t minutes, so $p^{(t+1)} = Mp^{(t)}$ where M is the circulant matrix with $M_{i,i-1} = M_{i,i+1} = \frac{1}{2}$ (indices mod n) and 0 elsewhere.

Eigenvalues of a circulant. M's eigenvectors are $v_j^{(k)} = \omega^{jk}$ for $\omega = e^{2\pi i/n}$, $k = 0, \dots, n-1$, with eigenvalues

$$\lambda_k = \frac{1}{2}(\omega^k + \omega^{-k}) = \cos\left(\frac{2\pi k}{n}\right).$$

The total $\sum_i p_i$ is conserved exactly ($\lambda_0 = 1$, eigenvector all-ones — averaging redistributes but never creates or destroys sand). Every other eigenvalue satisfies $|\lambda_k| < 1$, *unless* n is even and $k = n/2$: then $\cos(2\pi k/n) = \cos\pi = -1$, giving eigenvalue $\lambda_{n/2} = -1$ with eigenvector $(1, -1, 1, -1, \dots)$.

(a) $n = 3$ **(odd).** All nonzero-index eigenvalues ($\lambda_1 = \lambda_2 = \cos(2\pi/3) = -\frac{1}{2}$) have magnitude < 1, so those components of $p^{(t)}$ decay geometrically to 0. Writing $S = p_1^{(0)} + p_2^{(0)} + p_3^{(0)}$ for the (conserved) total,

$$\boxed{p_i^{(t)} \to S/3 \text{ for every } i}$$

— the piles equalize (e.g. from $(10, 0, 0)$ they converge to $(10/3, 10/3, 10/3)$).

(b) General n**.** If n is *odd*, no eigenvalue equals -1 (as $n/2$ isn't an integer), so exactly as above, all non-total modes decay and $p_i^{(t)} \to S/n$ for every i: the piles always equalize.

If n is *even*, the mode $\lambda_{n/2} = -1$ does *not* decay: writing $c = \frac{1}{n}\sum_j (-1)^j p_j^{(0)}$ for the initial alternating component, all other modes still decay to 0, so

$$\boxed{p_i^{(t)} \;\longrightarrow\; \frac{S}{n} + c\,(-1)^{i+t}}$$

— the piles approach a persistent period-2 alternation between "high" piles ($S/n+c$) and "low" piles ($S/n - c$) on alternating seats and alternating minutes, never settling to a fixed vector unless $c = 0$ (verified directly: $n = 4$ starting from $(10, 0, 0, 0)$ settles into the exact repeating pattern $(0, 5, 0, 5), (5, 0, 5, 0), (0, 5, 0, 5), \dots$ forever). □

Example 1.27 (Eigenvalues of a rank-one outer product)**.** Let v be a nonzero column vector in $\mathbb{R}^n$. Find the eigenvalues of vv^T.

Solution. vv^T has rank 1 (every column is a scalar multiple of v). Two direct computations pin down all n eigenvalues:

v **itself is an eigenvector.** $(vv^T)v = v(v^Tv) = \|v\|^2 v$, so v is an eigenvector with eigenvalue $\|v\|^2$.

Everything orthogonal to v **is killed.** For $w \perp v$ (i.e. $v^Tw = 0$), $(vv^T)w = v(v^Tw) = 0$, so w is an eigenvector with eigenvalue 0. The orthogonal complement of $\text{span}(v)$ has dimension $n - 1$, giving $n - 1$ independent eigenvectors for eigenvalue 0.

Together with v's eigenvector, this accounts for all n eigenvalues (in fact a full eigenbasis, since $\mathbb{R}^n = \operatorname{span}(v) \oplus v^\perp$ orthogonally):

$$\boxed{\text{eigenvalues } \|v\|^2 \text{ (multiplicity 1) and 0 (multiplicity } n-1)}$$

(consistent with $\operatorname{tr}(vv^T) = v^T v = \|v\|^2$, matching the sum of these eigenvalues). □

Example 1.28 (Rank of $E - A$ from $A^2 = E$ and $\operatorname{rank}(E + A) = 7$)**.** Let A be a 9×9 matrix over a field of characteristic $\neq 2$, with $A^2 = E$ (the identity). Given $\operatorname{rank}(E+A) = 7$, find $\operatorname{rank}(E-A)$.

Solution. Since $A^2 = E$, the minimal polynomial of A divides $x^2 - 1 = (x-1)(x+1)$, which has distinct roots because the characteristic is $\neq 2$. So A is diagonalizable with eigenvalues only ± 1: let p be the multiplicity of $+1$ and q the multiplicity of -1, $p + q = 9$.

In a diagonalizing basis, $E + A$ has diagonal entries $1 + 1 = 2$ (on the p eigenvectors for $+1$) and $1 + (-1) = 0$ (on the q eigenvectors for -1). Since the characteristic is $\neq 2$, the entry 2 is nonzero in the field, so $\operatorname{rank}(E + A) = p$. Given $\operatorname{rank}(E + A) = 7$, this gives $p = 7$, hence $q = 9 - 7 = 2$.

Likewise $E - A$ has diagonal entries $1 - 1 = 0$ (multiplicity $p = 7$) and $1 - (-1) = 2 \neq 0$ (multiplicity $q = 2$) in that same basis, so

$$\boxed{\operatorname{rank}(E - A) = q = 2}.$$

□

Example 1.29 (Eigenvalues of the operator $M \mapsto AMB$)**.** Let A, B be 2×2 matrices, and let F act on the space of 2×2 matrices by $F(M) = AMB$. Suppose A has two distinct eigenvalues λ_1, λ_2 and B has two distinct eigenvalues μ_1, μ_2. Find the eigenvalues of F when (a) A, B are diagonal; (b) A, B are arbitrary.

Solution. **(a) Diagonal case.** Let $A = \operatorname{diag}(\lambda_1, \lambda_2)$, $B = \operatorname{diag}(\mu_1, \mu_2)$, and let E_{ij} denote the matrix unit (1 in position (i,j), 0 elsewhere). Left-multiplying by diagonal A scales row i by λ_i, and right-multiplying by diagonal B scales column j by μ_j; since E_{ij}'s only nonzero entry is in row i, column j,

$$F(E_{ij}) = AE_{ij}B = \lambda_i \mu_j E_{ij}.$$

So each matrix unit E_{ij} is an eigenvector of F, with eigenvalue $\lambda_i \mu_j$. Ranging over $i, j \in \{1, 2\}$ gives all 4 eigenvalues, accounting for the full 4-dimensional space of 2×2 matrices:

$$\lambda_1\mu_1, \quad \lambda_1\mu_2, \quad \lambda_2\mu_1, \quad \lambda_2\mu_2.$$

(b) General case, via similarity. Since A, B each have 2 distinct eigenvalues, they're diagonalizable: $A = PDP^{-1}$, $B = QD'Q^{-1}$ with $D = \operatorname{diag}(\lambda_1, \lambda_2)$, $D' = \operatorname{diag}(\mu_1, \mu_2)$. Define the invertible linear map T on the space of 2×2 matrices by $T(M) = P^{-1}MQ$. Then

$$T(F(M)) = P^{-1}(AMB)Q = P^{-1}(PDP^{-1}MQD'Q^{-1})Q = D(P^{-1}MQ)D'.$$

The right side is $D \cdot T(M) \cdot D' = \Psi(T(M))$, where $\Psi(N) = DND'$ is exactly the diagonal-case operator from part (a). So $T \circ F = \Psi \circ T$, i.e. $F = T^{-1}\Psi T$: F is similar to Ψ (as linear operators on the same 4-dimensional space), hence has the *same* eigenvalues as Ψ.

$$\boxed{\text{eigenvalues of } F \text{ are } \lambda_1\mu_1,\ \lambda_1\mu_2,\ \lambda_2\mu_1,\ \lambda_2\mu_2 \text{ in both (a) and (b)}}$$

(As a direct check: if $Av = \lambda v$ and $w^T B = \mu w^T$ for a row vector w^T, i.e. w is an eigenvector of B^T for μ, then $M = vw^T$ satisfies $F(M) = A(vw^T)B = (Av)(w^T B) = \lambda\mu\, vw^T$ — an explicit rank-1 eigenvector for each of the 4 eigenvalue products, matching the similarity argument.) □

1.4 Bilinear Forms, Euclidean Spaces, and Hermitian Structure

Definitions and formulas.

- A bilinear form on $\mathbb{R}^n$ is a map $B(x,y) = x^T Ay$ that is linear in each argument. If $A = A^T$, then the associated quadratic form is $Q(x) = x^T Ax$. The quadratic form records what the bilinear form does when both inputs are the same vector.
- Over $\mathbb{R}$, a symmetric bilinear form and its quadratic form determine each other through the polarization identity
$$B(x,y) = \frac{1}{2}(Q(x+y) - Q(x) - Q(y)).$$
This is the precise connection between bilinear and quadratic viewpoints.
- Under a change of basis with matrix C, the matrix of a bilinear form transforms by congruence:
$$A_{\text{new}} = C^T A_{\text{old}} C.$$
Congruence is the correct notion because both inputs of the form are being re-expressed in the new coordinates.
- A symmetric matrix is positive definite if $x^T Ax > 0$ for all $x \neq 0$. Sylvester's criterion says this holds if and only if all leading principal minors are positive. Positive definiteness means the quadratic form behaves like a genuine squared length.
- In Euclidean space,
$$|\langle u,v\rangle| \le \|u\|\,\|v\|,$$
and if $G = (\langle v_i, v_j\rangle)$ is the Gram matrix, then the squared volume of the parallelotope spanned by $v_1, \dots, v_k$ is $\det(G)$. The Gram matrix packages all pairwise inner products at once.
- Gram–Schmidt orthogonalization turns a basis into an orthogonal basis. Orthogonal projections onto $\operatorname{span}\{q_1, \dots, q_k\}$ are computed by
$$P = Q(Q^T Q)^{-1} Q^T,$$
where $Q = [q_1 \ \cdots \ q_k]$. Orthogonal coordinates simplify both geometry and computation.
- In complex vector spaces, Hermitian matrices satisfy $A = A^*$ and have real eigenvalues with orthonormal eigenvectors. They are the complex analog of real symmetric matrices.

Example 1.30 (Why $A^T A$ is always positive semidefinite)**.** Prove that for every real matrix A, the matrix $A^T A$ is positive semidefinite.

Solution. Take any vector $x \in \mathbb{R}^n$. Then
$$x^T A^T Ax = (Ax)^T (Ax) = \|Ax\|_2^2 \ge 0.$$
Therefore $x^T A^T Ax \ge 0$ for all x, which is exactly the definition of positive semidefinite. Equality holds if and only if $Ax = 0$. □

Example 1.31 (A bilinear-form matrix that is similar to an idempotent)**.** Let $Q : V \times W \to \mathbb{R}$ be a bilinear map. In bases α of V and β of W, write
$$Q(v,w) = v_\alpha^T Q_{\alpha,\beta} w_\beta.$$
Now regard $Q_{\alpha,\beta}$ as the matrix of a linear operator on a 4-dimensional real space. Suppose that in some other basis this operator has matrix B satisfying
$$B^2 = B.$$

It is also known that in some pair of bases ω, γ for the bilinear map,

$$Q_{\omega,\gamma} = \begin{pmatrix} 2 & 0 & 2 & 3 \\ 0 & 0 & 0 & 2 \\ 4 & 0 & 4 & 5 \\ 2 & 0 & 2 & 2 \end{pmatrix}.$$

Find the maximum and minimum possible dimensions of an eigenspace of this operator.

Solution. For a bilinear map, changing the basis in V and the basis in W acts by left and right multiplication with invertible matrices:

$$Q_{\omega,\gamma} = C^T Q_{\alpha,\beta} D.$$

Therefore the rank of the matrix is invariant under such changes of bases.

Now inspect the given matrix

$$M = \begin{pmatrix} 2 & 0 & 2 & 3 \\ 0 & 0 & 0 & 2 \\ 4 & 0 & 4 & 5 \\ 2 & 0 & 2 & 2 \end{pmatrix}.$$

Its second column is zero and its third column equals its first column, so $\operatorname{rank}(M) \leq 2$. On the other hand, the first and fourth columns are linearly independent, so $\operatorname{rank}(M) = 2$. Hence

$$\operatorname{rank}(Q_{\alpha,\beta}) = 2.$$

Now switch to the operator point of view. The matrix $Q_{\alpha,\beta}$ is similar to B, and similarity preserves rank, so

$$\operatorname{rank}(B) = 2.$$

Because $B^2 = B$, the operator is idempotent. For any idempotent operator T,

$$\operatorname{im}(T) = E_1(T), \qquad \ker(T) = E_0(T).$$

Indeed, if $y = Tx$, then $Ty = T^2x = Tx = y$, so every vector in the image is a 1-eigenvector. Conversely, if $Ty = y$, then $y = Tx$ with $x = y$, so $y \in \operatorname{im}(T)$. The kernel is exactly the eigenspace for eigenvalue 0.

Therefore for B we get

$$\dim E_1(B) = \operatorname{rank}(B) = 2, \qquad \dim E_0(B) = 4 - \operatorname{rank}(B) = 2.$$

So both eigenspaces have the same dimension. Hence the maximum and minimum possible dimensions of an eigenspace are both

$$\boxed{2}.$$

Equivalently, the only possible answer is

$$\boxed{\min = \max = 2}.$$

□

Example 1.32 (Matrix-valued quadratic form induced by a positive definite matrix)**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive definite. Define

$$q(X) = \operatorname{tr}(X^T A X), \qquad X \in \mathbb{R}^{n\times m}.$$

Is q a positive definite quadratic form on the vector space $\mathbb{R}^{n\times m}$?

Solution. Yes. The expression is quadratic because for any scalar c,

$$q(cX) = \operatorname{tr}((cX)^T A(cX)) = c^2 \operatorname{tr}(X^T AX) = c^2 q(X).$$

The associated bilinear form is

$$B(X,Y) = \operatorname{tr}(X^T AY),$$

since, using the symmetry of A,

$$q(X+Y) = q(X) + q(Y) + 2\operatorname{tr}(X^T AY).$$

To check positive definiteness, write the columns of X as

$$X = [x_1 \ \cdots \ x_m].$$

Then

$$q(X) = \operatorname{tr}(X^T AX) = \sum_{j=1}^{m} x_j^T A x_j.$$

Because A is positive definite, each term satisfies $x_j^T A x_j \geq 0$, and it is strictly positive whenever $x_j \neq 0$. Therefore if $X \neq 0$, at least one column x_j is nonzero, so

$$q(X) > 0.$$

Hence q is a positive definite quadratic form on $\mathbb{R}^{n\times m}$.

There is also a useful norm interpretation. Since A is positive definite, it has a symmetric square root $A^{1/2}$, and

$$q(X) = \operatorname{tr}(X^T A^{1/2} A^{1/2} X) = \operatorname{tr}((A^{1/2}X)^T (A^{1/2}X)) = \|A^{1/2}X\|_F^2.$$

So q is literally the squared Frobenius norm after the linear change of variables $X \mapsto A^{1/2}X$. □

Example 1.33 (When do two matrices represent the same bilinear form?)**.** For which $a \in \mathbb{R}$ can

$$A = \begin{pmatrix} 1 & 4-a-a^2 \\ 2 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} -a-1 & 3 \\ 3 & -5 \end{pmatrix}$$

be matrices of the *same* bilinear form $V \times V \to \mathbb{R}$ in two different bases?

Solution. Two matrices represent the same bilinear form in different bases iff they are *congruent*: $B = C^T AC$ for some invertible C. Write $A = S + K$ as symmetric plus antisymmetric parts; since $(C^T AC)^T$-splitting commutes with congruence, $C^T AC = C^T SC + C^T KC$ is again a symmetric-plus-antisymmetric split, so congruence acts separately on S and K — in particular it preserves $\operatorname{rank} K$ (for a 2×2 antisymmetric matrix, $K = 0$ or $\operatorname{rank} K = 2$).

Here B is symmetric for every a, so $K_B = 0$, forcing $K_A = 0$ too: A must be symmetric, i.e. $4 - a - a^2 = 2$, i.e. $a^2 + a - 2 = (a-1)(a+2) = 0$, so $a \in \{-2, 1\}$. In both cases $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$ (the same matrix, since $4 - a - a^2 = 2$ at both roots): $\det A = -5 < 0$, so A has signature $(1,1)$.

By Sylvester's law of inertia, two real symmetric matrices are congruent iff they have the same rank and signature, so we just need B's signature at each candidate:

- $a = -2$: $B = \begin{pmatrix} 1 & 3 \\ 3 & -5 \end{pmatrix}$, $\det B = -5 - 9 = -14 < 0$, signature $(1,1)$ — matches A.
- $a = 1$: $B = \begin{pmatrix} -2 & 3 \\ 3 & -5 \end{pmatrix}$, $\det B = 10 - 9 = 1 > 0$ with $\operatorname{tr} B = -7 < 0$, so both eigenvalues are negative: signature $(0,2)$ — does *not* match A.

So only $a = -2$ works:

$$\boxed{a = -2}.$$

$\square$

Example 1.34 (Recovering a column of an orthogonal projection matrix)**.** The matrix

$$\frac{1}{6}\begin{pmatrix} 5 & -2 & ? \\ -2 & 2 & ? \\ -1 & -2 & ? \end{pmatrix}$$

is known to be the matrix of the orthogonal projection onto some plane through the origin in $\mathbb{R}^3$. Fill in the third column.

Solution. An orthogonal projection matrix P is characterized by $P^T = P$ (symmetric) and $P^2 = P$ (idempotent). Symmetry alone pins down two of the three unknown entries: the $(1,3)$ entry must equal the given $(3,1)$ entry $-\frac{1}{6}$, and the $(2,3)$ entry must equal the given $(3,2)$ entry $-\frac{2}{6}$. So the column is $\frac{1}{6}(-1,-2,c)^T$ for some remaining unknown c.

For the last entry, use that a projection onto a 2-dimensional subspace has $\operatorname{rank} P = 2$, and for an idempotent matrix $\operatorname{rank} P = \operatorname{tr} P$ (its eigenvalues are all 0 or 1, and the trace counts the 1's). So

$$\operatorname{tr} P = \frac{5+2+c}{6} = 2 \quad \Longrightarrow \quad c = 5.$$

So the completed matrix is

$$P = \frac{1}{6}\begin{pmatrix} 5 & -2 & -1 \\ -2 & 2 & -2 \\ -1 & -2 & 5 \end{pmatrix}, \qquad \boxed{\text{third column } = \tfrac{1}{6}(-1,-2,5)^T}.$$

(One checks directly that this P is symmetric and satisfies $P^2 = P$, confirming it is indeed a valid orthogonal projection.) $\square$

Example 1.35 (Orthogonal skew-symmetric matrices and dimension parity)**.** Does there exist a real 2019×2019 orthogonal matrix that is also skew-symmetric? What about 2018×2018?

Solution. Let Q be orthogonal ($Q^T Q = I$) and skew-symmetric ($Q^T = -Q$). Combining, $-Q \cdot Q = I$, i.e. $Q^2 = -I$.

Odd size. Any real matrix of odd size has at least one real eigenvalue (its characteristic polynomial has odd degree, hence a real root by the Intermediate Value Theorem). If $Q^2 = -I$, every eigenvalue λ (real or complex) satisfies $\lambda^2 = -1$, i.e. $\lambda = \pm i$ — never real. So no such Q can have odd size:

$$\boxed{\text{no orthogonal skew-symmetric } 2019 \times 2019 \text{ matrix exists}}.$$

Even size. For $2018 = 2 \times 1009$, take the block-diagonal matrix with 1009 copies of $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ along the diagonal. Each 2×2 block is orthogonal (it is a rotation by $90°$) and skew-symmetric, so the whole block-diagonal matrix is too:

$$\boxed{\text{yes, such a } 2018 \times 2018 \text{ matrix exists}}.$$

$\square$

Example 1.36 (A determinant identity for two (unrelated) orthogonal matrices)**.** Let A, B be $n \times n$ orthogonal matrices (not necessarily orthogonal to each other — just each individually orthogonal). Prove

$$\det(A^T B - B^T A) = \det(A + B) \cdot \det(A - B).$$

Solution. Let $M = A^T B$. Since A, B are orthogonal, M is orthogonal too, and $M^{-1} = B^{-1}A = B^T A$. So

$$A^T B - B^T A = M - M^{-1}.$$

Also, since A is invertible, $A + B = A(I + A^{-1}B) = A(I + M)$ and $A - B = A(I - M)$, so

$$\det(A + B)\det(A - B) = \det(A)^2 \det(I + M)\det(I - M) = \det(A)^2 \det(I - M^2)$$

(using $(I + M)(I - M) = I - M^2$, valid regardless of commutativity since it's the same M on both sides).

For the left side, $M - M^{-1} = M^{-1}(M^2 - I)$, so

$$\det(A^T B - B^T A) = \det(M)^{-1}\det(M^2 - I) = \det(M)^{-1}(-1)^n \det(I - M^2).$$

Case 1: M **has no real eigenvalue.** As a real orthogonal matrix, M's eigenvalues are either ± 1 or come in complex-conjugate pairs $e^{\pm i\theta}$ on the unit circle. If none are real, all n eigenvalues pair up, forcing n even, and $\det(M) = \prod \lambda_i = 1$ (each conjugate pair contributes $e^{i\theta}e^{-i\theta} = 1$). So $\det(M)^{-1}(-1)^n = 1 \cdot 1 = 1 = \det(A)^2$ (always, since $\det(A) = \pm 1$), giving $\det(A^T B - B^T A) = \det(I - M^2) = \det(A)^2 \det(I - M^2) = \det(A + B)\det(A - B)$: the identity holds.

Case 2: M **has a real eigenvalue** $\lambda = \pm 1$**.** Then M^{-1} has eigenvalue $1/\lambda = \lambda$ too, so $M - M^{-1}$ has eigenvalue $\lambda - \lambda = 0$: $M - M^{-1}$ is singular, so the left side is 0. Also $I - M^2$ has eigenvalue $1 - \lambda^2 = 0$ at the same eigenvector, so $\det(I - M^2) = 0$, making the right side 0 too. Both sides vanish, so the identity holds trivially.

Either way,

$$\boxed{\det(A^T B - B^T A) = \det(A + B)\det(A - B)}$$

(numerically verified for random 4×4 orthogonal A, B across all four sign combinations of $\det A, \det B$). □

Example 1.37 (A quadratic form built from an SPD matrix, on the space of matrices)**.** Let $A \in \mathrm{Mat}_n(\mathbb{R})$ be symmetric positive definite. Is the quadratic form $q(X) = \mathrm{tr}(X^T AX)$ on $\mathrm{Mat}_n(\mathbb{R})$ positive definite?

Solution. Yes. Write X by its columns $x_1, \dots, x_n \in \mathbb{R}^n$. Then $(X^T AX)_{ij} = x_i^T Ax_j$, so the diagonal entries are $x_i^T Ax_i$, and

$$q(X) = \mathrm{tr}(X^T AX) = \sum_{i=1}^{n} x_i^T Ax_i.$$

Since A is positive definite, each term $x_i^T Ax_i \geq 0$, with equality iff $x_i = 0$. So $q(X) \geq 0$ always, and $q(X) = 0$ forces every column $x_i = 0$, i.e. $X = 0$. Hence

$$\boxed{\text{yes, } q \text{ is positive definite}}.$$

□

Example 1.38 (Making a fixed matrix orthogonal to a subspace)**.** For $n > 1$, does there exist an inner product on the space of $n \times n$ real matrices for which the all-ones matrix J is orthogonal to every upper-triangular matrix?

Solution. Yes, for every $n > 1$. Let $W \subset \mathrm{Mat}_n(\mathbb{R})$ be the subspace of upper-triangular matrices, of dimension $\frac{n(n+1)}{2} < n^2$. Since $n > 1$, J has a nonzero entry below the diagonal (e.g. $J_{21} = 1$), so $J \notin W$; in particular J is linearly independent from W.

Extend a basis $w_1, \dots, w_r$ of W (where $r = \frac{n(n+1)}{2}$) by adjoining J and then further vectors to a full basis of $\mathrm{Mat}_n(\mathbb{R})$. Declare this basis *orthonormal* by fiat: define an inner product by taking this basis to an orthonormal basis (i.e. decree $\langle u, v\rangle$ to be the standard dot product of the coordinate vectors of u, v in this chosen basis). This is a bona fide inner product (it is bilinear, symmetric, and positive definite, since it is literally the standard inner product transported along a linear isomorphism), and by construction every w_i is orthogonal to J in it. Since $W = \mathrm{span}(w_1, \dots, w_r)$, linearity of the inner product in each argument gives $\langle J, X\rangle = 0$ for every $X \in W$. So

$$\boxed{\text{yes — such an inner product exists for every } n > 1}$$

(the same argument shows any fixed vector not lying in a given proper subspace can always be made orthogonal to that subspace by a suitable choice of inner product). □

Example 1.39 (A symmetric matrix always maps one given vector to another)**.** Let x, y be nonzero vectors in $\mathbb{R}^n$. Does there always exist a symmetric matrix A with $y = Ax$?

Solution. Yes, always. Define

$$A = \frac{xy^T + yx^T}{\|x\|^2} - \frac{x \cdot y}{\|x\|^4}\, xx^T.$$

This is manifestly symmetric, being a real linear combination of the symmetric matrices $xy^T + yx^T$ and xx^T. Computing Ax term by term: $xy^Tx = (x \cdot y)x$, $yx^Tx = \|x\|^2 y$, and $xx^Tx = \|x\|^2 x$, so

$$Ax = \frac{(x \cdot y)x + \|x\|^2 y}{\|x\|^2} - \frac{x \cdot y}{\|x\|^4} \cdot \|x\|^2 x = \frac{(x \cdot y)x}{\|x\|^2} + y - \frac{(x \cdot y)x}{\|x\|^2} = y.$$

So

$$\boxed{\text{yes — such a symmetric } A \text{ always exists}}$$

(this holds regardless of whether x, y are parallel, orthogonal, or neither — verified directly above, and checked numerically on random pairs). □

Example 1.40 (Signature of an evaluation-product quadratic form)**.** On the space of real polynomials of degree at most n ($n \geq 1$), consider the quadratic form $Q(f) = f(1)f(2)$. Find its signature (the number of $+1$'s and -1's in its normal form).

Solution. Write $Q(f) = L_1(f)L_2(f)$ where $L_1(f) = f(1)$, $L_2(f) = f(2)$ are linear functionals on the $(n+1)$-dimensional space P_n. Since $n \geq 1$, $\dim P_n \geq 2$ and L_1, L_2 are linearly independent (e.g. the polynomial $f(x) = x - 2$ has $L_1(f) = -1 \neq 0 = L_2(f)$, and $f(x) = x - 1$ has the reverse), so there exist $e_1, e_2 \in P_n$ with $L_1(e_1) = 1, L_2(e_1) = 0$ and $L_1(e_2) = 0, L_2(e_2) = 1$ (e.g. take e_1, e_2 proportional to $x - 2$ and $x - 1$ respectively, rescaled).

The form on $\mathrm{span}(e_1, e_2)$**.** For $f = ae_1 + be_2$, $Q(f) = L_1(f)L_2(f) = (a \cdot 1 + b \cdot 0)(a \cdot 0 + b \cdot 1) = ab$. Substituting $u = a + b$, $v = a - b$ (an invertible linear change of coordinates) gives $ab = \frac{u^2 - v^2}{4}$ — a form with exactly one $+1$ and one -1 in its normal form.

The rest of the space contributes nothing. On the $(n-1)$-dimensional subspace $K = \{f \in P_n : f(1) = 0, f(2) = 0\}$ (the common kernel of L_1, L_2), $Q(f) = L_1(f)L_2(f) = 0 \cdot 0 = 0$ identically, and K is complementary to $\mathrm{span}(e_1, e_2)$ (since L_1, L_2 restricted to $\mathrm{span}(e_1, e_2)$ are already an isomorphism onto $\mathbb{R}^2$, matching dimensions add up: $2 + (n-1) = n + 1 = \dim P_n$).

So in the basis $\{e_1, e_2\} \cup$ (a basis of K), Q is already in normal form with one $+1$, one -1, and $n - 1$ zeros. Hence

$$\boxed{\text{signature } (1,1) \text{ (one } +1\text{, one } -1\text{, and } n-1 \text{ zero directions)}}$$

(confirmed numerically: diagonalizing the Gram matrix of Q in the monomial basis for $n = 1, 2, 3, 4$ gives exactly one positive and one negative eigenvalue in every case, with the rest exactly zero). $\square$

Example 1.41 (Every path between definite forms of opposite type passes through a degenerate one)**.** Let A, B be symmetric bilinear forms on a real 2-dimensional space, with A positive definite and B negative definite. Prove that every continuous path of symmetric bilinear forms connecting A to B contains a degenerate one.

Solution. Represent the path as a continuous family of symmetric 2×2 matrices $C(t)$, $t \in [0, 1]$, $C(0) = A$, $C(1) = B$. Both $\det C(t)$ and $\operatorname{tr} C(t)$ are continuous functions of t (polynomials in the matrix entries).

Suppose, for contradiction, that $C(t)$ is never degenerate, i.e. $\det C(t) \neq 0$ for all t. A non-degenerate 2×2 symmetric matrix has both eigenvalues nonzero, and $\det C(t)$ is their product: if the eigenvalues have the same sign, $\det C(t) > 0$; if opposite signs, $\det C(t) < 0$. Since $\det A > 0$ (A positive definite: both eigenvalues positive) and $\det B > 0$ (B negative definite: both eigenvalues negative, product still positive), and $\det C(t)$ is continuous and never 0, the Intermediate Value Theorem forces $\det C(t) > 0$ for *every* $t \in [0, 1]$ (it can't jump from positive to negative without passing through 0).

So at every t, $C(t)$ is non-degenerate with $\det C(t) > 0$, meaning its two eigenvalues always share a sign — so $\operatorname{tr} C(t) \neq 0$ throughout (a zero trace with same-signed nonzero eigenvalues is impossible). But $\operatorname{tr} C(t)$ is also continuous, with $\operatorname{tr} A > 0$ (sum of two positive eigenvalues) and $\operatorname{tr} B < 0$ (sum of two negative eigenvalues). A continuous function that is never 0 but takes both a positive and a negative value violates the Intermediate Value Theorem — contradiction.

So the assumption fails: some $C(t)$ must be degenerate. Hence

$$\boxed{\text{every such path contains a degenerate bilinear form}}.$$

$\square$

Example 1.42 (Vanishing against all traceless matrices forces a scalar matrix)**.** Let A be a square matrix such that $\operatorname{tr}(AX) = 0$ for every matrix X with $\operatorname{tr}(X) = 0$. Prove $A = \lambda I$ for some scalar λ.

Solution. Recall $\operatorname{tr}(AE_{ij}) = A_{ji}$, where E_{ij} is the matrix unit with a single 1 in position (i, j) (a direct computation: $(AE_{ij})_{kl} = A_{ki}\,[l = j]$, so its trace is the $k = j$ term, A_{ji}).

Off-diagonal entries vanish. For $i \neq j$, E_{ij} itself has trace 0 (all its diagonal entries are 0). So the hypothesis with $X = E_{ij}$ gives $\operatorname{tr}(AE_{ij}) = A_{ji} = 0$. This holds for every $i \neq j$, so every off-diagonal entry of A is 0.

Diagonal entries are all equal. For $i \neq j$, $X = E_{ii} - E_{jj}$ has trace 0. The hypothesis gives

$$0 = \operatorname{tr}\left(A(E_{ii} - E_{jj})\right) = \operatorname{tr}(AE_{ii}) - \operatorname{tr}(AE_{jj}) = A_{ii} - A_{jj}.$$

So $A_{ii} = A_{jj}$ for every i, j — all diagonal entries share a common value λ.

Combining both, A is diagonal with every diagonal entry equal to λ:

$$\boxed{A = \lambda I}.$$

$\square$

2 Calculus and Real Analysis

Interview analysis questions often look innocent but test whether you can move between definitions, expansions, and clean arguments. The core techniques are asymptotics, derivatives, optimization, integrals, and series.

2.1 Sequences, Limits, Continuity, and Asymptotics

Definitions and formulas.

- A sequence (a_n) converges to L if for every $\varepsilon > 0$ there exists N such that $|a_n - L| < \varepsilon$ for all $n \geq N$.
- A bounded monotone sequence converges. Every bounded sequence has a convergent subsequence (Bolzano–Weierstrass).
- Important limits:

$$\lim_{x\to 0} \frac{\sin x}{x} = 1, \qquad \lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}, \qquad \lim_{x\to\infty} \left(1+\frac{1}{x}\right)^x = e.$$

- A function is continuous at a if $\lim_{x\to a} f(x) = f(a)$. Continuous functions on compact intervals are bounded and attain maxima and minima.
- Asymptotic notation:

$$f(x) = O(g(x)) \iff \frac{f(x)}{g(x)} \text{ is bounded}, \qquad f(x) = o(g(x)) \iff \frac{f(x)}{g(x)} \to 0.$$

- Taylor expansions near 0 that should be memorized:

$$\sin x = x - \frac{x^3}{6} + O(x^5), \qquad \cos x = 1 - \frac{x^2}{2} + O(x^4), \qquad \log(1+x) = x - \frac{x^2}{2} + O(x^3).$$

Example 2.1 (A standard small-x limit)**.** Compute

$$\lim_{x\to 0} \frac{\sin x - x}{x^3}.$$

Solution. Use the Taylor expansion

$$\sin x = x - \frac{x^3}{6} + O(x^5).$$

Then

$$\sin x - x = -\frac{x^3}{6} + O(x^5),$$

so

$$\frac{\sin x - x}{x^3} = -\frac{1}{6} + O(x^2).$$

Therefore

$$\boxed{\lim_{x\to 0} \frac{\sin x - x}{x^3} = -\frac{1}{6}}.$$

□

Example 2.2 (Chaining two equivalent infinitesimals)**.** Suppose $\lim_{x\to 0} f(x)/\sin x = 2$. Find $\lim_{x\to 0} \ln(1+3x)/f(x)$.

Solution. Write the target limit as a product of three limits that each exist:

$$\frac{\ln(1+3x)}{f(x)} = \frac{\ln(1+3x)}{x} \cdot \frac{x}{\sin x} \cdot \frac{\sin x}{f(x)}.$$

As $x \to 0$: $\ln(1+3x)/x \to 3$, $x/\sin x \to 1$, and $\sin x/f(x) \to 1/2$ (the reciprocal of the given limit, valid since $f(x)/\sin x \to 2 \neq 0$ forces $f(x) \neq 0$ near 0). Hence

$$\boxed{\lim_{x\to 0} \frac{\ln(1+3x)}{f(x)} = 3 \cdot 1 \cdot \frac{1}{2} = \frac{3}{2}}.$$

□

Example 2.3 (Convergence forced by a strict averaging bound)**.** A sequence (a_n) has all $a_n \in (0,1)$ and satisfies $a_{n+1} < \frac{1}{2}(a_n + a_{n-1})$ for all $n \geq 1$. Must a_n converge? Find the set of all possible limits over all such sequences.

Solution. **Convergence.** Let $M_n = \max(a_n, a_{n-1})$. Since $a_{n+1} < \frac{1}{2}(a_n + a_{n-1}) \leq M_n$, we get $M_{n+1} = \max(a_{n+1}, a_n) \leq \max(M_n, a_n) = M_n$: the sequence (M_n) is non-increasing, and bounded below by 0, so it converges to some $L = \inf_n M_n \geq 0$ (with $M_n \geq L$ for every n).

Since $a_n \leq M_n$ for all n, $\limsup a_n \leq L$. Suppose toward a contradiction that $\liminf a_n = L' < L$; set $\delta = L - L' > 0$ and $\varepsilon = \delta/4$. Choose N with $M_n < L + \varepsilon$ for all $n \geq N$ (possible since $M_n \to L$), and pick $n_k \geq N$ (infinitely many, since L' is a liminf) with $a_{n_k} < L' + \varepsilon = L - 3\delta/4$. Then

$$a_{n_k+1} < \frac{a_{n_k} + a_{n_k-1}}{2} < \frac{(L - 3\delta/4) + (L+\varepsilon)}{2} = \frac{2L - \delta/2}{2} = L - \frac{\delta}{4},$$

using $a_{n_k-1} \leq M_{n_k} < L+\varepsilon$. So $M_{n_k+1} = \max(a_{n_k+1}, a_{n_k}) < \max(L-\delta/4, L-3\delta/4) = L-\delta/4 < L$, contradicting $M_n \geq L$ for all n. Hence $\liminf a_n = L = \limsup a_n$, so

$$\boxed{a_n \text{ always converges}}.$$

The set of possible limits. Since $a_n \in (0,1)$ throughout, $0 \leq L < 1$ (strictly, as $L \leq M_1 < 1$). Conversely, every $L \in [0,1)$ is attainable: fix L, pick any $c \in (L,1)$, and set $a_0 = a_1 = c$, $a_n = L + \frac{c-L}{n}$ for $n \geq 1$. Then $a_n \to L$, each $a_n \in (L,c] \subset (0,1)$, and the required inequality holds for $n \geq 2$: it reduces (after cancelling the positive factor $c - L$) to $\frac{1}{n+1} < \frac{1}{2}\left(\frac{1}{n} + \frac{1}{n-1}\right) = \frac{2n-1}{2n(n-1)}$, equivalent to $2n(n-1) < (2n-1)(n+1)$, i.e. $0 < 3n - 1$, true for all $n \geq 1$. So

$$\boxed{\text{the set of possible limits is exactly } [0,1)}.$$

□

Example 2.4 (A doubling functional equation does not force linearity)**.** Is every odd continuous function f satisfying $f(2x) = 2f(x)$ for all x necessarily linear?

Solution. No. Build a counterexample from a single "seed" piece. Pick any continuous $\varphi : [1,2] \to \mathbb{R}$ with $\varphi(1) = 1$, $\varphi(2) = 2$, but φ not linear in between — e.g. $\varphi(x) = x + \sin(\pi(x-1))$, which satisfies $\varphi(1) = 1$, $\varphi(2) = 2$ but $\varphi(1.5) = 1.5 + 1 = 2.5 \neq 1.5$.

For $x > 0$, define $f(x) = 2^n \varphi(x/2^n)$ where n is the integer with $x/2^n \in [1,2)$ (equivalently $n = \lfloor \log_2 x \rfloor$); set $f(0) = 0$ and $f(x) = -f(-x)$ for $x < 0$ (forcing oddness).

This satisfies $f(2x) = 2f(x)$ for all $x > 0$: if $x/2^n \in [1,2)$ then $(2x)/2^{n+1} = x/2^n$ is the same ratio, so $f(2x) = 2^{n+1}\varphi((2x)/2^{n+1}) = 2 \cdot 2^n \varphi(x/2^n) = 2f(x)$.

It is continuous at every dyadic point 2^k: approaching from below (block $n = k-1$) gives $2^{k-1}\varphi(2^-) = 2^{k-1} \cdot 2 = 2^k$; approaching from at/above (block $n = k$) gives $2^k \varphi(1) = 2^k \cdot 1 = 2^k$ — they agree, since $\varphi(2) = 2\varphi(1)$ was built into the choice of endpoints.

It is continuous at 0: for $x \in [2^n, 2^{n+1})$ with $n \to -\infty$ as $x \to 0^+$, $|f(x)| = 2^n |\varphi(x/2^n)| \leq 2^n \max_{[1,2]} |\varphi| \to 0$.

Since $f(1.5) = \varphi(1.5) = 2.5 \neq 1.5$, this f is odd, continuous, satisfies the doubling identity, and is *not* linear:

$$\boxed{\text{no — such } f \text{ need not be linear}}.$$

□

Example 2.5 (An integral recursion converging to π)**.** Fix $a_0 \in (0, 2\pi)$ and define a sequence by

$$a_{n+1} = \int_0^{a_n} \left(1 + \frac{1}{4}\cos^{2n+1} t\right) dt.$$

Show (a_n) converges, and find its limit.

Solution. Write $a_{n+1} = a_n + \frac{1}{4}G_n(a_n)$ where $G_n(x) = \int_0^x \cos^{2n+1} t\, dt$.

An exact vanishing identity. For any integer ℓ, substituting $t = \ell\pi + s$ gives $\cos(\ell\pi + s) = (-1)^\ell \cos s$, so $\cos^{2n+1}(\ell\pi + s) = (-1)^\ell \cos^{2n+1} s$ (the exponent is odd); hence

$$\int_{\ell\pi}^{(\ell+1)\pi} \cos^{2n+1} t\, dt = (-1)^\ell \int_0^\pi \cos^{2n+1} s\, ds = (-1)^\ell \left[\int_0^\pi \cos^{2n+1}(s)\, ds\right] = 0,$$

the last integral vanishing because substituting $s \mapsto \pi - s$ negates $\cos^{2n+1}$ while preserving the interval $[0, \pi]$. In particular $G_n(0) = G_n(\pi) = G_n(2\pi) = 0$.

Monotone, bounded convergence. On $[0, \pi]$, define $h_n(x) = (\pi - x) - G_n(x)$. Then $h_n(0) = \pi > 0$, $h_n(\pi) = 0$, and $h_n'(x) = -1 - \cos^{2n+1}(x) \le 0$ (as $\cos^{2n+1} x \ge -1$), so h_n is non-increasing on $[0, \pi]$ with minimum value 0 at $x = \pi$: hence $h_n(x) \ge 0$, i.e.

$$0 \le G_n(x) \le \pi - x \qquad \text{for } x \in [0, \pi].$$

An identical computation on $[\pi, 2\pi]$ (with $H_n(x) = G_n(x) - (\pi - x)$, $H_n'(x) = \cos^{2n+1}(x) + 1 \ge 0$, $H_n(\pi) = 0$) gives the mirror bound $G_n(x) \ge \pi - x$ for $x \in [\pi, 2\pi]$.

Consequently, if $a_n \in (0, \pi)$: $G_n(a_n) \ge 0$ gives $a_{n+1} \ge a_n$, and $G_n(a_n) \le \pi - a_n$ gives $a_{n+1} \le a_n + \frac{1}{4}(\pi - a_n) < \pi$ — so a_n increases, staying below π. Symmetrically, if $a_n \in (\pi, 2\pi)$: $a_{n+1} \le a_n$, staying above π. And if $a_n = \pi$, $G_n(\pi) = 0$ so $a_{n+1} = \pi$ exactly. So in every case (a_n) is monotonic and bounded between a_0 and π, hence converges to some limit L.

Identifying $L = \pi$. Suppose, for contradiction, $a_0 < \pi$ and $L < \pi$ (the case $a_0 > \pi$, $L > \pi$ is symmetric). Since $a_n \uparrow L$, eventually $a_n \ge L/2 > 0$, so $G_n(a_n) \ge G_n(L/2)$ for these n (as G_n is increasing on $[0, \pi/2]$ when $L/2 \le \pi/2$, or more simply because G_n is increasing up to $\pi/2$ and a_n stays in a fixed sub-interval bounded away from 0). The classical Wallis-integral asymptotic $\int_0^{\pi/2} \cos^{2n+1} t\, dt = \dfrac{(2n)!!}{(2n+1)!!} \sim \sqrt{\dfrac{\pi}{4n}}$ shows that this near-0 contribution to G_n does *not* decay fast enough to be summable in n (it is $\Theta(1/\sqrt{n})$, and $\sum 1/\sqrt{n} = \infty$) whenever the domain of integration stays bounded away from 0 — which it does here, since $a_n \ge L/2$ for all large n. So $\sum_n G_n(a_n) = \infty$, forcing $\sum_n (a_{n+1} - a_n) = \infty$ too, contradicting that this telescoping sum equals the finite quantity $L - a_0$. Hence L cannot be strictly less than π; by the mirrored argument L cannot exceed π either. So

$$\boxed{a_n \to \pi \text{ for every } a_0 \in (0, 2\pi)}$$

(consistent with $a_0 = \pi$ itself being an exact fixed point of the recursion). □

Example 2.6 (Can a continuous function hit every value exactly twice? Exactly three times?)**.** (a) Does there exist a continuous $f : \mathbb{R} \to \mathbb{R}$ that attains every value in its range exactly twice? (b) Exactly three times?

Solution. **(a) No.** Suppose f is continuous and exactly two-to-one. Since f is not injective, pick $a < b$ with $f(a) = f(b) =: y_0$. On $[a, b]$, f cannot be constant (that would give infinitely many preimages of y_0), so its max M or min on $[a, b]$ differs from y_0; by replacing f with $-f$ if needed, assume $M > y_0$, attained at some $c \in (a, b)$.

Step 1: $f < y_0$ outside $[a, b]$. For $y \in (y_0, M)$, the Intermediate Value Theorem on $[a, c]$ and on $[c, b]$ each supplies a preimage of y, giving 2 preimages already — so (exactly two-to-one) *no* preimage of any such y lies outside $[a, b]$: $f(x) \notin (y_0, M)$ for $x \notin [a, b]$. Now $f((-\infty, a))$ is a connected set (continuous image of an interval) avoiding the gap (y_0, M), so it lies entirely in

$(-\infty, y_0]$ or entirely in $[M, \infty)$; the latter is impossible since $f(x) \to f(a) = y_0 < M$ as $x \to a^-$. So $f(x) < y_0$ for all $x < a$ (it cannot equal y_0 either, as a, b are already y_0's only two preimages), and symmetrically $f(x) < y_0$ for all $x > b$.

Step 2: M is the global maximum, attained twice inside (a, b). Since $f < y_0 < M$ outside $[a, b]$ and $f \le M$ on $[a, b]$, M is f's global maximum. Being exactly two-to-one, M has a second preimage $c_2 \neq c$, and by Step 1, $c_2 \in (a, b)$ too. Say $c < c_2$.

Step 3: contradiction. For $y = M - \varepsilon$ (small $\varepsilon > 0$), IVT on $[a, c]$ and on $[c_2, b]$ again supplies 2 preimages — so, by the two-to-one count, *no* preimage of $M - \varepsilon$ lies in $[c, c_2]$, for every small $\varepsilon > 0$. So $f(x) \notin (y_0, M)$ for $x \in [c, c_2]$, and since also $f \le M$ there, $f(x) \in (-\infty, y_0] \cup \{M\}$ on $[c, c_2]$. But $f(c) = f(c_2) = M$, and if f ever dropped to $\le y_0$ somewhere in between, it would have to cross back up through the forbidden gap (y_0, M) to return to M — impossible. So $f \equiv M$ on the entire interval $[c, c_2]$, giving *infinitely many* preimages of M — contradicting two-to-one. So

$$\boxed{\text{no continuous exactly-two-to-one } f : \mathbb{R} \to \mathbb{R} \text{ exists}}.$$

(b) Yes. Define f using $n = \lfloor x/3 \rfloor$ and $r = x - 3n \in [0, 3)$:

$$f(x) = \begin{cases} n + r, & 0 \le r < 1 \\ n + 2 - r, & 1 \le r < 2 \\ n + (r - 2), & 2 \le r < 3. \end{cases}$$

This is continuous (the three pieces agree at $r = 1, 2$, and the blocks agree at integer multiples of 3: block n ends at value $n + 1$, matching block $n + 1$'s start). On each block $[3n, 3n + 3)$, f traces $n \to n + 1 \to n \to n + 1$ — three monotonic strands, each covering $[n, n + 1]$ exactly once — so every value strictly inside $(n, n + 1)$ has exactly 3 preimages, all within block n. A boundary value $y = n+1$ gets its 3 preimages at $x = 3n+1$ (rise-then-fall peak within block n), $x = 3n + 3$ (block n's end = block n+1's start), and $x = 3n + 5$ (the peak of block n+1's middle strand) — verified directly by evaluating f at these points. So

$$\boxed{\text{yes — an explicit continuous exactly-three-to-one } f : \mathbb{R} \to \mathbb{R} \text{ exists}}$$

(confirmed by a computer search over a wide window of x and y values, finding exactly 3 preimages in every case checked). □

Example 2.7 (Matching values one unit apart, from a boundary condition)**.** Let f be continuous with $f(0) = f(2)$. Show that $f(x) = f(x - 1)$ for some $x \in [0, 2]$.

Solution. Define $g(x) = f(x) - f(x - 1)$ for $x \in [1, 2]$ (so $x - 1 \in [0, 1] \subset [0, 2]$, keeping g well-defined and continuous on $[1, 2]$).

At the endpoints:

$$g(2) = f(2) - f(1), \qquad g(1) = f(1) - f(0) = f(1) - f(2) = -g(2)$$

(using $f(0) = f(2)$ in the last step). So $g(1) = -g(2)$: either both are 0, or $g(1)$ and $g(2)$ have opposite signs. Either way, by the Intermediate Value Theorem applied to the continuous function g on $[1, 2]$, there is some $x \in [1, 2] \subseteq [0, 2]$ with $g(x) = 0$, i.e.

$$\boxed{f(x) = f(x - 1) \text{ for some } x \in [0, 2]}.$$

□

Example 2.8 (Solving a limiting trigonometric equation)**.** Solve $\lim_{n\to\infty} \cos(nx) = 1$ (find all real x for which the limit exists and equals 1).

Solution. If $x = 2\pi k$ for an integer k, then $nx = 2\pi nk$ is always an integer multiple of 2π, so $\cos(nx) = 1$ for every n, and the limit is trivially 1.

Conversely, suppose $x \neq 2\pi k$ for any integer k, i.e. $x/(2\pi) \notin \mathbb{Z}$.

Case $x/(2\pi)$ irrational. By Weyl equidistribution, $\{nx/(2\pi)\}$ (the fractional part) is equidistributed on $[0,1)$, so $\cos(nx)$ takes values densely across $[-1,1]$ and does not converge at all — in particular not to 1.

Case $x/(2\pi) = p/q$ rational in lowest terms, $q \geq 2$. Then $\cos(nx)$ is periodic in n with period q (since $nx \bmod 2\pi$ depends only on $n \bmod q$), and as n ranges over one period, $np \bmod q$ takes all q distinct residues $0, 1, \ldots, q-1$ (as $\gcd(p,q) = 1$) — so $\cos(nx)$ cycles through $q \geq 2$ genuinely different values infinitely often, and cannot converge to a single limit.

So the limit exists and equals 1 exactly when $x/(2\pi)$ is an integer:

$$\boxed{x = 2\pi k,\ k \in \mathbb{Z}}.$$

□

Example 2.9 (No continuous functional square root of $1 - x^3$)**.** Does there exist a continuous function $f : \mathbb{R} \to \mathbb{R}$ with $f(f(x)) = 1 - x^3$ for all x?

Solution. No. Let $g(x) = 1 - x^3$; since $-x^3$ is strictly decreasing, g is a strictly decreasing bijection $\mathbb{R} \to \mathbb{R}$.

f must be a continuous bijection. If $f \circ f = g$ is injective, f itself must be injective (if $f(a) = f(b)$ then $g(a) = f(f(a)) = f(f(b)) = g(b)$, so $a = b$ since g is injective). And since $g = f \circ f$ is surjective, so is f (every y in the range of g is f applied to $f(x)$ for suitable x, i.e. $y = f(f(x))$ is in the range of f; as g is onto $\mathbb{R}$, f is too). So f is a continuous bijection $\mathbb{R} \to \mathbb{R}$, hence — a standard fact for continuous injections on an interval — f is *strictly monotonic*.

Neither monotonicity type is possible. If f is strictly increasing, then $f \circ f$ (increasing composed with increasing) is strictly increasing. If f is strictly decreasing, then $f \circ f$ (decreasing composed with decreasing) is *also* strictly increasing. Either way, $f \circ f$ must be strictly increasing. But $f \circ f = g$ is strictly *decreasing* — a contradiction. So

$$\boxed{\text{no continuous } f \text{ with } f(f(x)) = 1 - x^3 \text{ exists}}.$$

□

Example 2.10 (Limit of a quadratic-cubic recursion)**.** Let $a_0 = -\frac{1}{2}$ and $a_{n+1} = \dfrac{a_n^2(a_n - 3)}{4}$. Find $\lim_{n\to\infty} a_n$.

Solution. Let $f(x) = \dfrac{x^2(x-3)}{4}$. We show by induction that $a_n \in (-1,0)$ for every n, with $|a_n|$ strictly decreasing to 0.

The invariant. Suppose $x \in (-1,0)$. Then $x^2 \in (0,1)$ and $x - 3 \in (-4,-3)$, so $f(x) = x^2(x-3)/4 < 0$, and

$$|f(x)| = \frac{x^2(3-x)}{4} < \frac{x^2 \cdot 4}{4} = x^2 < |x|$$

(using $3 - x < 4$ for $x > -1$, and $x^2 < |x|$ since $|x| < 1$). Also $|f(x)| < x^2 < 1$, so $f(x) \in (-1,0)$ too — the invariant is preserved, and moreover $|f(x)| < x^2$ strictly.

Applying it. Since $a_0 = -\frac{1}{2} \in (-1,0)$, induction gives $a_n \in (-1,0)$ for all n, with $|a_{n+1}| < a_n^2$ at every step. So $|a_n|$ is strictly decreasing and bounded below by 0, hence converges to some $L \in [0,1)$.

Identifying L. Taking $n \to \infty$ in $|a_{n+1}| < a_n^2$ gives $L \leq L^2$. For $L \in [0,1)$, $L \leq L^2$ forces $L = 0$ (since $L^2 < L$ strictly whenever $L \in (0,1)$). So $|a_n| \to 0$, i.e.

$$\boxed{\lim_{n\to\infty} a_n = 0}$$

(numerically, a_n reaches 0 to machine precision within about 10 iterations from $a_0 = -\frac{1}{2}$, consistent with the quadratic-rate convergence $|a_{n+1}| < a_n^2$). □

Example 2.11 (A limit that collapses to a maximum)**.** For $x \geq 0$, sketch the graph of

$$f(x) = \lim_{n\to\infty} \sqrt[n]{1 + x^n + \left(\frac{x^2}{2}\right)^n}.$$

Solution. **The general fact.** For non-negative $a_1, \dots, a_k$, $\left(a_1^n + \cdots + a_k^n\right)^{1/n} \to \max(a_1, \dots, a_k)$ as $n \to \infty$: writing $M = \max_i a_i$, $M \leq (a_1^n + \cdots + a_k^n)^{1/n} \leq (kM^n)^{1/n} = k^{1/n}M \to M$.

Here $f(x) = \max\left(1,\, x,\, \dfrac{x^2}{2}\right)$.

Finding the crossover points. Compare the three functions pairwise on $x \geq 0$: $1 = x$ at $x = 1$; $x = x^2/2$ at $x = 2$; $1 = x^2/2$ at $x = \sqrt{2}$ (between the other two crossings, so it never becomes the relevant boundary). On $[0,1]$: $x \leq 1$ and $x^2/2 \leq \frac{1}{2} \leq 1$, so the max is 1. On $[1,2]$: $x \geq 1$, and $x \geq x^2/2 \iff x \leq 2$, so the max is x. On $[2,\infty)$: $x^2/2 \geq x$ (since $x \geq 2$) and $x^2/2 \geq 2 > 1$, so the max is $x^2/2$.

So

$$\boxed{f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ x, & 1 \leq x \leq 2 \\ x^2/2, & x \geq 2 \end{cases}}$$

— a graph that is flat at height 1, then rises linearly with slope 1 from $(1,1)$ to $(2,2)$, then continues as an upward-opening parabola matching both the value and slope ($f' = x$, giving slope 2 at $x = 2$ from the right, versus slope 1 from the left — so the graph has a corner at $x = 2$, though it is continuous there and also at $x = 1$). □

Example 2.12 (Limit of a Cesàro-type recurrence with vanishing perturbation)**.** The sequence (c_n) satisfies $c_{n+1} = \left(1 - \frac{1}{n}\right)c_n + \beta_n$ for $n \geq 1$, where (β_n) is any sequence with $n^2\beta_n \to 0$ as $n \to \infty$. Find $\lim_{n\to\infty} c_n$.

Solution. **An explicit formula.** At $n = 1$, the recurrence gives $c_2 = \left(1 - \frac{1}{1}\right)c_1 + \beta_1 = \beta_1$, independent of c_1. Claim: for all $n \geq 2$,

$$c_n = \frac{1}{n-1}\sum_{k=1}^{n-1} k\beta_k.$$

This holds at $n = 2$ (both sides equal β_1). Assuming it for n,

$$c_{n+1} = \left(1 - \frac{1}{n}\right)c_n + \beta_n = \frac{n-1}{n}\cdot\frac{1}{n-1}\sum_{k=1}^{n-1} k\beta_k + \beta_n = \frac{1}{n}\left(\sum_{k=1}^{n-1} k\beta_k + n\beta_n\right) = \frac{1}{n}\sum_{k=1}^{n} k\beta_k,$$

which is exactly the claimed formula at $n+1$. This proves the formula by induction (and shows c_1's value never matters).

Bounding c_n. Since $n^2\beta_n \to 0$, the sequence $n^2\beta_n$ is bounded: there is a constant A with $|\beta_k| \leq A/k^2$ for all $k \geq 1$. Then

$$|c_n| \leq \frac{1}{n-1}\sum_{k=1}^{n-1} k|\beta_k| \leq \frac{1}{n-1}\sum_{k=1}^{n-1}\frac{A}{k} = \frac{A}{n-1}\sum_{k=1}^{n-1}\frac{1}{k}.$$

The harmonic sum satisfies $\sum_{k=1}^{n-1}\frac{1}{k} \leq 1 + \ln(n-1)$, so

$$|c_n| \leq \frac{A(1 + \ln(n-1))}{n-1} \longrightarrow 0$$

as $n \to \infty$, since $\ln(n-1) = o(n-1)$. Hence

$$\boxed{\lim_{n\to\infty} c_n = 0}.$$

□

2.2 Trigonometric and Hyperbolic Identities for Reference

Useful identities.

- Pythagorean identities:

$$\sin^2 x + \cos^2 x = 1, \qquad 1 + \tan^2 x = \sec^2 x, \qquad 1 + \cot^2 x = \csc^2 x.$$

- Angle-sum formulas:

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b,$$
$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b.$$

- Double-angle and half-angle formulas:

$$\sin 2x = 2 \sin x \cos x,$$
$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1,$$
$$\sin^2 \frac{x}{2} = \frac{1 - \cos x}{2}, \qquad \cos^2 \frac{x}{2} = \frac{1 + \cos x}{2}.$$

- Product-to-sum formulas:

$$\sin a \sin b = \frac{1}{2}(\cos(a-b) - \cos(a+b)),$$
$$\cos a \cos b = \frac{1}{2}(\cos(a-b) + \cos(a+b)),$$
$$\sin a \cos b = \frac{1}{2}(\sin(a+b) + \sin(a-b)).$$

- Hyperbolic identities:

$$\sinh t = \frac{e^t - e^{-t}}{2}, \qquad \cosh t = \frac{e^t + e^{-t}}{2}, \qquad \cosh^2 t - \sinh^2 t = 1.$$

- Derivatives of hyperbolic functions:

$$\frac{d}{dt}\sinh t = \cosh t, \qquad \frac{d}{dt}\cosh t = \sinh t.$$

- Useful substitution pattern:

$$x = e^t \quad \Longrightarrow \quad x - \frac{1}{x} = 2\sinh t.$$

  In some texts, especially older or Eastern European ones, sinh is denoted by sh and cosh by ch.

Example 2.13 (A power-reduction integral)**.** Evaluate

$$\int_0^{\pi/2} \sin^4 x \, dx.$$

Solution. Use the half-angle identity

$$\sin^2 x = \frac{1 - \cos 2x}{2}.$$

Then

$$\sin^4 x = \left(\frac{1 - \cos 2x}{2}\right)^2 = \frac{1}{4}\left(1 - 2\cos 2x + \cos^2 2x\right).$$

Since

$$\cos^2 2x = \frac{1 + \cos 4x}{2},$$

we get

$$\sin^4 x = \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x.$$

Therefore

$$\int_0^{\pi/2} \sin^4 x\,dx = \int_0^{\pi/2} \left(\frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x\right) dx.$$

The cosine terms integrate to 0 on $[0, \pi/2]$, so

$$\int_0^{\pi/2} \sin^4 x\,dx = \frac{3}{8} \cdot \frac{\pi}{2} = \frac{3\pi}{16}.$$

Hence

$$\boxed{\int_0^{\pi/2} \sin^4 x\,dx = \frac{3\pi}{16}}.$$

□

2.3 Standard Substitutions for Reference

Useful substitution patterns.

- For integrands involving $1 + x^2$ or rational functions of x and $1 + x^2$, use

  $$x = \tan t, \qquad dx = \sec^2 t\,dt, \qquad 1 + x^2 = \sec^2 t.$$

  This turns expressions involving $1 + x^2$ into trigonometric squares and is especially useful for integrals such as

  $$\int \frac{dx}{1 + x^2}, \qquad \int \frac{x^2}{(1 + x^2)^2}\,dx.$$

- For radicals of the form $\sqrt{1 + x^2}$ or, more generally, $\sqrt{a^2 + x^2}$, a hyperbolic substitution is often cleaner than a trigonometric one:

  $$x = \sinh t, \qquad dx = \cosh t\,dt, \qquad \sqrt{1 + x^2} = \cosh t.$$

  More generally,

  $$x = a\sinh t, \qquad dx = a\cosh t\,dt, \qquad \sqrt{a^2 + x^2} = a\cosh t.$$

  This is useful because $\cosh^2 t - \sinh^2 t = 1$ has the same algebraic role as $1 + \sinh^2 t = \cosh^2 t$.

- For radicals of the form $\sqrt{a^2 - x^2}$, the standard circular substitution is

  $$x = a\sin t, \qquad dx = a\cos t\,dt, \qquad \sqrt{a^2 - x^2} = a\cos t.$$

  For radicals of the form $\sqrt{x^2 - a^2}$, useful choices are

  $$x = a\sec t \quad \text{or} \quad x = a\cosh t.$$

- For rational functions of $\sin x$ and $\cos x$, use the tangent half-angle substitution

$$u = \tan \frac{x}{2}.$$

Then

$$\sin x = \frac{2u}{1+u^2}, \qquad \cos x = \frac{1-u^2}{1+u^2}, \qquad dx = \frac{2}{1+u^2}\,du.$$

This converts any rational expression in $\sin x$ and $\cos x$ into a rational function of u.

- A useful decision rule is: if the integrand contains a quadratic expression under a square root, choose a substitution that turns that quadratic into a square; if the integrand is rational in $\sin x$ and $\cos x$, use $u = \tan(x/2)$; if exponentials create expressions like $x - 1/x$, try $x = e^t$ and rewrite the result in terms of $\sinh t$ or $\cosh t$.

Example 2.14 (Hyperbolic substitution in a definite integral)**.** Evaluate

$$\int_0^1 \sqrt{1+x^2}\,dx.$$

Solution. Because the integrand contains $\sqrt{1+x^2}$, use the substitution

$$x = \sinh t, \qquad dx = \cosh t\,dt, \qquad \sqrt{1+x^2} = \cosh t.$$

When $x = 0$, we have $t = 0$. When $x = 1$, we have $t = \operatorname{arsinh}(1)$.

Thus

$$\int_0^1 \sqrt{1+x^2}\,dx = \int_0^{\operatorname{arsinh}(1)} \cosh^2 t\,dt.$$

Using

$$\cosh^2 t = \frac{1+\cosh 2t}{2},$$

we obtain

$$\int \cosh^2 t\,dt = \frac{1}{2}(t + \sinh t \cosh t).$$

Therefore

$$\int_0^1 \sqrt{1+x^2}\,dx = \frac{1}{2}(t + \sinh t \cosh t)\Big|_0^{\operatorname{arsinh}(1)}.$$

At $t = \operatorname{arsinh}(1)$, we have $\sinh t = 1$ and $\cosh t = \sqrt{2}$. Hence

$$\int_0^1 \sqrt{1+x^2}\,dx = \frac{1}{2}(\operatorname{arsinh}(1) + \sqrt{2}).$$

Since

$$\operatorname{arsinh}(1) = \log(1+\sqrt{2}),$$

the final answer is

$$\boxed{\int_0^1 \sqrt{1+x^2}\,dx = \frac{\sqrt{2}+\log(1+\sqrt{2})}{2}}.$$

□

2.4 One-Variable Differential Calculus

Definitions and formulas.

- If f is differentiable at x, then

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

  Differentiability implies continuity.

- Rolle's theorem: if $f(a) = f(b)$ and f is continuous on $[a, b]$ and differentiable on (a, b), then $f'(c) = 0$ for some $c \in (a, b)$.

- Lagrange's mean value theorem:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

  for some $c \in (a, b)$.

- Taylor's formula around a is

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + R_n(x),$$

  with Lagrange remainder

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}$$

  for some ξ between a and x.

- Critical point tests: if $f'(x_0) = 0$ and $f''(x_0) > 0$, then x_0 is a local minimum; if $f''(x_0) < 0$, it is a local maximum.

- L'Hopital's rule handles $0/0$ and ∞/∞ forms when the relevant hypotheses are satisfied.

Example 2.15 (Optimization on $(0, \infty)$)**.** Find the global minimum of

$$f(x) = x - \log x, \qquad x > 0.$$

Solution. Differentiate:

$$f'(x) = 1 - \frac{1}{x} = \frac{x-1}{x}.$$

The only critical point is $x = 1$. The second derivative is

$$f''(x) = \frac{1}{x^2} > 0,$$

so $x = 1$ is a strict local minimum. Since $f(x) \to \infty$ as $x \downarrow 0$ and also $f(x) \to \infty$ as $x \to \infty$, this local minimum is global. The minimum value is

$$f(1) = 1.$$

□

Example 2.16 (A forced inequality on a long enough interval)**.** Let f be a real-valued differentiable function on $[a, b]$ with $b - a \geq 4$. Prove there is a point $x_0 \in (a, b)$ with

$$f'(x_0) < 1 + f(x_0)^2.$$

Solution. Suppose, toward a contradiction, that $f'(x) \geq 1 + f(x)^2$ for *every* $x \in (a, b)$. Let $g(x) = \arctan(f(x))$, differentiable with

$$g'(x) = \frac{f'(x)}{1 + f(x)^2} \geq 1$$

for all $x \in (a, b)$ (dividing the assumed inequality by the positive quantity $1 + f(x)^2$ preserves it). By the Mean Value Theorem, there is $\xi \in (a, b)$ with

$$g(b) - g(a) = g'(\xi)(b - a) \geq 1 \cdot (b - a) \geq 4.$$

But arctan takes values only in the open interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, so

$$g(b) - g(a) < \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi < 4.$$

This contradicts $g(b) - g(a) \geq 4$. Hence the assumption was false, and

$$\boxed{\text{some } x_0 \in (a, b) \text{ satisfies } f'(x_0) < 1 + f(x_0)^2}.$$

□

Example 2.17 (A harmonic-mean pair from the Mean Value Theorem)**.** Let f be a smooth real function with $f(0) = 0, f(1) = 1$. Prove there exist *distinct* $x_1, x_2 \in [0, 1]$ with

$$\frac{1}{f'(x_1)} + \frac{1}{f'(x_2)} = 2.$$

Solution. Let $h(x) = f'(x) - 1$; since $f(1) - f(0) = 1$, $\int_0^1 h(x)\,dx = 0$. If $h \equiv 0$ (i.e. $f' \equiv 1$), any two distinct points work. Otherwise, split into cases on the boundary values $h(0), h(1)$.

Case 1: $h(0) = 0$ **or** $h(1) = 0$**.** By the ordinary Mean Value Theorem there is $c \in (0, 1)$ with $f'(c) = 1$. If also $f'(0) = 1$ (or $f'(1) = 1$), take $x_1 = 0, x_2 = c$ (or $x_1 = c, x_2 = 1$): distinct points, each with $f' = 1$, giving $1 + 1 = 2$.

Case 2: $h(0)$ **and** $h(1)$ **have the** ***same*** **strict sign.** Say $h(0) > 0$ and $h(1) > 0$ (the other sign is symmetric). Since $\int_0^1 h = 0$ and $h(0) > 0$, h cannot stay ≥ 0 throughout (that would force $\int h > 0$, using $h(0) > 0$ and continuity), so $h(c^*) < 0$ for some $c^* \in (0, 1)$. By the Intermediate Value Theorem, h has a root in $(0, c^*)$ (since $h(0) > 0 > h(c^*)$) and another root in $(c^*, 1)$ (since $h(c^*) < 0 < h(1)$) — two *distinct* points x_1, x_2 with $f'(x_1) = f'(x_2) = 1$, giving $1 + 1 = 2$ again.

Case 3: $h(0), h(1)$ **have strictly opposite signs.** For $c \in (0, 1)$ with $f(c) \in (0, 1)$, apply the Mean Value Theorem on $[0, c]$ and $[c, 1]$ to get $x_1(c) \in (0, c)$, $x_2(c) \in (c, 1)$ with $f'(x_1(c)) = f(c)/c$ and $f'(x_2(c)) = (1 - f(c))/(1 - c)$, so

$$G(c) := \frac{1}{f'(x_1(c))} + \frac{1}{f'(x_2(c))} = \frac{c}{f(c)} + \frac{1 - c}{1 - f(c)}.$$

As $c \to 0^+$, $G(c) \to \frac{1}{f'(0)} + 1 = \frac{1}{f'(0)} + 1$, i.e. $2 + h(0)/(1 + h(0))$-type limit — concretely $G(0^+) = 1 + \frac{1}{1+h(0)}$; as $c \to 1^-$, $G(1^-) = 1 + \frac{1}{1+h(1)}$. Since $h(0)$ and $h(1)$ have opposite signs, one of $\frac{1}{1+h(0)}, \frac{1}{1+h(1)}$ exceeds 1 and the other is less than 1, so $G(0^+)$ and $G(1^-)$ lie on opposite sides of 2. As G is continuous on $(0, 1)$ (for the generic case where f stays within $(0, 1)$ on the open interval), the Intermediate Value Theorem gives c_0 with $G(c_0) = 2$ exactly, and $x_1 = x_1(c_0) < c_0 < x_2(c_0) = x_2$ are the required distinct points.

In every case,

$$\boxed{\text{distinct } x_1, x_2 \in [0, 1] \text{ with } \frac{1}{f'(x_1)} + \frac{1}{f'(x_2)} = 2 \text{ exist}}.$$

□

Example 2.18 (A weighted coefficient sum forces a real root)**.** Suppose $a_0 + \frac{a_1}{2} + \frac{a_2}{3} + \cdots + \frac{a_n}{n+1} = 0$. Prove the polynomial $p(x) = a_0 + a_1 x + \cdots + a_n x^n$ has a real root.

Solution. Let $F(x) = a_0 x + \frac{a_1}{2}x^2 + \frac{a_2}{3}x^3 + \cdots + \frac{a_n}{n+1}x^{n+1}$, an antiderivative of p with $F(0) = 0$. The hypothesis says exactly $F(1) = 0$. Since F is differentiable (a polynomial) with $F(0) = F(1) = 0$, Rolle's Theorem gives $x_0 \in (0,1)$ with $F'(x_0) = 0$. But $F'(x) = p(x)$, so

$$\boxed{p(x_0) = 0 \text{ for some } x_0 \in (0,1)}.$$

□

2.5 Multivariable Calculus and Constrained Optimization

Definitions and formulas.

- For $f : \mathbb{R}^n \to \mathbb{R}$, the gradient is

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n} \right),$$

  and the directional derivative along a unit vector u is $D_u f = \nabla f \cdot u$.

- The Hessian matrix is

$$H_f = \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right)_{i,j}.$$

  At a critical point, a positive definite Hessian implies a local minimum; a negative definite Hessian implies a local maximum.

- To optimize $f(x)$ subject to $g(x) = 0$, introduce the Lagrangian

$$\mathcal{L}(x, \lambda) = f(x) - \lambda g(x)$$

  and solve $\nabla_x \mathcal{L} = 0$ together with $g(x) = 0$.

- In optimization algorithms, gradient descent updates

$$x_{k+1} = x_k - \eta \nabla f(x_k).$$

- The implicit function theorem says that if $F(x, y) = 0$ and $F_y(x_0, y_0) \neq 0$, then locally one can write y as a differentiable function of x.

Example 2.19 (Lagrange multipliers)**.** Minimize

$$f(x, y) = x^2 + y^2$$

subject to the linear constraint

$$x + 2y = 1.$$

Solution. Set

$$\mathcal{L}(x, y, \lambda) = x^2 + y^2 - \lambda(x + 2y - 1).$$

Then

$$2x - \lambda = 0, \qquad 2y - 2\lambda = 0, \qquad x + 2y = 1.$$

From the first two equations,

$$x = \frac{\lambda}{2}, \qquad y = \lambda.$$

Substitute into the constraint:

$$\frac{\lambda}{2} + 2\lambda = 1 \quad \Longrightarrow \quad \frac{5\lambda}{2} = 1 \quad \Longrightarrow \quad \lambda = \frac{2}{5}.$$

Hence

$$x = \frac{1}{5}, \qquad y = \frac{2}{5}.$$

The minimum value is

$$f\left(\frac{1}{5}, \frac{2}{5}\right) = \frac{1}{25} + \frac{4}{25} = \frac{1}{5}.$$

Because the objective is strictly convex and the constraint is affine, this minimizer is unique. □

Example 2.20 (An extremal problem with infimum zero)**.** Let M be the set of continuous, non-increasing functions $f : [0, 1] \to \mathbb{R}$ with $f(1) = 0$ and $f \not\equiv 0$. Find

$$\inf_{f \in M} \sup_{x \in [0,1]} \frac{x f(x)}{\int_0^1 f(t)\,dt}.$$

Solution. Since f is non-increasing with $f(1) = 0$, automatically $f \geq 0$ on $[0, 1]$. Under $f \mapsto cf$ $(c > 0)$ both the numerator's supremum and the denominator scale by c, so the ratio is scale-invariant; we may search among a convenient family.

For an integer $n \geq 2$, define

$$f_n(x) = \min\left(n, \frac{1}{x}\right) - 1 \quad (x > 0), \qquad f_n(0) := n - 1,$$

i.e. $f_n(x) = n-1$ on $[0, 1/n]$ and $f_n(x) = \frac{1}{x} - 1$ on $[1/n, 1]$. This is continuous and non-increasing (the two pieces agree and are each non-increasing), with $f_n(1) = 0$, so $f_n \in M$.

Numerator. On $[0, 1/n]$, $xf_n(x) = x(n-1)$ increases to $\frac{n-1}{n}$ at $x = 1/n$. On $[1/n, 1]$, $xf_n(x) = x\left(\frac{1}{x} - 1\right) = 1 - x$ decreases from the same value $\frac{n-1}{n}$. Hence

$$\sup_x xf_n(x) = \frac{n-1}{n} < 1.$$

Denominator.

$$\int_0^1 f_n = \int_0^{1/n} (n-1)\,dt + \int_{1/n}^1 \left(\frac{1}{t} - 1\right) dt = \frac{n-1}{n} + \big[\ln t - t\big]_{1/n}^1 = \frac{n-1}{n} + \left(\ln n + \frac{1}{n} - 1\right) = \ln n.$$

Ratio.

$$\frac{\sup_x xf_n(x)}{\int_0^1 f_n} = \frac{(n-1)/n}{\ln n} \xrightarrow[n\to\infty]{} 0,$$

since the numerator stays bounded below 1 while the denominator grows without bound. The ratio is always strictly positive for $f \in M$, but by taking n large it can be made smaller than any $\varepsilon > 0$, so the infimum is not attained and equals

$$\boxed{0}.$$

□

Example 2.21 (Reading a gradient field: degenerate directions and gradient descent)**.** A plotted gradient field of a smooth $f(x, y)$ shows a periodic pattern: near the vertical grid lines the field is essentially vertical, near the horizontal grid lines it is essentially horizontal, and the pattern is symmetric under swapping the roles of x and y and under reflections — the signature of a field built from a product of two matching one-dimensional oscillations, $f(x, y) = g(x)g(y)$ with g satisfying $g'' = -c^2 g$ for some constant c (i.e. g is a sinusoid). (a) Along which lines is the Hessian of f singular, and how many families are there? (b) Is it possible to have $\nabla f \neq 0$ at a point from which gradient descent never reaches a minimum of f?

Solution. **(a)** Take $f(x, y) = g(x)g(y)$ with $g'' = -c^2 g$ (so g is, up to scale and phase, $\cos(cx)$). Then

$$f_{xx} = g''(x)g(y) = -c^2 f, \qquad f_{yy} = g(x)g''(y) = -c^2 f, \qquad f_{xy} = g'(x)g'(y).$$

So

$$\begin{aligned}\det H &= f_{xx}f_{yy} - f_{xy}^2 = c^4 g(x)^2 g(y)^2 - g'(x)^2 g'(y)^2 \\ &= (c^2 g(x)g(y) - g'(x)g'(y))(c^2 g(x)g(y) + g'(x)g'(y)).\end{aligned}$$

With $g(x) = \cos(cx)$ (so $g'(x) = -c\sin(cx)$), $c^2 g(x)g(y) \mp g'(x)g'(y) = c^2(\cos(cx)\cos(cy) \mp \sin(cx)\sin(cy)) = c^2\cos(c(x \mp y))$. Hence

$$\det H = c^4 \cos\left(c(x - y)\right)\cos\left(c(x + y)\right),$$

which vanishes exactly on the lines $x - y = \text{const}$ and $x + y = \text{const}$ (specifically where $c(x \mp y) = \frac{\pi}{2} + k\pi$). These are two families of parallel lines, with slopes

$$\boxed{a = 1 \text{ and } a = -1}$$

(each family repeating with period $\pi\sqrt{2}/c$ along its perpendicular direction) — exactly the "more than one line" the problem describes, and this conclusion is forced by the request for the degenerate locus to be *straight lines*: for a product $g(x)g(y)$, $\det H$ factors into a function of $x - y$ times a function of $x + y$ only when $g'' = -c^2 g$.

(b) Yes. Besides the checkerboard of local maxima/minima at $g'(x) = g'(y) = 0$, f has genuine saddle points where $g(x) = g(y) = 0$ (e.g. $x = y = \pi/(2c)$ for $g = \cos(cx)$): there $f_{xx} = f_{yy} = 0$ but $f_{xy} = g'(x)g'(y) \neq 0$, so $\det H < 0$. On the diagonal $x = y$, $\nabla f = (-g(x)g'(x), -g(x)g'(x))$ always has equal components, so the diagonal is invariant under the gradient flow, and the induced one-dimensional flow $\dot{x} = g(x)g'(x)$ has the saddle at $x = \pi/(2c)$ as an *attracting* fixed point from the max at $x = 0$: any point (t, t) with $0 < t < \pi/(2c)$ has $\nabla f \neq 0$ (it vanishes only at $t = 0, \pi/(2c)$), yet gradient descent started there converges to the saddle, not to a minimum. So

$$\boxed{\text{yes, such points exist}}$$

— they lie on the stable manifold (here, a straight diagonal segment) of a saddle point. ☐

Example 2.22 (A fixed-radius mean-value property forces a constant function)**.** Let $f : \mathbb{R}^2 \to \mathbb{R}$ be bounded and smooth, and suppose the average of f over every circle of radius 1 equals f at the circle's center. Prove f is constant.

Solution. **The mechanism, when a maximum is attained.** Suppose first that f attains its supremum M at some point p_0 (this case already shows the key idea). On the unit circle around p_0, $f \leq M$ everywhere, and the average over that circle equals $f(p_0) = M$ by hypothesis — an average of values $\leq M$ can only equal M if every value equals M. So $f \equiv M$ on the *entire* unit circle around p_0. But every point q on that circle is then itself a maximizer, so the same argument applies at q: $f \equiv M$ on the unit circle around q too. As q ranges over the original circle, these new circles sweep out an annulus (indeed, all points within distance 2 of p_0), and iterating — using points on those new circles as fresh centers — reaches every point of $\mathbb{R}^2$ in finitely many hops of length 1. Hence $f \equiv M$ everywhere.

The general case. Boundedness alone does not guarantee the supremum is attained (the domain $\mathbb{R}^2$ is not compact), so the argument above needs strengthening to cover every bounded smooth f satisfying the hypothesis. This can be done — the complete statement is a genuine (if more delicate) fact, closely related to the classical theory of mean-value operators: writing the hypothesis as $f = f * \mu$ for μ the uniform probability measure on the unit circle, and noting that the Fourier transform of μ is the Bessel function $J_0(|\xi|)$, which equals 1 only at $\xi = 0$, forces f's Fourier transform to be supported at the origin alone — meaning f is a polynomial, and a bounded polynomial on $\mathbb{R}^2$ must be constant. Fleshing out this Fourier-analytic route rigorously (or finding a fully elementary substitute covering the unattained-supremum case) is not carried out in detail here, but the conclusion is

$$\boxed{f \text{ is constant}}$$

in every case, with the attained-maximum argument above already illustrating why: the mean-value-at-radius-1 property is rigid enough to propagate any extremal value across the whole plane. □

2.6 Vector Fields and Vector Calculus

Definitions and formulas.

- A vector field on $\mathbb{R}^n$ assigns a vector $F(x)$ to each point x. In coordinates, one writes

$$F = (F_1, \dots, F_n).$$

  Vector fields model forces, velocities, gradients of scalar objectives, and fluxes.

- If ϕ is a scalar field, then its gradient

$$\nabla\phi = \left(\frac{\partial\phi}{\partial x_1}, \dots, \frac{\partial\phi}{\partial x_n}\right)$$

  points in the direction of steepest increase. A field of the form $F = \nabla\phi$ is called *conservative*.

- For $F = (P, Q, R)$ in $\mathbb{R}^3$, the divergence and curl are

$$\nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}, \qquad \nabla \times F = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right).$$

  Divergence measures local source or sink strength, while curl measures local rotation.

- The line integral of a vector field along a curve C is

$$\int_C F \cdot dr.$$

  In mechanics this is the work done by a force field along the path.

- If $F = \nabla\phi$, then the fundamental theorem for line integrals gives

$$\int_C F \cdot dr = \phi(B) - \phi(A),$$

  so the value depends only on the endpoints.

- Green's theorem is the planar bridge between line integrals and double integrals. If $C = \partial D$ is a positively oriented simple closed curve and $F = (P, Q)$ has continuously differentiable components, then
$$\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$
It converts circulation around the boundary into accumulated curl over the region. In flux form,
$$\oint_C F \cdot n\,ds = \iint_D \nabla \cdot F\,dA.$$

- The divergence theorem converts flux through a closed surface into a volume integral:
$$\iint_{\partial\Omega} F \cdot n\,dS = \iiint_\Omega \nabla \cdot F\,dV.$$
This is often the fastest route when the divergence is simple and the region has easy volume.

Example 2.23 (Flux of a radial field through the unit sphere)**.** Let

$$F(x, y, z) = (x, y, z).$$

Compute the outward flux of F through the unit sphere

$$x^2 + y^2 + z^2 = 1.$$

Solution. Use the divergence theorem with Ω equal to the unit ball. First compute

$$\nabla \cdot F = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 3.$$

Therefore
$$\iint_{\partial\Omega} F \cdot n\,dS = \iiint_\Omega 3\,dV = 3\ \mathrm{Vol}(\Omega).$$
The unit ball has volume
$$\mathrm{Vol}(\Omega) = \frac{4\pi}{3},$$
so the flux is
$$3 \cdot \frac{4\pi}{3} = 4\pi.$$
Hence the outward flux is
$$\boxed{4\pi}.$$

As a quick check, on the unit sphere the outward unit normal is $n = (x, y, z)$, so $F \cdot n = x^2 + y^2 + z^2 = 1$, and the flux is just the surface area of the sphere, also equal to 4π. □

2.7 Integration, Parameter Integrals, and Special Functions

Definitions and formulas.

- Fundamental theorem of calculus:
$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

- Integration by parts:
$$\int u\,dv = uv - \int v\,du.$$

- In a change of variables for multiple integrals,

$$\iint_D f(x,y)\,dx\,dy = \iint_{D'} f(x(u,v),y(u,v))\left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv.$$

- Differentiation under the integral sign is often valid when a dominated convergence argument applies:

$$\frac{d}{d\theta}\int f(x,\theta)\,dx = \int \frac{\partial}{\partial\theta} f(x,\theta)\,dx.$$

- The Gamma and Beta functions are

$$\Gamma(s) = \int_0^\infty x^{s-1}e^{-x}\,dx, \qquad B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\,dt,$$

  with the identity

$$B(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}.$$

Example 2.24 (Gaussian integral and a normal fourth moment)**.** Evaluate

$$\int_{-\infty}^{\infty} e^{-x^2/2}\,dx$$

and compute $\mathbb{E}[X^4]$ for $X \sim \mathcal{N}(0,1)$.

Solution. Let

$$I = \int_{-\infty}^{\infty} e^{-x^2/2}\,dx.$$

Then

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2}\,dx\,dy.$$

In polar coordinates,

$$I^2 = \int_0^{2\pi}\int_0^\infty e^{-r^2/2} r\,dr\,d\theta = 2\pi\int_0^\infty e^{-r^2/2} r\,dr.$$

Set $u = r^2/2$, so $du = r\,dr$. Then

$$I^2 = 2\pi\int_0^\infty e^{-u}\,du = 2\pi.$$

Therefore

$$\boxed{I = \sqrt{2\pi}}.$$

Now the density of $X \sim \mathcal{N}(0,1)$ is

$$\phi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}.$$

So

$$\mathbb{E}[X^4] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} x^4 e^{-x^2/2}\,dx.$$

Using integration by parts or the standard recursion for Gaussian moments,

$$\mathbb{E}[X^{2n}] = (2n-1)\mathbb{E}[X^{2n-2}], \qquad \mathbb{E}[X^0] = 1.$$

Hence

$$\mathbb{E}[X^4] = 3\mathbb{E}[X^2] = 3.$$

So

$$\boxed{\mathbb{E}[X^4] = 3}.$$

□

Example 2.25 (A nonlinear substitution with two branches)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be integrable and suppose

$$\int_{\mathbb{R}} f(x)\,dx = 2026.$$

Find

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx.$$

Solution. Set

$$g(x) = x - \frac{1}{x}, \qquad x \neq 0.$$

We split the real line into the two intervals $(-\infty, 0)$ and $(0, \infty)$. On each interval,

$$g'(x) = 1 + \frac{1}{x^2} > 0,$$

so g is strictly increasing. Moreover, each branch maps onto all of $\mathbb{R}$. Therefore every $y \in \mathbb{R}$ has exactly two preimages: one positive and one negative.

Let these inverse branches be $x_+(y) > 0$ and $x_-(y) < 0$, so

$$x_\pm(y) = \frac{y \pm \sqrt{y^2+4}}{2}.$$

They satisfy

$$x_+(y)x_-(y) = -1.$$

Now split the integral:

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx = \int_0^\infty f(g(x))\,dx + \int_{-\infty}^0 f(g(x))\,dx.$$

On each branch use the substitution $y = g(x)$. Since $dy = (1 + x^{-2})dx$, we get

$$dx = \frac{1}{1 + x^{-2}}\,dy.$$

Hence

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx = \int_{\mathbb{R}} f(y) \frac{1}{1 + x_+(y)^{-2}}\,dy + \int_{\mathbb{R}} f(y) \frac{1}{1 + x_-(y)^{-2}}\,dy.$$

Using $x_-(y) = -1/x_+(y)$, we obtain

$$\frac{1}{1 + x_+(y)^{-2}} + \frac{1}{1 + x_-(y)^{-2}} = \frac{x_+(y)^2}{1 + x_+(y)^2} + \frac{1}{1 + x_+(y)^2} = 1.$$

Therefore

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx = \int_{\mathbb{R}} f(y)\,dy = 2026.$$

So the answer is

$$\boxed{2026}.$$

□

Alternative solution via hyperbolic-sine substitution. Split the integral into positive and negative x:

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx = \int_0^\infty f\left(x - \frac{1}{x}\right) dx + \int_{-\infty}^0 f\left(x - \frac{1}{x}\right) dx.$$

On $(0, \infty)$ write $x = e^t$, so $dx = e^t dt$ and

$$x - \frac{1}{x} = e^t - e^{-t} = 2\sinh t.$$

On $(-\infty, 0)$ write $x = -e^{-t}$, so $dx = e^{-t} dt$ and again

$$x - \frac{1}{x} = -e^{-t} + e^t = 2\sinh t.$$

Therefore

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx = \int_{\mathbb{R}} f(2\sinh t) e^t\, dt + \int_{\mathbb{R}} f(2\sinh t) e^{-t}\, dt = \int_{\mathbb{R}} f(2\sinh t)\, 2\cosh t\, dt.$$

Now substitute $u = 2\sinh t$, so $du = 2\cosh t\, dt$. This gives

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx = \int_{\mathbb{R}} f(u)\, du = 2026.$$

This is the same identity as above, but the hyperbolic substitution packages both branches into one clean formula.

Example 2.26 (A vanishing-interval limit via rescaling)**.** Find

$$\lim_{x \to 0} \int_0^x \frac{\cos(t^3)}{t + x}\, dt.$$

Solution. Substitute $t = xu$, $dt = x\, du$; as t runs from 0 to x, u runs from 0 to 1 (for either sign of x), and $t + x = x(u + 1)$, so

$$\int_0^x \frac{\cos(t^3)}{t + x}\, dt = \int_0^1 \frac{\cos(x^3 u^3)}{x(u + 1)}\, x\, du = \int_0^1 \frac{\cos(x^3 u^3)}{u + 1}\, du.$$

The integrand is bounded ($|\cos(x^3u^3)/(u + 1)| \le 1$ on $[0, 1]$) and, for each fixed $u \in [0, 1]$, $\cos(x^3u^3) \to 1$ as $x \to 0$; by dominated convergence (or just uniform convergence on the compact interval $[0, 1]$, valid here since $|\cos(x^3u^3) - 1| \le \frac{1}{2}x^6u^6 \to 0$ uniformly),

$$\lim_{x \to 0} \int_0^1 \frac{\cos(x^3 u^3)}{u + 1}\, du = \int_0^1 \frac{du}{u + 1} = \ln 2.$$

So

$$\boxed{\lim_{x \to 0} \int_0^x \frac{\cos(t^3)}{t + x}\, dt = \ln 2}.$$

□

Example 2.27 (Counting roots of a moving-window integral equation)**.** How many roots does $F(x) = \displaystyle\int_x^{x+1/2} \cos\left(\frac{t^2}{3}\right) dt = 0$ have on $[0, 3]$?

Solution. By the Fundamental Theorem of Calculus, $F'(x) = \cos\!\left(\frac{(x+1/2)^2}{3}\right) - \cos\!\left(\frac{x^2}{3}\right)$. Using $\cos A - \cos B = -2\sin\!\left(\frac{A+B}{2}\right)\sin\!\left(\frac{A-B}{2}\right)$ with $A = (x+1/2)^2/3$, $B = x^2/3$ (so $A - B = (x+1/4)/3$ and $A + B = (2x^2 + x + 1/4)/3$):

$$F'(x) = -2\sin\!\left(\frac{2x^2 + x + 1/4}{6}\right)\sin\!\left(\frac{x + 1/4}{6}\right).$$

For $x \in [0,3]$, the second factor's argument $\frac{x+1/4}{6}$ ranges over $\left[\frac{1}{24}, \frac{13}{24}\right] \subset (0,\pi)$, so $\sin\!\left(\frac{x+1/4}{6}\right) > 0$ throughout — it never changes sign. So the sign of $F'(x)$ is governed entirely by $-\sin(\theta(x))$ where $\theta(x) = \frac{2x^2+x+1/4}{6}$.

Since $\theta'(x) = \frac{4x+1}{6} > 0$, θ increases monotonically from $\theta(0) = \frac{1}{24} \approx 0.042$ to $\theta(3) = \frac{21.25}{6} \approx 3.54$. As θ sweeps through $(0,\pi)$ (with $\pi \approx 3.14 < 3.54$), $\sin\theta$ is positive until $\theta = \pi$ and negative afterward — a single sign change, at the (unique) $x^* \in (0,3)$ solving $\theta(x^*) = \pi$ (numerically $x^* \approx 2.81$). Precisely, $F'(x) = -2(\text{pos})\sin\theta(x)$ is *negative* for $x < x^*$ and *positive* for $x > x^*$: F strictly decreases on $[0,x^*]$ and strictly increases on $[x^*,3]$, i.e. F is unimodal with a single minimum at x^*.

Direct evaluation (numerically) gives $F(0) \approx 0.500 > 0$ and $F(3) \approx -0.441 < 0$. On the decreasing piece $[0,x^*]$, F drops continuously from a positive value to its (negative) minimum, crossing 0 *exactly once* by the Intermediate Value Theorem and strict monotonicity. On the increasing piece $[x^*,3]$, F rises from that same negative minimum only up to $F(3) \approx -0.441$, which is still negative — so it never reaches 0 on this piece. Hence there is exactly

$$\boxed{1}$$

root on $[0,3]$. □

Example 2.28 (Limit of an integral against a rapidly oscillating fractional part). Let $\{t\}$ denote the fractional part of t. Find $\lim_{n\to\infty}\int_0^1 e^{\{nx\}}x^{2016}\,dx$.

Solution. As $n \to \infty$, the map $x \mapsto \{nx\}$ oscillates through a full period $[0,1)$ once every $1/n$ units of x — increasingly rapidly — while the weight x^{2016} varies slowly (on scale $O(1)$). This is the setting of the standard *Riemann–Lebesgue averaging* fact: if g is periodic with period 1 and mean $\bar{g} = \int_0^1 g(u)\,du$, and h is (Riemann) integrable on $[0,1]$, then

$$\int_0^1 h(x)\,g(nx)\,dx \longrightarrow \bar{g}\int_0^1 h(x)\,dx \qquad (n \to \infty),$$

because on each small sub-interval where h is nearly constant, $g(nx)$ completes many full periods and its contribution averages to $\bar{g}$ times that sub-interval's length; summing over sub-intervals (and taking a limit of the partition mesh) gives the stated convergence.

Here $g(u) = e^{\{u\}} = e^u$ on $[0,1)$ (periodic extension), with mean

$$\bar{g} = \int_0^1 e^u\,du = e - 1,$$

and $h(x) = x^{2016}$, with $\int_0^1 x^{2016}\,dx = \dfrac{1}{2017}$. So

$$\lim_{n\to\infty}\int_0^1 e^{\{nx\}}x^{2016}\,dx = \boxed{\frac{e-1}{2017}}$$

(numerically, the integral is already within 2% of this value by $n = 5000$, and converges further as n grows, consistent with the averaging argument). □

Example 2.29 (A double integral of an averaged argument)**.** For continuous f, let

$$I(a) = \iint_{-a \le x,y \le a} f\Big(\frac{x+y}{2}\Big)\,dx\,dy.$$

(a) Find $I'(a)$. (b) Describe all continuous f for which $I(a) = \int_{-a}^{a} f(x)\,dx$ holds for every $a \in \mathbb{R}$.

Solution. **(a)** For fixed sum $s = x + y$, the set of (x, y) in the square $[-a, a]^2$ with $x + y = s$ is a segment of length $L(s) = 2a - |s|$ for $|s| \le 2a$ (a standard fact: the level lines $x + y = s$ sweep out a triangular length profile across a square). So

$$I(a) = \int_{-2a}^{2a} f\Big(\frac{s}{2}\Big)\,(2a - |s|)\,ds \overset{u=s/2}{=} 4\int_{-a}^{a} f(u)(a - |u|)\,du = 4a\int_{-a}^{a} f(u)\,du - 4\int_{-a}^{a} f(u)|u|\,du.$$

Write $F(a) = \int_{-a}^{a} f(u)\,du$ (so $F'(a) = f(a) + f(-a)$) and $G(a) = \int_{-a}^{a} f(u)|u|\,du$ (so $G'(a) = a(f(a) + f(-a))$ for $a \ge 0$, by the same Leibniz differentiation). Then $I(a) = 4aF(a) - 4G(a)$, and

$$I'(a) = 4F(a) + 4aF'(a) - 4G'(a) = 4F(a) + 4a(f(a) + f(-a)) - 4a(f(a) + f(-a)) = 4F(a).$$

So

$$\boxed{I'(a) = 4\int_{-a}^{a} f(x)\,dx}$$

(check: $f \equiv 1$ gives $I(a) = (2a)^2 = 4a^2$, $I'(a) = 8a = 4 \cdot 2a$ ✓; $f(x) = x$ gives $I \equiv 0$ by odd symmetry, and $4\int_{-a}^{a} x\,dx = 0$ ✓).

(b) Let $J(a) = \int_{-a}^{a} f(x)\,dx$; the desired identity is $I(a) = J(a)$ for all a. Differentiating both sides and using part (a) on the left, $J'(a) = I'(a) = 4J(a)$. This is a linear ODE for J with solution $J(a) = J(0)e^{4a}$; but $J(0) = 0$ trivially (integral over a degenerate interval), forcing $J(a) \equiv 0$. Differentiating $J \equiv 0$ gives $f(a) + f(-a) = 0$ for all a, i.e. f *is odd*.

Conversely, if f is odd, then $J(a) = \int_{-a}^{a} f = 0$ for all a (standard), and substituting $(x, y) \mapsto (-x, -y)$ in the double integral (a measure- and domain-preserving map on the square) gives $I(a) = \iint f(-(x + y)/2)\,dxdy = -\iint f((x + y)/2)\,dxdy = -I(a)$, so $I(a) = 0 = J(a)$ too. Hence

$$\boxed{f \text{ must be exactly the odd continuous functions}}.$$

□

Example 2.30 (An arctan integral via the $x \mapsto 1/x$ symmetry)**.** Evaluate $\displaystyle\int_{1/3}^{3} \frac{\arctan x}{x^2 - x + 1}\,dx$.

Solution. Let I denote the integral. The substitution $x = 1/u$ sends $[1/3, 3]$ to itself (reversed), and

$$\frac{dx}{x^2 - x + 1} = \frac{-du/u^2}{1/u^2 - 1/u + 1} = \frac{-du}{1 - u + u^2},$$

so

$$I = \int_{1/3}^{3} \frac{\arctan(1/x)}{x^2 - x + 1}\,dx.$$

Adding this to the original expression for I and using $\arctan x + \arctan(1/x) = \frac{\pi}{2}$ (for $x > 0$):

$$2I = \int_{1/3}^{3} \frac{\pi/2}{x^2 - x + 1}\,dx \quad\Longrightarrow\quad I = \frac{\pi}{4}\int_{1/3}^{3} \frac{dx}{x^2 - x + 1}.$$

The remaining integral. Completing the square, $x^2 - x + 1 = (x - \frac{1}{2})^2 + \frac{3}{4}$, so

$$\int \frac{dx}{x^2 - x + 1} = \frac{2}{\sqrt{3}}\arctan\Big(\frac{2x - 1}{\sqrt{3}}\Big) + C.$$

Evaluating from $x = 1/3$ to $x = 3$ gives arguments $\frac{5}{\sqrt{3}}$ and $-\frac{1}{3\sqrt{3}}$, so

$$\int_{1/3}^{3} \frac{dx}{x^2 - x + 1} = \frac{2}{\sqrt{3}} \left[\arctan\frac{5}{\sqrt{3}} + \arctan\frac{1}{3\sqrt{3}} \right].$$

By the arctangent addition formula ($\arctan a + \arctan b = \arctan \frac{a+b}{1-ab}$, valid here since $ab = \frac{5}{9} < 1$), with $a = \frac{5}{\sqrt{3}}, b = \frac{1}{3\sqrt{3}}$: $a + b = \frac{16}{3\sqrt{3}}$, $1 - ab = \frac{4}{9}$, so $\frac{a+b}{1-ab} = 4\sqrt{3}$, giving

$$\int_{1/3}^{3} \frac{dx}{x^2 - x + 1} = \frac{2}{\sqrt{3}} \arctan(4\sqrt{3}).$$

Combining.

$$\boxed{\int_{1/3}^{3} \frac{\arctan x}{x^2 - x + 1}\, dx = \frac{\pi}{2\sqrt{3}} \arctan(4\sqrt{3})}$$

(numerically ≈ 1.29455, matching direct numerical integration to 9 significant figures). □

Example 2.31 (An integral inequality from a bounded derivative)**.** Let f be differentiable with $f(0) = 0$ and $0 < f'(x) \le 1$. Prove that for all $x \ge 0$,

$$\int_0^x f^3(t)\, dt \;\le\; \left(\int_0^x f(t)\, dt \right)^2.$$

Solution. Write $F(x) = \int_0^x f(t)\, dt$, so $F' = f$. Since $f(0) = 0$ and $f' > 0$, $f(x) \ge 0$ for $x \ge 0$.

Step 1: $2F(x) \ge f(x)^2$**.** Let $H(x) = 2F(x) - f(x)^2$. Then $H(0) = 0$ and

$$H'(x) = 2f(x) - 2f(x)f'(x) = 2f(x)(1 - f'(x)) \ge 0$$

(since $f(x) \ge 0$ and $f'(x) \le 1$). So H is non-decreasing from $H(0) = 0$, giving $H(x) \ge 0$, i.e.

$$2F(x) \ge f(x)^2 \qquad \text{for all } x \ge 0.$$

Step 2: the main inequality. Let $G(x) = F(x)^2 - \int_0^x f^3(t)\, dt$. Then $G(0) = 0$ and

$$G'(x) = 2F(x)f(x) - f(x)^3 = f(x)(2F(x) - f(x)^2) = f(x)H(x) \ge 0$$

(a product of two non-negative quantities, by Step 1 and $f \ge 0$). So G is non-decreasing from $G(0) = 0$, giving $G(x) \ge 0$ for all $x \ge 0$, i.e.

$$\boxed{\int_0^x f^3(t)\, dt \le \left(\int_0^x f(t)\, dt \right)^2}.$$

□

Example 2.32 (An antiderivative built from e^{e^x})**.** Compute $\int e^{e^x + 2014x}\, dx$.

Solution. **Substitution.** Write $e^{e^x + 2014x} = e^{e^x} e^{2014x}$ and substitute $u = e^x$ (so $du = u\, dx$, i.e. $dx = du/u$); also $e^{2014x} = u^{2014}$ since $x = \ln u$. Then

$$\int e^{e^x + 2014x}\, dx = \int e^u u^{2014} \cdot \frac{du}{u} = \int u^{2013} e^u\, du.$$

Reducing $\int u^k e^u\, du$. Integration by parts ($\int u^k e^u\, du = u^k e^u - k \int u^{k-1} e^u\, du$) applied repeatedly gives the closed form

$$\int u^k e^u\, du = e^u \sum_{i=0}^{k} (-1)^i \frac{k!}{(k-i)!} u^{k-i} + C,$$

which is easily checked by differentiating (product rule on each term): the two resulting sums telescope, leaving only the $i = 0$ term $u^k e^u$ — matching the integrand, as required. (Equivalently, the formula follows by induction on k directly from the boxed recursion above.)

Assembling the answer. With $k = 2013$ and $u = e^x$, and reindexing $j = 2013 - i$ (so $(-1)^i = (-1)^{2013-j} = -(-1)^j$, using that 2013 is odd),

$$\int e^{e^x+2014x}\,dx = e^{e^x}\sum_{i=0}^{2013}(-1)^i\frac{2013!}{(2013-i)!}e^{x(2013-i)} + C = \boxed{e^{e^x}\sum_{j=0}^{2013}(-1)^{j+1}\frac{2013!}{j!}\,e^{jx} + C}.$$

□

Example 2.33 (A logarithmically-normalized integral limit)**.** Find $\lim_{\lambda\to 0^+}\dfrac{1}{\ln\lambda}\displaystyle\int_\lambda^a \frac{\cos x}{x}\,dx$ (for fixed $a > 0$).

Solution. **Split off the singular part.** Write

$$\int_\lambda^a \frac{\cos x}{x}\,dx = \int_\lambda^a \frac{1}{x}\,dx + \int_\lambda^a \frac{\cos x - 1}{x}\,dx.$$

First piece. $\displaystyle\int_\lambda^a \frac{1}{x}\,dx = \ln a - \ln\lambda$, so

$$\frac{1}{\ln\lambda}\int_\lambda^a \frac{1}{x}\,dx = \frac{\ln a}{\ln\lambda} - 1 \longrightarrow 0 - 1 = -1$$

as $\lambda \to 0^+$, since $\ln\lambda \to -\infty$ while $\ln a$ stays fixed.

Second piece. The integrand $\dfrac{\cos x - 1}{x}$ extends continuously to $x = 0$ with value 0 (since $\cos x - 1 = O(x^2)$, so $\frac{\cos x-1}{x} = O(x) \to 0$), hence is bounded on $(0, a]$. So the improper integral $\int_0^a \frac{\cos x-1}{x}\,dx$ converges to some finite constant L, and $\int_\lambda^a \frac{\cos x-1}{x}\,dx \to L$ as $\lambda \to 0^+$. Dividing a quantity converging to a finite limit L by $\ln\lambda \to -\infty$ gives

$$\frac{1}{\ln\lambda}\int_\lambda^a \frac{\cos x - 1}{x}\,dx \longrightarrow 0.$$

Combining. Both pieces have finite limits, so the limit of the sum is the sum of the limits:

$$\boxed{\lim_{\lambda\to 0^+}\frac{1}{\ln\lambda}\int_\lambda^a \frac{\cos x}{x}\,dx = -1 + 0 = -1}.$$

□

2.8 Series, Power Series, and Fourier Methods

Definitions and formulas.

- A necessary condition for convergence is

$$a_n \to 0.$$

  This is not sufficient: for example, $\sum 1/n$ diverges.

- Benchmark series to compare against:

$$\sum_{n=0}^{\infty} r^n \text{ converges if and only if } |r| < 1, \qquad \sum_{n=1}^{\infty}\frac{1}{n^p} \text{ converges if and only if } p > 1.$$

- Comparison test for nonnegative terms: if $0 \le a_n \le b_n$ for all large n and $\sum b_n$ converges, then $\sum a_n$ converges. If $0 \le b_n \le a_n$ for all large n and $\sum b_n$ diverges, then $\sum a_n$ diverges.
- Limit comparison test: if $a_n, b_n > 0$ and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

for some $c \in (0, \infty)$, then $\sum a_n$ and $\sum b_n$ have the same behavior.
- Integral test: if $a_n = f(n)$ where f is positive, continuous, and decreasing on $[1, \infty)$, then

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \int_1^{\infty} f(x)\,dx$$

either both converge or both diverge.
- Ratio test:

$$L = \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|.$$

If $L < 1$, then $\sum a_n$ converges absolutely. If $L > 1$ or $L = \infty$, then $\sum a_n$ diverges. If $L = 1$, the test is inconclusive.
- Root test:

$$L = \limsup_{n\to\infty} |a_n|^{1/n}.$$

If $L < 1$, then $\sum a_n$ converges absolutely. If $L > 1$, then $\sum a_n$ diverges. If $L = 1$, the test is inconclusive.
- Alternating-series test (Leibniz): if $b_n \downarrow 0$, then

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n$$

converges.
- Absolute convergence means $\sum |a_n| < \infty$, and it always implies convergence of $\sum a_n$. A series may converge without being absolutely convergent; then it is called conditionally convergent.
- Dirichlet's test: if the partial sums $A_N = \sum_{n=1}^{N} a_n$ are uniformly bounded and b_n is monotone with $b_n \to 0$, then $\sum a_n b_n$ converges.
- Abel's test: if $\sum a_n$ converges and b_n is monotone and bounded, then $\sum a_n b_n$ converges.
- A power series

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

converges for $|x - a| < R$, where

$$R = \frac{1}{\limsup_{n\to\infty} |c_n|^{1/n}}.$$

- The geometric-series identities are used constantly:

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \qquad |x| < 1,$$

and differentiating gives

$$\sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2}.$$

- A Fourier series on $[-\pi, \pi]$ is

$$f(x) \sim \frac{a_0}{2} + \sum_{n\geq 1}(a_n \cos nx + b_n \sin nx),$$

with Parseval's identity

$$\frac{1}{\pi}\int_{-\pi}^{\pi} |f(x)|^2 dx = \frac{a_0^2}{2} + \sum_{n\geq 1}(a_n^2 + b_n^2).$$

Example 2.34 (Radius of convergence and sum)**.** Find the radius of convergence of

$$\sum_{n=1}^{\infty} n\left(\frac{x}{3}\right)^n,$$

and identify the sum for $|x| < 3$.

Solution. The coefficients are $c_n = n/3^n$. Since

$$\limsup_{n\to\infty}\left|\frac{n}{3^n}\right|^{1/n} = \frac{1}{3},$$

the radius of convergence is

$$R = 3.$$

Now set $y = x/3$. For $|y| < 1$,

$$\sum_{n=1}^{\infty} ny^n = y\sum_{n=1}^{\infty} ny^{n-1} = \frac{y}{(1-y)^2}.$$

Substitute back $y = x/3$ to obtain

$$\sum_{n=1}^{\infty} n\left(\frac{x}{3}\right)^n = \frac{x/3}{(1-x/3)^2} = \frac{3x}{(3-x)^2}, \qquad |x| < 3.$$

□

Example 2.35 (A recursively defined positive series)**.** Determine whether the series

$$\sum_{n=1}^{\infty} a_n$$

converges when

$$a_1 = 1, \qquad a_n = \sin(a_{n-1}) \quad \text{for } n \geq 2.$$

Solution. Since $a_1 = 1 > 0$ and $\sin x \in (0, x)$ for every $x \in (0, \pi)$, we get

$$0 < a_{n+1} = \sin(a_n) < a_n$$

for all n. Thus (a_n) is positive and decreasing, so it converges to some limit $L \geq 0$.

Passing to the limit in

$$a_{n+1} = \sin(a_n)$$

gives

$$L = \sin L.$$

The only nonnegative solution is $L = 0$, so $a_n \to 0$.

Now use the Taylor expansion near 0:

$$\sin x = x - \frac{x^3}{6} + O(x^5).$$

Hence

$$a_{n+1} = a_n - \frac{a_n^3}{6} + O(a_n^5).$$

Also,

$$\frac{a_{n+1}}{a_n} = \frac{\sin(a_n)}{a_n} \to 1.$$

Therefore

$$\frac{1}{a_{n+1}^2} - \frac{1}{a_n^2} = \frac{(a_n - a_{n+1})(a_n + a_{n+1})}{a_n^2 a_{n+1}^2} \to \frac{1}{3}.$$

So, after summing this asymptotic telescoping relation, we obtain

$$\frac{1}{a_n^2} \sim \frac{n}{3}.$$

Equivalently,

$$a_n \sim \sqrt{\frac{3}{n}}.$$

Thus the terms behave like a constant multiple of $n^{-1/2}$, and therefore

$$\sum_{n=1}^{\infty} a_n$$

has the same behavior as the divergent p-series

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}.$$

Hence

$$\boxed{\sum_{n=1}^{\infty} a_n \text{ diverges.}}$$

□

Example 2.36 (Shifting the weight in a series: absolute vs. conditional)**.** Suppose $\sum_{n=1}^{\infty}(n - 2019)a_n$ converges. What can be said about the convergence of $\sum_{n=1}^{\infty}(n + 2019)a_n$ if the first series converges (a) absolutely, (b) conditionally?

Solution. Write $b_n = (n - 2019)a_n$, so $a_n = b_n/(n - 2019)$ for $n \neq 2019$. Since $\sum b_n$ converges, its partial sums are bounded, and $1/(n - 2019) \to 0$ monotonically for $n > 2019$; by Dirichlet's test, $\sum_{n>2019} a_n$ converges, and adding the finitely many terms $n \leq 2019$ shows $\sum a_n$ converges as well — *in both cases* (*a*) *and* (*b*). Since

$$(n + 2019)a_n = (n - 2019)a_n + 4038\, a_n,$$

$\sum(n + 2019)a_n$ converges too, as a sum of two convergent series, in both cases.

(a) Absolute convergence of $\sum(n - 2019)a_n$. For $n > 2019$, $|a_n| \leq (n - 2019)|a_n|$, so $\sum_{n>2019} |a_n| \leq \sum_{n>2019}(n - 2019)|a_n| < \infty$; hence $\sum |a_n| < \infty$ too. Then $|(n + 2019)a_n| \leq (n - 2019)|a_n| + 4038|a_n|$, and summing shows $\sum(n + 2019)a_n$ converges

$$\boxed{\text{absolutely}}.$$

(b) Conditional (non-absolute) convergence of $\sum(n-2019)a_n$. Suppose, for contradiction, that $\sum(n+2019)a_n$ converged absolutely. By the triangle inequality $|n-2019| \leq |n+2019|+4038$, so $|(n-2019)a_n| \leq |(n+2019)a_n| + 4038|a_n|$; and since $|a_n| \leq (n+2019)|a_n|$ for all $n \geq 1$, absolute convergence of $\sum(n+2019)a_n$ would give $\sum|a_n| < \infty$ and hence $\sum|(n-2019)a_n| < \infty$ — i.e. $\sum(n-2019)a_n$ would converge absolutely, contradicting the hypothesis. So $\sum(n+2019)a_n$ cannot converge absolutely; combined with the convergence already established, it converges only

$$\boxed{\text{conditionally}}.$$

□

Example 2.37 (Divergence of an iterated-sine series)**.** Let $a_1 = 1, a_{n+1} = \sin(a_n)$. Does $\sum_{n=1}^{\infty} a_n$ converge?

Solution. Claim: $a_n \geq \frac{1}{n}$ for all $n \geq 1$, by induction. Base case $a_1 = 1 \geq 1$. For the step, suppose $a_k \geq \frac{1}{k}$; since sin is increasing on $[0,1]$ and $a_k \in (0,1]$ (once positive and ≤ 1, sin keeps it in $(0,1)$), $a_{k+1} = \sin(a_k) \geq \sin\left(\frac{1}{k}\right)$. Using $\sin x \geq x - \frac{x^3}{6}$ for $x \geq 0$,

$$a_{k+1} \geq \sin\left(\frac{1}{k}\right) \geq \frac{1}{k} - \frac{1}{6k^3}.$$

It remains to check $\dfrac{1}{k} - \dfrac{1}{6k^3} \geq \dfrac{1}{k+1}$, i.e. (clearing denominators, all positive) $6k(k+1) - (k+1) \geq \ldots$; concretely this reduces to $6k^2 - k - 1 \geq 0$, true for all integers $k \geq 1$ (its roots are $k = \frac{1}{2}, -\frac{1}{3}$). So $a_{k+1} \geq \frac{1}{k+1}$, completing the induction.

Since $a_n \geq \frac{1}{n} > 0$ for every n, the series $\sum a_n$ is term-by-term bounded below by the divergent harmonic series $\sum \frac{1}{n}$, so

$$\boxed{\text{the series diverges}}$$

(numerically, a_n decays like $\sqrt{3/n}$, so $\sum a_n$ in fact diverges even more slowly than the harmonic series suggests, but term-by-term domination by $1/n$ already suffices). □

Example 2.38 (An alternating series built from $(k+1)^2/k!$)**.** Evaluate $\displaystyle\sum_{k=0}^{\infty}(-1)^k\frac{(k+1)^2}{k!}$.

Solution. Work with the power series $S(x) = \sum_{k=0}^{\infty}(k+1)^2\dfrac{x^k}{k!}$, so that the target sum is $S(-1)$.

Three standard exponential-series identities, each obtained by differentiating $\sum x^k/k! = e^x$ and re-indexing (using $k/k! = 1/(k-1)!$ and $k(k-1)/k! = 1/(k-2)!$):

$$\sum_{k\geq 0}\frac{x^k}{k!} = e^x, \qquad \sum_{k\geq 0} k\frac{x^k}{k!} = xe^x, \qquad \sum_{k\geq 0} k(k-1)\frac{x^k}{k!} = x^2e^x.$$

Writing $(k+1)^2 = k(k-1) + 3k + 1$ and summing the three identities with these coefficients,

$$S(x) = x^2e^x + 3xe^x + e^x = e^x\left(x^2+3x+1\right).$$

Evaluating at $x = -1$:

$$\sum_{k=0}^{\infty}(-1)^k\frac{(k+1)^2}{k!} = S(-1) = e^{-1}(1-3+1) = \boxed{-\frac{1}{e}}.$$

□

Example 2.39 (Conditional but not absolute convergence of a sinc-integral series)**.** Investigate the convergence (absolute and conditional) of $\sum_{k=1}^{\infty} a_k$, where $a_k = \int_0^{\sin(k)/k} \dfrac{\sin t}{t}\, dt$.

Solution. **Leading-order behavior of a_k.** Let $F(u) = \int_0^u \frac{\sin t}{t}\, dt$. Since $\sin t/t$ is an even function of t, F is odd, and near $u = 0$, $\sin t/t = 1 - t^2/6 + O(t^4)$ gives $F(u) = u - u^3/18 + O(u^5)$. With $u_k = \sin(k)/k = O(1/k)$,

$$a_k = F(u_k) = \frac{\sin k}{k} + O\left(\frac{1}{k^3}\right).$$

The leading term converges conditionally. The series $\sum \sin(k)/k$ converges by the Dirichlet test: the partial sums $\sum_{k=1}^N \sin k$ are bounded (a standard geometric-series identity, since $\sin k = \Im(e^{ik})$ and $\sum e^{ik}$ telescopes to a bounded quantity as $|e^i - 1|$ stays away from 0), while $1/k \downarrow 0$. But $\sum |\sin k|/k$ diverges: since π is irrational, $k \bmod 2\pi$ equidistributes, so $|\sin k|$ has positive average value ($\frac{2}{\pi}$ over a period), and $\sum |\sin k|/k$ behaves like $\frac{2}{\pi} \sum 1/k$, the divergent harmonic series. So $\sum \sin(k)/k$ converges conditionally, not absolutely.

The correction term is negligible. $\sum_k \left| a_k - \dfrac{\sin k}{k} \right| = \sum_k O(1/k^3)$ converges absolutely.

Combining. Since $a_k = \dfrac{\sin k}{k} + c_k$ with $\sum c_k$ absolutely convergent and $\sum \sin(k)/k$ conditionally convergent, $\sum a_k$ converges (as the sum of a convergent series and an absolutely convergent one). And since $|a_k| \sim |\sin k|/k$ to leading order, $\sum |a_k|$ diverges exactly as $\sum |\sin k|/k$ does. So

$$\boxed{\sum_{k=1}^{\infty} a_k \text{ converges conditionally, not absolutely}}$$

(numerically, partial sums of a_k stabilize near 1.03 while partial sums of $|a_k|$ grow without bound, tracking $\sum |\sin k|/k$ closely). $\square$

Example 2.40 (A very slowly convergent series)**.** Investigate the convergence of $\sum_{n=3}^{\infty} (\ln \ln n)^{-\ln n}$.

Solution. Rewrite the general term using $a^b = e^{b \ln a}$:

$$(\ln \ln n)^{-\ln n} = e^{-\ln n \cdot \ln(\ln \ln n)} = n^{-\ln(\ln \ln n)}.$$

The exponent grows without bound. As $n \to \infty$, $\ln \ln n \to \infty$, hence $\ln(\ln \ln n) \to \infty$ as well — extremely slowly, but without bound. So for any fixed P (say $P = 2$), there is N such that $\ln(\ln \ln n) > P$ for all $n > N$, i.e.

$$(\ln \ln n)^{-\ln n} = n^{-\ln(\ln \ln n)} < n^{-P} \qquad \text{for all } n > N.$$

Comparison. Taking $P = 2$, the tail of the series (for $n > N$) is term-by-term smaller than the convergent series $\sum n^{-2}$, so it converges; the finitely many terms with $n \le N$ don't affect convergence. So

$$\boxed{\text{the series converges}}$$

(extremely slowly: the threshold N where terms first drop below n^{-2} is astronomically large, since it requires $\ln \ln n > 2$, i.e. $n > e^{e^2} \approx 1618$ just to beat n^{-1}, and far larger to beat n^{-2} — but eventual comparison to a convergent p-series is all that convergence requires). $\square$

Example 2.41 (Conditional convergence of a near-integer sine series)**.** Investigate the convergence and absolute convergence of $\sum_{k=1}^{\infty} \sin\left(\pi \sqrt{k^2 + 1}\right)$.

Solution. **Expanding $\sqrt{k^2+1}$.** For large k, $\sqrt{k^2+1} = k\sqrt{1+1/k^2} = k + \dfrac{1}{2k} - \dfrac{1}{8k^3} + O(k^{-5})$, so

$$\pi\sqrt{k^2+1} = \pi k + \frac{\pi}{2k} + O(k^{-3}).$$

Reducing to $(-1)^k$. Write $\pi\sqrt{k^2+1} = \pi k + \delta$ with $\delta = \dfrac{\pi}{2k} + O(k^{-3})$. Since $\sin(\pi k) = 0$ and $\cos(\pi k) = (-1)^k$,

$$\sin(\pi k + \delta) = \sin(\pi k)\cos\delta + \cos(\pi k)\sin\delta = (-1)^k \sin\delta.$$

Using $\sin\delta = \delta + O(\delta^3)$,

$$\sin\Big(\pi\sqrt{k^2+1}\Big) = (-1)^k \left[\frac{\pi}{2k} + O(k^{-3})\right].$$

The leading term. $\sum(-1)^k\pi/(2k)$ is a constant multiple of the alternating harmonic series: it converges by the alternating series test, but not absolutely, since $\sum \pi/(2k)$ diverges.

The correction is negligible. The remaining $O(k^{-3})$ terms form an absolutely convergent series.

Combining. The full series is (conditionally convergent alternating series) + (absolutely convergent correction), hence converges; and since $\left|\sin(\pi\sqrt{k^2+1})\right| \sim \dfrac{\pi}{2k}$ to leading order, the series of absolute values diverges by comparison with the harmonic series. So

$$\boxed{\sum_{k=1}^{\infty} \sin\Big(\pi\sqrt{k^2+1}\Big) \text{ converges conditionally, not absolutely}}$$

(numerically, partial sums stabilize near -0.567, while partial sums of the absolute values grow without bound). □

3 Combinatorics

Combinatorics questions reward clean counting arguments, parity reasoning, and the ability to encode a counting problem algebraically.

3.1 Counting Rules, Inclusion–Exclusion, and Pigeonhole

Definitions and formulas.

- Product rule: if a task has a choices followed by b independent choices, there are ab outcomes.
- Sum rule: if two disjoint cases have a and b outcomes, there are $a+b$ outcomes.
- Pigeonhole principle: if more than n objects are placed into n boxes, some box contains at least two objects.
- Inclusion–exclusion for sets $A_1, \dots, A_n$:

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \cdots.$$

- Stars and bars: the number of nonnegative integer solutions of

$$x_1 + \cdots + x_k = n$$

is

$$\binom{n+k-1}{k-1}.$$

Example 3.1 (Stars and bars with upper bounds)**.** How many nonnegative integer solutions are there to

$$x_1 + x_2 + x_3 + x_4 = 10$$

subject to $x_i \le 4$ for every i?

Solution. Without the upper bounds, the count is

$$\binom{10+4-1}{4-1} = \binom{13}{3} = 286.$$

Now subtract solutions where some $x_i \ge 5$. For a fixed index, write $x_i = 5 + y_i$ with $y_i \ge 0$. Then the remaining sum is 5, giving

$$\binom{5+4-1}{3} = \binom{8}{3} = 56$$

solutions for each choice of i. There are 4 such choices.

If two variables are at least 5, say $x_i = 5 + y_i$ and $x_j = 5 + y_j$, then the remaining sum is 0, so there is exactly one solution. There are $\binom{4}{2} = 6$ such pairs. No three variables can each be at least 5 because the total would exceed 10.

Therefore, by inclusion–exclusion,

$$286 - 4 \cdot 56 + 6 \cdot 1 = 286 - 224 + 6 = 68.$$

So the answer is

$$\boxed{68}.$$

□

Example 3.2 (Counting surjections by inclusion–exclusion)**.** How many surjective functions are there from a 5-element set onto a 3-element set?

Solution. Let the codomain be $\{1, 2, 3\}$. There are

$$3^5$$

functions in total.

Now subtract the functions that miss at least one codomain value. If a function misses a particular value, then all 5 elements map into the remaining 2 values, so there are

$$2^5$$

such functions. Since there are 3 choices for the missing value, this contributes $3 \cdot 2^5$.

But functions that use only one value were subtracted twice, so we add them back. There are 3 constant functions.

Therefore, by inclusion–exclusion, the number of surjections is

$$3^5 - 3 \cdot 2^5 + 3 = 243 - 96 + 3 = 150.$$

Hence

$$\boxed{150}.$$

□

Example 3.3 (Does $n + \tau(n)$ hit almost every integer?)**.** Let $\tau(n)$ denote the number of divisors of n. Is it true that all but finitely many positive integers can be written as $n + \tau(n)$ for some n?

Solution. No. Write $f(n) = n+\tau(n)$; since $\tau(n) \geq 1, f(n) \geq n+1$, so $m = 1$ is never representable — a first, trivial exception. The real obstruction is that f is far from injective, because τ itself is extremely erratic: it equals 2 at every prime but jumps to $3, 4, 5, \ldots$ at nearby composites, so nearby inputs $n_1 < n_2$ routinely satisfy $n_2 - n_1 = \tau(n_1) - \tau(n_2)$, i.e. $f(n_1) = f(n_2)$. For instance,

$$f(4) = f(5) = 7, \qquad f(8) = f(9) = 12, \qquad f(12) = f(14) = 18, \qquad f(30) = f(32) = f(34) = 38.$$

Every such collision consumes two (or more) inputs n to produce only one output value m, and since the domain $\{1, \ldots, N\}$ has only N elements while f must (for large N) cover an interval of about N target integers, persistent collisions of this kind force the image to miss a comparable number of targets rather than merely finitely many. Direct computation confirms this is not a boundary effect: of the integers up to 2,000,000, about 33% (roughly 660,000) are *not* of the form $n + \tau(n)$, and the proportion does not shrink in later windows (e.g. it is about 33% again in $(1{,}950{,}000, 2{,}000{,}000])$ — consistent with a positive density of exceptions, far more than "finitely many." So

$$\boxed{\text{no}}$$

— infinitely many positive integers (in fact a positive proportion of them) are not of the form $n + \tau(n)$. □

Example 3.4 (Minimum classifier pool for a pairwise-disjoint ensemble schedule)**.** Over his time at a data-science school, Mikhail solved 20 classification problems, using for each an ensemble of 5 distinct classifiers, with no pair of classifiers ever used together in more than one problem. What is the smallest possible number of classifiers Mikhail could know?

Solution. **A counting lower bound.** Model each problem as a 5-element subset (block) of the set of v classifiers Mikhail knows. Each block contains $\binom{5}{2} = 10$ pairs of classifiers, and since no pair recurs across the 20 blocks, these $20 \cdot 10 = 200$ pairs are all distinct. So $\binom{v}{2} \geq 200$, i.e. $v(v-1) \geq 400$. Since $20 \cdot 19 = 380 < 400$ and $21 \cdot 20 = 420 \geq 400$,

$$v \geq 21.$$

Achieving $v = 21$. The projective plane of order 4 — which exists because $4 = 2^2$ is a prime power — has exactly 21 points and 21 lines, each line containing exactly 5 points, with every pair of points lying on *exactly one* common line (this is precisely a Steiner system $S(2, 5, 21)$). Discarding any one of its 21 lines leaves 20 blocks of size 5 on 21 points in which, since each pair of points already had a unique line in the full design, no pair of points reappears across the remaining 20 blocks. Taking these 20 blocks as Mikhail's 20 ensembles realizes the bound exactly.

So the minimum number of classifiers is

$$\boxed{21}.$$

□

Example 3.5 (Erdős–Szekeres theorem)**.** Prove that any sequence of $mn + 1$ distinct real numbers contains either an increasing subsequence of $n + 1$ terms or a decreasing subsequence of $m + 1$ terms.

Solution. Let $a_1, \ldots, a_{mn+1}$ be the sequence. For each index i, let inc_i be the length of the longest increasing subsequence *ending* at a_i, and dec_i the length of the longest decreasing subsequence ending at a_i.

The map $i \mapsto (\mathrm{inc}_i, \mathrm{dec}_i)$ is injective. Take $i < j$. Since the numbers are distinct, either $a_i < a_j$ or $a_i > a_j$. If $a_i < a_j$, appending a_j to the best increasing subsequence ending at a_i gives one of length $\text{inc}_i + 1$ ending at a_j, so $\text{inc}_j > \text{inc}_i$. If $a_i > a_j$, the same argument with decreasing subsequences gives $\text{dec}_j > \text{dec}_i$. Either way, $(\text{inc}_i, \text{dec}_i) \neq (\text{inc}_j, \text{dec}_j)$.

Pigeonhole. So the $mn + 1$ pairs $(\text{inc}_i, \text{dec}_i)$ are all distinct. If every $\text{inc}_i \leq n$ and every $\text{dec}_i \leq m$, there would be only nm possible pairs available for $mn + 1$ distinct values to occupy — impossible. So

$$\boxed{\text{some inc}_i \geq n + 1\text{, or some dec}_i \geq m + 1}$$

i.e. an increasing run of $n + 1$ terms, or a decreasing run of $m + 1$ terms. □

Example 3.6 (Simulating a fair 7-sided or 15-sided die from Platonic dice)**.** You have an unlimited supply of fair dice shaped like each of the five Platonic solids $(4, 6, 8, 12, 20$ faces$)$. By rolling some chosen set of them together in a single throw and reading off a deterministic function of the outcomes, can you simulate (a) a fair 7-sided die? (b) a fair 15-sided die?

Solution. Rolling dice with $d_1, \dots, d_m$ faces together (independently) produces a uniformly random outcome among $N = d_1 d_2 \cdots d_m$ equally likely joint results. A single deterministic function of this joint outcome can be uniform on a target set of size s if and only if $s \mid N$ (the s target values must partition the N equally-likely joint outcomes into s groups of equal size N/s; conversely if $s \mid N$, any grouping into equal-sized blocks works).

(a) Impossible. The available face counts are $4 = 2^2$, $6 = 2 \cdot 3$, $8 = 2^3$, $12 = 2^2 \cdot 3$, $20 = 2^2 \cdot 5$ — none divisible by the prime 7. Any product of these numbers (in any combination, with repetition) is therefore also never divisible by 7 (a prime divides a product only if it divides one of the factors). Since $7 \nmid N$ for every possible choice of dice, no deterministic combination can be uniform on 7 outcomes:

$$\boxed{\text{no — a fair 7-sided die cannot be simulated this way}}.$$

(b) Possible. Roll one 6-sided die and one 20-sided die together: $N = 6 \times 20 = 120 = 15 \times 8$, so $15 \mid N$. Group the 120 equally-likely joint outcomes into 15 blocks of 8 (in any fixed way, e.g. label the outcomes $0, \dots, 119$ and take the result mod 15). So

$$\boxed{\text{yes — a } d_6 \text{ and a } d_{20} \text{ rolled together simulate it}}.$$

□

Example 3.7 (Cutting a Square into n Squares for Any $n > 5$)**.** Prove that for every integer $n > 5$, a square sheet of paper can be cut into exactly n square pieces (of possibly different sizes).

Solution. **The key move (a "+3" step).** Given any dissection of the square into some number k of square pieces, pick any one piece and cut it with two perpendicular lines through its center, parallel to its sides, splitting it into 4 equal smaller squares. This replaces 1 piece by 4 pieces, so the total piece count goes from k to $k + 3$. Hence: *if k squares is achievable, so is $k + 3$.*

Three base cases (one for each residue mod 3**).**

- $n = 6$: Divide the square into a 3×3 grid of 9 equal unit squares. Merge one 2×2 block of 4 adjacent unit squares back into a single square of twice the side length (this is legal — four unit squares arranged in a 2×2 pattern exactly tile a square of double side). This leaves $9 - 4 = 5$ unit squares plus 1 merged square: 6 squares total.

- $n = 7$: Divide the square into a 2×2 grid of 4 equal squares, then apply the +3 move once to any one of them (splitting it into 4 smaller squares): $4 - 1 + 4 = 7$ squares total.

- $n = 8$: Divide the square into a 4×4 grid of 16 equal unit squares. Merge one 3×3 block of 9 adjacent unit squares (placed in a corner, say) into a single square of triple side length. This leaves $16 - 9 = 7$ unit squares plus 1 merged square: 8 squares total.

Induction. Every integer $n \geq 6$ can be written as $6 + 3k$, $7 + 3k$, or $8 + 3k$ for some integer $k \geq 0$, according to its residue mod 3. Starting from the corresponding base case (6, 7, or 8 squares) and applying the +3 move k times produces a valid dissection into exactly n squares. Since the base cases and the induction step are both constructive, this proves the claim for all $n > 5$. $\square$

Example 3.8 (Counting Sub-Rectangles of a Checkerboard with At Most Four Black Cells)**.** An $N \times M$ chessboard is colored in the usual alternating (checkerboard) pattern. Consider all axis-aligned sub-rectangles of the board (formed by grid lines). Find the number of such sub-rectangles containing at most 4 black cells, for (a) $N = M = 8$; (b) $N = 99, M = 101$; (c) general $N, M \geq 8$.

Solution. **Step 1: black-cell count of an $a \times b$ sub-rectangle.** If a sub-rectangle has a rows and b columns, its number of black cells depends only on a, b, and — when a, b are *both odd* — on the color of its top-left cell:

- If a or b is even, every row (or column) contributes equal black/white counts up to swapping, and in fact the black-cell count is always exactly $\frac{ab}{2}$, regardless of position.
- If a and b are both odd, the rectangle's four corners share one color (say the color of the top-left cell); that color appears on $\frac{ab+1}{2}$ cells and the other on $\frac{ab-1}{2}$ cells.

So the number of black cells lies in $[\lfloor ab/2 \rfloor, \lceil ab/2 \rceil]$, and both endpoints occur (the upper one exactly when a, b are both odd *and* the top-left corner is black).

Step 2: which shapes $a \times b$ can ever have ≤ 4 black cells?

- If $ab \leq 8$: then $\lceil ab/2 \rceil \leq 4$, so *every* placement of an $a \times b$ rectangle has ≤ 4 black cells — the shape always qualifies, independent of position.
- If $ab = 9$ (so $(a, b) \in \{(1, 9), (9, 1), (3, 3)\}$): black count is 4 or 5 depending on the corner color; only the placements with 4 black cells (top-left corner *white*) qualify — exactly half the placements of that shape.
- If $ab \geq 10$: then $\lfloor ab/2 \rfloor \geq 5$, so *no* placement qualifies.

Step 3: counting placements. An $a \times b$ rectangle has $(N - a + 1)(M - b + 1)$ possible positions on the board. For a fixed shape, write $R = N - a + 1$, $C = M - b + 1$ for the number of row- and column-positions of its top-left corner. Since the board is checkerboard-colored, the color of the top-left corner cycles with the parity of (row index + column index) of the position; a direct count shows that among the $R \times C$ grid of positions, exactly

$$W(R, C) = \left\lceil \frac{R}{2} \right\rceil \left\lceil \frac{C}{2} \right\rceil + \left\lfloor \frac{R}{2} \right\rfloor \left\lfloor \frac{C}{2} \right\rfloor$$

of them place the corner on one fixed color (the count for the other color is $RC - W(R, C)$).

Step 4: the shapes with $ab \leq 8$. There are exactly 20 pairs (a, b) with $a, b \geq 1$ and $ab \leq 8$:

$$(1, 1), \ldots, (1, 8),\ (2, 1), \ldots, (2, 4),\ (3, 1), (3, 2),\ (4, 1), (4, 2),\ (5, 1), (6, 1), (7, 1), (8, 1).$$

Every placement of every such shape qualifies, contributing

$$S_1(N, M) = \sum_{ab \leq 8} (N - a + 1)(M - b + 1).$$

Carrying out this finite sum (and simplifying, valid once $N, M \geq 8$ so all 20 shapes fit) gives the closed form

$$S_1(N, M) = 20(N + 1)(M + 1) - 56(N + M + 2) + 103.$$

Step 5: the shapes with $ab = 9$. Each of $(a,b) = (1,9),(9,1),(3,3)$ contributes only its "white-corner" placements:

$$S_2(N,M) = W(N,\, M-8) + W(N-8,\, M) + W(N-2,\, M-2),$$

where a term is included only when the corresponding shape fits the board (e.g. the first term requires $M \geq 9$).

Total: the answer is $S_1(N,M) + S_2(N,M)$.

(a) $N = M = 8$**.** Here only $(3,3)$ fits among the $ab = 9$ shapes (since $M = 8 < 9$ rules out $(1,9)$ and $N = 8 < 9$ rules out $(9,1)$). We get $S_1(8,8) = 20 \cdot 9 \cdot 9 - 56 \cdot 18 + 103 = 1620 - 1008 + 103 = 715$, and $S_2(8,8) = W(6,6) = 3 \cdot 3 + 3 \cdot 3 = 18$. Total:

$$715 + 18 = \boxed{733}.$$

(This matches a direct brute-force count over all sub-rectangles of the 8×8 board.)

(b) $N = 99, M = 101$**.** $S_1(99,101) = 20 \cdot 100 \cdot 102 - 56 \cdot 202 + 103 = 204000 - 11312 + 103 = 192791$. For S_2: $W(99,93) = 50 \cdot 47 + 49 \cdot 46 = 2350 + 2254 = 4604$; $W(91,101) = 46 \cdot 51 + 45 \cdot 50 = 2346 + 2250 = 4596$; $W(97,99) = 49 \cdot 50 + 48 \cdot 49 = 2450 + 2352 = 4802$. Sum: $4604 + 4596 + 4802 = 14002$. Total:

$$192791 + 14002 = \boxed{206793}.$$

(c) General $N, M \geq 8$**.** The answer is

$$20(N+1)(M+1) - 56(N+M+2) + 103 + W(N,M-8) + W(N-8,M) + W(N-2,M-2)$$

(dropping any W-term whose shape does not fit the board), with $W(R,C) = \lceil R/2 \rceil \lceil C/2 \rceil + \lfloor R/2 \rfloor \lfloor C/2 \rfloor$. □

Example 3.9 (Two players can collude to make the third lose a stone-removal game)**.** Three players take turns (in a fixed cyclic order) removing between 1 and m stones ($m > 1$) from a pile of known size; the player who removes the last stone loses. Prove that if the pile is initially large enough, any two of the players can agree in advance on a joint strategy that always forces the third player to lose.

Solution. Call the players A, B, C (turn order $A, B, C, A, B, C, \ldots$), with A and C the colluders trying to force B to take the last stone.

Reduction to a 2-player game. Since B never moves between C's move and the following A move, that consecutive pair of colluder moves acts as one flexible combined move: if C takes $c \in [1,m]$ and A then takes $a \in [1,m]$, the pair removes any total $s = c + a$, and every integer $s \in [2, 2m]$ is achievable this way (e.g. $c = 1, a = s-1$ when $s \leq m+1$; $c = s-m, a = m$ when $s > m+1$). So the whole game is equivalent to a standard 2-player subtraction game between $X = B$ (who removes $x \in [1,m]$ each turn) and $Y =$ the coalition (who removes $y \in [2, 2m]$ each turn, capped by the remaining pile), alternating turns, last mover loses.

Base facts. With 1 stone left, whoever must move is forced to take it and lose. With 2 stones left and Y to move: C must take 1 or 2; taking 2 makes C lose immediately, while taking 1 leaves 1 stone for A, who is then forced to lose. Either way *some* colluder loses — so a pile of 2 facing Y is a forced loss for the coalition; a pile of 1 is a forced loss for whoever moves.

Claim (by strong induction on the pile size n).

- X (to move) wins for $2 \leq n \leq m+2$, and loses for $n = 1$ or $n \geq m+3$.
- Y (to move) wins for all $n \geq 3$ (with the pile large enough that the pair-move range $[2, 2m]$ isn't otherwise restricted), and loses for $n \in \{1, 2\}$.

Why X wins on $[2, m+2]$: for $2 \le n \le m+1$, X takes $n-1 \le m$ stones, leaving 1 for Y (a forced Y-loss). For $n = m+2$, X takes m stones, leaving 2 for Y (also a forced Y-loss, by the base fact above). *Why X loses for* $n \ge m+3$: any move $x \in [1, m]$ leaves $n - x \ge 3$ for Y, and by the induction hypothesis Y wins from every pile ≥ 3 — so no move helps X. *Why Y wins for* $n \ge 3$: if $n \le 2m+1$, Y takes $s = n-1 \in [2, 2m]$, leaving exactly 1 for X (forced loss). If $n \ge 2m+2$ (so in particular $n \ge m+5$ once $m \ge 3$; the case $m = 2$ has one isolated exception at $n = 6$, checked directly), Y takes the minimum $s = 2$, leaving $n - 2 \ge m+3$, which by the induction hypothesis is again an X-losing pile.

Conclusion. Since the real game begins with A's move — i.e. Y moving first — the coalition wins whenever the initial pile N satisfies $N \ge 3$ (in fact for every N except the single small case $N = 1, 2$, and, only when $m = 2$, $N = 6$). In particular, for every sufficiently large initial pile,

$$\boxed{A \text{ and } C \text{ can jointly force } B \text{ to take the last stone}}.$$

(This full recursive game value was verified by exhaustive computation for $m = 2, 3, 4, 5, 6$ and piles up to several hundred, confirming the coalition wins for all $n \ge 3$ apart from the noted small exceptions.) ☐

Example 3.10 (0/1 matrices whose square is the all-ones matrix)**.** For which natural numbers N does there exist an $N \times N$ matrix A with entries in $\{0, 1\}$ such that $A^2 = J$, the all-ones matrix?

Solution. **Answer: exactly the perfect squares,** $N = k^2$ **for a positive integer** k**.**

Necessity. Suppose $A^2 = J$. Taking traces, $\mathrm{tr}(A^2) = \mathrm{tr}(J) = N$ (every diagonal entry of J is 1). For any complex matrix, the eigenvalues of A^2 (with algebraic multiplicity, via the Jordan form of A) are exactly the squares of the eigenvalues of A; since J's eigenvalues are N (multiplicity 1, eigenvector $\mathbf{1}$) and 0 (multiplicity $N-1$), every eigenvalue λ of A satisfies $\lambda^2 \in \{0, N\}$, i.e. $\lambda \in \{0, \sqrt{N}, -\sqrt{N}\}$ (necessarily real, since λ^2 is a non-negative real number). Let p, q be the algebraic multiplicities of $\sqrt{N}, -\sqrt{N}$ respectively. Then

$$N = \mathrm{tr}(A^2) = p \cdot N + q \cdot N + 0 \quad \Longrightarrow \quad p + q = 1,$$

so $(p, q) = (1, 0)$ or $(0, 1)$. Also $\mathrm{tr}(A) = p\sqrt{N} - q\sqrt{N}$ is a non-negative integer (a sum of 0/1 diagonal entries). If $(p, q) = (0, 1)$, $\mathrm{tr}(A) = -\sqrt{N} < 0$ for $N \ge 1$ — impossible. So $(p, q) = (1, 0)$, giving $\mathrm{tr}(A) = \sqrt{N}$, a non-negative integer — hence N is a perfect square.

Sufficiency. For $N = k^2$, index rows and columns by $0, \dots, N-1$ and write each index in base k via $i = ak + r$, $j = ck + s$ $(0 \le a, r, c, s < k)$. Define

$$A_{ij} = 1 \iff a = s \qquad (\text{i.e. } \lfloor i/k \rfloor = j \bmod k).$$

For fixed i, j (giving fixed $a = \lfloor i/k \rfloor$ and $s = j \bmod k$), the sum $(A^2)_{ij} = \sum_{k'} A_{ik'} A_{k'j}$ has a nonzero term exactly when $k' = a'k + r'$ satisfies $r' = a$ (from $A_{ik'} = 1$) and $a' = s$ (from $A_{k'j} = 1$); this pins down $k' = sk + a$ *uniquely*. So exactly one term of the sum is 1 and the rest are 0, giving $(A^2)_{ij} = 1$ for every i, j, i.e. $A^2 = J$.

$$\boxed{\text{such an } A \text{ exists} \iff N \text{ is a perfect square}}$$

(Both directions were checked computationally: exhaustive search confirms no solution for $N = 2, 3, 5, 6, 7, 8$, and the explicit construction above was verified to give $A^2 = J$ for $N = 1, 4, 9, 16, 25$.) ☐

Example 3.11 (Fewest distinct pairwise midpoints of an n-point real set)**.** Let $M \subseteq \mathbb{R}$ be a set of n real numbers $(n \ge 2)$, and let $S_M = \left\{\frac{x+y}{2} : x, y \in M,\ x \ne y\right\}$. Find the smallest possible value of $|S_M|$.

Solution. **Lower bound.** Write $M = \{x_1 < x_2 < \cdots < x_n\}$. Consider the chain of midpoints

$$\frac{x_1 + x_2}{2} < \frac{x_1 + x_3}{2} < \cdots < \frac{x_1 + x_n}{2} < \frac{x_2 + x_n}{2} < \frac{x_3 + x_n}{2} < \cdots < \frac{x_{n-1} + x_n}{2}.$$

Each successive inequality is immediate: increasing the second index while x_1 is fixed increases the sum, and (at the switch-over point) $\frac{x_1+x_n}{2} < \frac{x_2+x_n}{2}$ since $x_1 < x_2$, after which increasing the first index while x_n is fixed keeps increasing the sum. This is a strictly increasing list, so its $(n-1) + (n-2) = 2n-3$ entries are pairwise distinct elements of S_M, giving $|S_M| \geq 2n-3$.

Sharpness. Take $M = \{1, 2, \ldots, n\}$ (an arithmetic progression). For $i < j$ in $\{1, \ldots, n\}$, the midpoint $\frac{i+j}{2}$ depends only on the sum $i+j$, and $i+j$ ranges exactly over the integers $3, 4, \ldots, 2n-1$ as i, j range over all valid pairs (every such sum is attainable: e.g. for $3 \leq s \leq n+1$ take $(i, j) = (1, s-1)$ suitably adjusted, and for $n+1 \leq s \leq 2n-1$ take $(i, j) = (s-n, n)$ suitably adjusted — in all cases some pair with $1 \leq i < j \leq n$ and $i + j = s$ exists). That is $2n - 1 - 3 + 1 = 2n - 3$ distinct sums, hence exactly $2n-3$ distinct midpoints.

Combining both parts,

$$\boxed{\min |S_M| = 2n-3}.$$

□

3.2 Permutations, Combinations, and Hypergeometric Counting

Definitions and formulas.

- Ordered selections without repetition:

$$A_n^k = \frac{n!}{(n-k)!}.$$

- Unordered selections without repetition:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

- Permutations with repeated letters are counted by multinomial coefficients:

$$\frac{n!}{n_1! \cdots n_r!}.$$

- Newton's binomial theorem:

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

- In the hypergeometric model, if a population has N items, K of them are marked, and we draw n items without replacement, then

$$\mathbb{P}(X = r) = \frac{\binom{K}{r}\binom{N-K}{n-r}}{\binom{N}{n}}.$$

Example 3.12 (Exactly two aces)**.** What is the probability that a 5-card poker hand contains exactly two aces?

Solution. There are $\binom{52}{5}$ equally likely hands. To get exactly two aces, choose 2 of the 4 aces and 3 of the 48 non-aces:

$$\mathbb{P}(\text{exactly two aces}) = \frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}}.$$

That is the final answer. Numerically it is approximately 0.03993. □

Example 3.13 (Expected number of fixed points)**.** If π is a uniformly random permutation of $\{1, 2, \ldots, n\}$, what is the expected number of indices i such that $\pi(i) = i$?

Solution. Let

$$X = \#\{i : \pi(i) = i\}$$

be the number of fixed points. For each i, define the indicator

$$I_i = \begin{cases} 1, & \text{if } \pi(i) = i, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = I_1 + \cdots + I_n.$$

For a uniformly random permutation, every value is equally likely to appear in position i, so

$$\mathbb{P}(\pi(i) = i) = \frac{1}{n}.$$

Hence

$$\mathbb{E}[I_i] = \frac{1}{n}.$$

By linearity of expectation,

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[I_i] = n \cdot \frac{1}{n} = 1.$$

Therefore the expected number of fixed points is

$$\boxed{1}.$$

□

Example 3.14 (Filling a $2 \times n$ table with no odd number under an even one)**.** In how many ways can n distinct even numbers and n distinct odd numbers be written into a $2 \times n$ table so that no cell holding an odd number is directly below a cell holding an even number? (An answer with at most one summation sign is expected.)

Solution. Read the constraint column by column: if the top of a column is even, the bottom cannot be odd, so it must be even too. So each column is one of three types: **EE** (both even), **OO** (both odd), or **OE** (top odd, bottom even) — the fourth combination, EO, is exactly what's forbidden.

Say k columns are type EE. Counting the n even numbers used: $2k$ in the EE columns plus one per OE column go to evens, and counting the n odd numbers: one per OE column's top plus $2 \times$ (#OO columns) — matching totals forces #OO $= k$ as well (the two counting equations both reduce to this). So there are k EE columns, k OO columns, and $n - 2k$ OE columns, for some $k = 0, 1, \ldots, \lfloor n/2 \rfloor$.

For fixed k: choose which k of the n column-positions are EE, which k are OO, and which $n - 2k$ are OE — a multinomial count of $\dfrac{n!}{k!\,k!\,(n-2k)!}$ ways. Independently, the n distinct even numbers must fill the n "even-designated" cells ($2k$ from EE columns, $n-2k$ bottoms of OE columns) — a bijection, $n!$ ways — and likewise the n odd numbers fill the n "odd-designated" cells in $n!$ ways.

Summing over k:

$$\boxed{(n!)^2 \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{k!\,k!\,(n-2k)!}}.$$

(Check: $n=1$ gives $(1!)^2 \cdot \frac{1!}{0!0!1!} = 1$, matching the only valid arrangement — odd on top, even on bottom. $n=2$ gives $(2!)^2 \left[\frac{2!}{0!0!2!} + \frac{2!}{1!1!0!}\right] = 4(1+2) = 12$, confirmed by direct enumeration.) ☐

Example 3.15 (The combinatorial number system)**.** Prove that for every non-negative integer n, there exist integers $0 \le x < y < z$ with $n = \binom{x}{1} + \binom{y}{2} + \binom{z}{3}$.

Solution. This is a greedy construction, using repeatedly the fact that $\binom{k}{r}$ is strictly increasing in k and the Pascal identity $\binom{k+1}{r} - \binom{k}{r} = \binom{k}{r-1}$.

Choosing z. If $n=0$, take $(x,y,z) = (0,1,2)$ directly (all three binomial coefficients vanish). Otherwise, let z be the *largest* integer with $\binom{z}{3} \le n$ (such z exists and is ≥ 3, since $\binom{3}{3} = 1 \le n$). Set $n_1 = n - \binom{z}{3} \ge 0$. By maximality of z, $\binom{z+1}{3} > n$, i.e.

$$n_1 = n - \binom{z}{3} < \binom{z+1}{3} - \binom{z}{3} = \binom{z}{2}.$$

Choosing y. Let y be the largest integer with $\binom{y}{2} \le n_1$. Since $n_1 < \binom{z}{2}$, this forces $y < z$ (if y were $\ge z$, we'd have $\binom{y}{2} \ge \binom{z}{2} > n_1$, contradicting $\binom{y}{2} \le n_1$). Set $n_2 = n_1 - \binom{y}{2} \ge 0$; by maximality of y, $\binom{y+1}{2} > n_1$, giving

$$n_2 = n_1 - \binom{y}{2} < \binom{y+1}{2} - \binom{y}{2} = \binom{y}{1} = y.$$

Choosing x. Set $x = n_2$. Then $0 \le x < y$ (just shown), and

$$\binom{x}{1} + \binom{y}{2} + \binom{z}{3} = n_2 + \binom{y}{2} + \binom{z}{3} = n_1 + \binom{z}{3} = n.$$

So (x,y,z) with $0 \le x < y < z$ satisfies $\binom{x}{1} + \binom{y}{2} + \binom{z}{3} = n$, as required

$$\boxed{\text{such } x,y,z \text{ always exist (constructed greedily, from the top binomial coefficient down)}}$$

(verified by direct computation of this construction for every $n = 0, \dots, 29$). ☐

3.3 Graphs, Trees, Planarity, and Tournaments

Definitions and formulas.

- Handshake lemma:
$$\sum_{v \in V} \deg(v) = 2|E|.$$
  In particular, the number of odd-degree vertices is even.
- A tree on n vertices has exactly $n-1$ edges. A connected graph with $n-1$ edges is a tree.
- A graph has an Eulerian circuit if and only if every vertex has even degree and the non-isolated part is connected.

- Euler's formula for connected planar graphs is

$$V - E + F = 2.$$

- Every tournament has a Hamiltonian path.

Example 3.16 (Parity-constrained graphs on 100 vertices)**.** How many simple graphs on 100 labeled vertices have an even number of edges and all degrees odd?

Solution. Let $m = \binom{100}{2} = 4950$ be the number of possible edges, and identify a graph with its edge-indicator vector $x \in \mathbb{F}_2^m$.

Define the incidence map over $\mathbb{F}_2$,

$$M : \mathbb{F}_2^m \to \mathbb{F}_2^{100},$$

where Mx records the parity of the degrees. For the complete graph K_{100}, the incidence matrix has rank 99 over $\mathbb{F}_2$, because the graph is connected and the image is exactly the set of parity vectors with even total sum.

The requirement "all degrees odd" means

$$Mx = (1, 1, \dots, 1).$$

This right-hand side is feasible because the sum of its coordinates is $100 \equiv 0 \pmod 2$. Therefore the set of solutions is an affine space of dimension

$$4950 - 99 = 4851.$$

So there are 2^{4851} graphs with all degrees odd.

Now impose the additional condition that the total number of edges is even. This is one more linear equation over $\mathbb{F}_2$:

$$\sum_e x_e = 0.$$

This equation is independent of the degree-parity equations: if it were a linear combination of them, there would exist bits $y_1, \dots, y_{100}$ such that $y_i + y_j = 1$ for every pair $i \neq j$, which is impossible for $100 \geq 3$.

Therefore the extra parity condition cuts the affine space in half. The final count is

$$\boxed{2^{4850}}.$$

□

Example 3.17 (More than n^2 edges on $2n$ vertices force a diamond)**.** Show that every graph on $2n$ vertices with $n^2 + 1$ edges contains a diamond, that is, a copy of K_4 with one edge removed.

Solution. We prove the stronger statement that every diamond-free graph on N vertices has at most

$$\left\lfloor \frac{N^2}{4} \right\rfloor$$

edges. Taking $N = 2n$ will then give the desired conclusion.

Proceed by induction on N. The claim is trivial for $N \leq 3$.

Now let G be a diamond-free graph on $N \geq 4$ vertices.

If G is triangle-free, then Mantel's theorem applies directly and gives

$$e(G) \leq \left\lfloor \frac{N^2}{4} \right\rfloor.$$

Mantel's theorem used here. Mantel's theorem states that every triangle-free graph on N vertices has at most

$$\left\lfloor \frac{N^2}{4} \right\rfloor$$

edges, with equality for the complete bipartite graph $K_{\lfloor N/2\rfloor,\lceil N/2\rceil}$. One short proof is this: if uv is an edge in a triangle-free graph, then u and v have no common neighbor, so

$$\deg(u) + \deg(v) \le N.$$

Summing over all edges gives

$$\sum_v \deg(v)^2 = \sum_{uv\in E} (\deg(u) + \deg(v)) \le Ne(G).$$

Now apply Cauchy to the degree sequence:

$$(2e(G))^2 = \left(\sum_v \deg(v)\right)^2 \le N \sum_v \deg(v)^2 \le N^2 e(G).$$

Cancelling $e(G)$ yields $e(G) \le N^2/4$, hence $e(G) \le \lfloor N^2/4 \rfloor$.

So the only remaining case is that G contains a triangle T. Write $T = \{a, b, c\}$. Since G is diamond-free, no vertex outside T can be adjacent to two vertices of T: if some outside vertex v were adjacent to, say, a and b, then the four vertices a, b, c, v would contain two triangles sharing the edge ab, which is exactly a diamond.

Therefore every vertex outside T sends at most one edge into T. As there are $N - 3$ outside vertices, the number of edges joining T to $V(G) \setminus T$ is at most $N - 3$. Adding the 3 edges of the triangle itself, we find that the total number of edges incident with T is at most

$$(N - 3) + 3 = N.$$

Hence

$$e(G) \le e(G - T) + N.$$

Now $G - T$ has $N - 3$ vertices and is still diamond-free, so by the induction hypothesis,

$$e(G - T) \le \left\lfloor \frac{(N-3)^2}{4} \right\rfloor.$$

Thus

$$e(G) \le \left\lfloor \frac{(N-3)^2}{4} \right\rfloor + N.$$

It remains to compare this with $\lfloor N^2/4 \rfloor$. If $N = 2m$, then

$$\left\lfloor \frac{(N-3)^2}{4} \right\rfloor + N = \left\lfloor \frac{(2m-3)^2}{4} \right\rfloor + 2m = m^2 - m + 2 \le m^2 = \left\lfloor \frac{N^2}{4} \right\rfloor$$

for $m \ge 2$. If $N = 2m + 1$, then

$$\left\lfloor \frac{(N-3)^2}{4} \right\rfloor + N = \frac{(2m-2)^2}{4} + (2m + 1) = m^2 + 2 \le m^2 + m = \left\lfloor \frac{N^2}{4} \right\rfloor$$

for $m \ge 2$. So in all cases,

$$e(G) \le \left\lfloor \frac{N^2}{4} \right\rfloor.$$

Now specialize to $N = 2n$. A diamond-free graph on $2n$ vertices has at most

$$\left\lfloor \frac{(2n)^2}{4} \right\rfloor = n^2$$

edges. Therefore any graph on $2n$ vertices with $n^2 + 1$ edges cannot be diamond-free. Hence it must contain a diamond.

So the answer is

$$\boxed{\text{every graph on } 2n \text{ vertices with } n^2 + 1 \text{ edges contains a diamond.}}$$

□

Example 3.18 (Hamiltonian path in a tournament)**.** In a tournament on n vertices, prove that there exists a directed path that visits every vertex exactly once.

Solution. Use induction on n. The claim is obvious for $n = 1$.

Assume it holds for tournaments on $n - 1$ vertices, and consider a tournament on vertices $v_1, \dots, v_n$. Remove one vertex, say v_n. By the induction hypothesis, the remaining tournament has a Hamiltonian path

$$u_1 \to u_2 \to \cdots \to u_{n-1}.$$

Now insert v_n into this path. If $v_n \to u_1$, place v_n at the beginning. If $u_{n-1} \to v_n$, place v_n at the end. Otherwise there exists an index k such that

$$u_k \to v_n \to u_{k+1}.$$

Indeed, scan the path from left to right until the direction of the edge involving v_n changes. Inserting v_n between u_k and u_{k+1} produces a Hamiltonian path through all n vertices.

Thus every tournament has a Hamiltonian path. □

Example 3.19 (Roads and cities as a finite incidence structure)**.** Suppose there are n roads and finitely many cities such that

- any two distinct cities lie on a road;
- any two distinct roads meet in exactly one city.

How many cities can there be?

Solution. Let the number of cities be m.

First note that any two cities actually lie on a *unique* road. Indeed, if two different roads both passed through the same two cities, then those two roads would meet in at least two cities, contradicting the assumption that any two roads meet in exactly one city.

So we have a finite incidence structure in which any two cities determine a unique road. A classical theorem of de Bruijn and Erdős says that in any finite linear space, the number of lines is at least the number of points. Applying it here gives

$$n \geq m.$$

Sketch of the de Bruijn–Erdős theorem used here. Write P for the set of points and L for the set of lines in a finite linear space, and let

$$v = |P|, \qquad b = |L|.$$

For a point p, let $d(p)$ be the number of lines through p, and for a line ℓ, let $s(\ell)$ be the number of points on ℓ.

The key inequality is that if $p \notin \ell$, then

$$s(\ell) \le d(p),$$

because each point of ℓ together with p determines a distinct line through p.

Now assume $b \le v$. A standard Hall-marriage argument, applied to the family of complements $P \setminus \ell$, gives distinct points $p_\ell \notin \ell$ for each line $\ell \in L$. Therefore

$$s(\ell) \le d(p_\ell) \qquad \text{for every } \ell \in L.$$

Hall's theorem used here. Hall's marriage theorem says that a finite family of sets $S_1, \dots, S_m$ has a system of distinct representatives if and only if for every index set $I \subseteq \{1, \dots, m\}$,

$$\left| \bigcup_{i \in I} S_i \right| \ge |I|.$$

Here we apply it to the sets $S_\ell = P \setminus \ell$. Why does Hall's condition hold? In a nontrivial linear space, every line misses at least one point, so the case $|I| = 1$ is clear. If $2 \le |I| < b$, then the corresponding distinct lines have common intersection of size at most 1, hence

$$\left| \bigcup_{\ell \in I} (P \setminus \ell) \right| = \left| P \setminus \bigcap_{\ell \in I} \ell \right| \ge v - 1 \ge b - 1 \ge |I|.$$

If $|I| = b$, then the union is all of P, so it has size $v \ge b$. Thus Hall's theorem produces distinct points $p_\ell \notin \ell$ as claimed.

Summing over all lines yields

$$\sum_{\ell \in L} s(\ell) \le \sum_{\ell \in L} d(p_\ell) \le \sum_{p \in P} d(p).$$

But both outer sums count the same thing, namely the total number of incidences point–line, so

$$\sum_{\ell \in L} s(\ell) = \sum_{p \in P} d(p).$$

Hence every inequality above is in fact an equality, which forces $b = v$ whenever $b \le v$. Therefore in all cases

$$b \ge v.$$

That is the de Bruijn–Erdős theorem in the form used here.

Now use duality: interchange the roles of cities and roads. The second assumption says that any two roads meet in exactly one city, so in the dual picture any two "points" determine a unique "line". Applying the same theorem to the dual incidence structure gives

$$m \ge n.$$

Combining the two inequalities yields

$$\boxed{m = n}.$$

So the number of cities must be exactly the number of roads. □

Example 3.20 (Expected Number of Cliques)**. Problem:** There are n students seeing each other for the first time. An amusement park offers a discount if a group of exactly k friends comes. Any two random students can become friends with probability p. On average, how many such valid groups of exactly k friends can be formed?

Solution: Let the n students be the vertices of a random graph $G(n, p)$, where each edge exists independently with probability p. A valid group of exactly k friends corresponds to a k-clique (a complete subgraph K_k) in the graph.

Let S be a specific subset of k students. We define an indicator random variable I_S such that $I_S = 1$ if S forms a k-clique, and 0 otherwise. A subset of size k contains exactly $\binom{k}{2}$ possible pairs. For all these pairs to be friends, every corresponding edge must be present. Thus,

$$P(I_S = 1) = p^{\binom{k}{2}}, \quad \text{and hence} \quad E[I_S] = p^{\binom{k}{2}}.$$

The total number of k-cliques X is the sum of I_S over all possible subsets S of size k. By linearity of expectation,

$$E[X] = E\left[\sum_{S \subset V, |S|=k} I_S\right] = \sum_{|S|=k} E[I_S] = \binom{n}{k} p^{\binom{k}{2}}.$$

Example 3.21 (Guaranteed Dominating Set via Maximal Independent Sets)**. Problem:** There are 100 people in a company. It is known that among any 10 people, there are three pairwise acquaintances. What is the smallest n that guarantees there are n people such that any of the remaining ones is acquainted with someone in these n?

Solution: We model the people as the vertices V of a graph G, where edges represent acquaintances. The condition states that any subset of 10 vertices contains a triangle (a K_3).

We are looking for the smallest n that guarantees the existence of a dominating set of size at most n. (A set $D \subseteq V$ is dominating if every vertex in $V \setminus D$ has at least one neighbor in D).

We first claim that the independence number satisfies $\alpha(G) \leq 8$. Indeed, suppose there were 9 pairwise nonadjacent vertices $v_1, \ldots, v_9$. Take any tenth vertex w. Then the induced subgraph on

$$\{v_1, \ldots, v_9, w\}$$

contains no triangle: a triangle cannot lie entirely inside $\{v_1, \ldots, v_9\}$ because that set is independent, and any triangle using w would require an edge between two of the vertices v_i, which also does not exist. This contradicts the hypothesis that every 10-vertex subset contains a triangle. Hence

$$\alpha(G) \leq 8.$$

Now, consider any *maximal* independent set I in G. By maximality, we cannot add any other vertex from $V \setminus I$ to I without creating an edge. This implies that every vertex in $V \setminus I$ must be adjacent to at least one vertex in I. Consequently, every maximal independent set is also a dominating set.

Since every independent set has size at most 8, a maximal independent set I has size $|I| \leq 8$. So G always has a dominating set of size at most 8.

This bound is sharp. Consider a graph consisting of one clique on 93 vertices together with 7 isolated vertices. Any set of 10 vertices must contain at least 3 vertices from the 93-clique, and those 3 vertices form a triangle. So the hypothesis is satisfied. On the other hand, every dominating set must contain all 7 isolated vertices, and it must also contain at least one vertex from the clique in order to dominate the clique vertices. Therefore every dominating set has size at least

$$7 + 1 = 8.$$

Hence the smallest value of n that is guaranteed is

$$\boxed{8}.$$

Example 3.22 (Minimum edges forced by a local domination condition)**.** A graph G has 40 vertices, and among any 5 of them there is always one adjacent to the other 4. What is the minimum possible number of edges of G?

Solution. Pass to the complement $H = \overline{G}$; since $|E(G)| + |E(H)| = \binom{40}{2} = 780$, minimizing $|E(G)|$ is the same as maximizing $|E(H)|$. A vertex v is G-adjacent to the other four vertices of a 5-set S exactly when v has no H-neighbor inside S, i.e. v is isolated in the induced subgraph $H[S]$. So the hypothesis on G says: every 5-element $S \subseteq V(H)$ contains a vertex isolated in $H[S]$. We must maximize $|E(H)|$ subject to this.

No component of H has 5 or more vertices. If a component C had $|C| \geq 5$, a spanning tree of C has a connected sub-tree on some 5 vertices S (e.g. the first 5 vertices visited by a breadth-first search). The tree edges among S are genuine edges of H, so every vertex of S has an H-neighbor inside S: no isolated vertex, contradicting the hypothesis.

H cannot have two distinct components of sizes ≥ 2 and ≥ 3. If C_1, C_2 are distinct components with $|C_1| \geq 2$, $|C_2| \geq 3$, take an edge $\{a, b\} \subseteq C_1$ and three vertices $\{c, d, e\} \subseteq C_2$ that induce a subgraph with no isolated point (any connected graph on ≥ 3 vertices contains such a triple, e.g. a vertex together with two of its tree-neighbors). Then $S = \{a, b, c, d, e\}$ has no isolated vertex in $H[S]$, a contradiction.

So the components of H fall into one of two patterns: (A) a single component of size $s \in \{2, 3, 4\}$ plus $40 - s$ isolated vertices, contributing at most $\binom{4}{2} = 6$ edges; or (B) every component has size at most 2 (a disjoint union of edges and isolated vertices), contributing at most $\lfloor 40/2 \rfloor = 20$ edges. Since $20 > 6$, the maximum is achieved by a perfect matching on all 40 vertices, $|E(H)| = 20$.

Thus

$$|E(G)| = 780 - 20 = \boxed{760},$$

attained by $G = \overline{H} = K_{40}$ minus a perfect matching (each vertex has exactly one non-neighbor, its “partner”). Indeed, in any 5 chosen vertices, since 5 is odd the partners cannot all pair up inside the set, so some vertex's partner lies outside it — that vertex is adjacent to the other four. □

Example 3.23 (The double-sweep diameter heuristic can fail forever)**.** For a connected graph, the following classical heuristic estimates the diameter: set A_0 = a fixed start vertex; run BFS from A_0 and let A_1 be a vertex at maximum distance from A_0; run BFS from A_1 and let A_2 be a vertex at maximum distance from A_1 (breaking all ties by smallest vertex label); continue to build $A_3, A_4, \dots$, and estimate the diameter as $\max_i \operatorname{dist}(A_i, A_{i+1})$. Exhibit a connected graph on at most 10 vertices, with A_0 a fixed labeled vertex, on which this process — however long it is run — never produces the true diameter.

Solution. This heuristic (“double sweep”) is exact on trees, so any counterexample must contain a cycle. Take the 5-vertex graph consisting of the 4-cycle $1-2-3-4-1$ with a pendant vertex 5 attached to vertex 2 (edges $\{1,2\}, \{2,3\}, \{3,4\}, \{1,4\}, \{2,5\}$), and start the process at $A_0 = 1$. Direct computation gives the distance matrix

	1	2	3	4	5
1	0	1	2	1	2
2	1	0	1	2	1
3	2	1	0	1	2
4	1	2	1	0	3
5	2	1	2	3	0

so the true diameter is 3, uniquely realized by the pair $(4, 5)$.

Now run the process. From $A_0 = 1$, the farthest vertices are 3 and 5, both at distance 2; the tie-break picks the smaller label, so $A_1 = 3$. From $A_1 = 3$, the farthest vertices are 1 and 5, again both at distance 2, and the tie-break sends us back to $A_2 = 1$. The process has entered the cycle $1 \to 3 \to 1 \to 3 \to \cdots$ and will repeat it forever, with $\operatorname{dist}(1,3) = \operatorname{dist}(3,1) = 2$ at every step. Hence

$$\max_i \operatorname{dist}(A_i, A_{i+1}) = 2 < 3 = \text{true diameter},$$

for every stopping point k — the diametral pair $(4, 5)$ is never even visited as a sweep center, so

$$\boxed{\text{no choice of } k \text{ recovers the diameter on this graph}}.$$

□

Example 3.24 (A local 5-cycle condition forces a large clique)**.** A graph G has 30 vertices, and for *every* 5 of them there is a 5-cycle of G passing through exactly those 5 vertices. Prove that G contains 10 pairwise adjacent vertices.

Solution. Work with the complement $H = \overline{G}$. A 5-cycle on a given 5-set S exists in G iff the induced subgraph $G[S]$ is Hamiltonian; a graph on 5 vertices with a vertex of degree ≤ 1 can never be Hamiltonian (a Hamiltonian cycle needs every vertex to have degree exactly 2 within it). Translating to H: for every 5-set S and every $v \in S$, the G-degree of v within S is $4 - \deg_{H[S]}(v)$, so we need $\deg_{H[S]}(v) \leq 2$ for every $v \in S$ (else v's G-degree within S drops to ≤ 1, killing Hamiltonicity).

Claim: H has maximum degree ≤ 2. Suppose some vertex v has $\deg_H(v) \geq 3$; pick 3 of its H-neighbors n_1, n_2, n_3 and any other vertex $w \notin \{v, n_1, n_2, n_3\}$ (possible since there are $30 \geq 5$ vertices). In $S = \{v, n_1, n_2, n_3, w\}$, v already has H-degree ≥ 3, so its G-degree within S is ≤ 1 — $G[S]$ is not Hamiltonian, contradicting the hypothesis. So every vertex of H has degree ≤ 2.

Finishing with a greedy independent set. A graph with maximum degree ≤ 2 on 30 vertices has an independent set of size $\geq \lceil 30/3 \rceil = 10$: repeatedly pick any remaining vertex, put it in the independent set, and delete it together with its (at most 2) neighbors; each round removes at most 3 vertices, so after $\lceil 30/3 \rceil = 10$ rounds every vertex has been deleted, and the picked vertices — pairwise non-adjacent by construction — number at least 10.

An independent set of size 10 in H is, by definition, a set of 10 vertices with no H-edges among them, i.e.

$$\boxed{10 \text{ pairwise } G\text{-adjacent vertices exist}}$$

(the bound is tight: 10 disjoint triangles in H, i.e. $G =$ the complement of 10 disjoint K_3's, realizes independence number exactly 10). □

Example 3.25 (A high minimum degree forces a large clique)**.** In a country of 100 cities, some pairs are connected by a direct flight. Every city has more than 90 direct flights. Prove that some 11 cities are pairwise connected by direct flights.

Solution. Let H be the complement graph (pairs *without* a direct flight). Each city has 99 other cities to potentially connect to; since its flight-degree exceeds 90 (so is at least 91), its H-degree is at most $99 - 91 = 8$.

A graph with maximum degree ≤ 8 on 100 vertices has an independent set of size at least $\lceil 100/9 \rceil = 12$: greedily pick any remaining vertex, add it to the independent set, and remove it together with its at most 8 neighbors (at most 9 vertices per round); after 12 rounds at most 108 vertex-removals have been used, more than enough to exhaust all 100 vertices, and the 12 chosen vertices are pairwise non-adjacent in H by construction.

An independent set in H of size $12 \geq 11$ is a set of (at least) 11 cities with no missing flight between any pair, i.e.

$$\boxed{11 \text{ pairwise connected cities exist}}$$

(indeed the argument gives 12, comfortably more than the 11 requested). □

Example 3.26 (A friendship-pattern graph is regular)**.** In a social network, any two friends have no common friends, and any two non-friends have exactly two common friends. Prove that every user has the same number of friends.

Solution. **Step 1: friends have equal degree.** Let a, b be friends. Since friends share no common friends, $N(a) \cap N(b) = \emptyset$ (neighborhoods excluding a, b themselves), so $N(a) \setminus \{b\}$ and $N(b) \setminus \{a\}$ are disjoint sets. For $x \in N(a) \setminus \{b\}$: is x adjacent to b? If it were, x would be a common friend of a, b — impossible, since a, b are friends. So x, b are non-friends, hence have exactly two common friends; one of them is a (since $x \sim a, a \sim b$), so there is exactly one *other* common friend of x, b, call it $\varphi(x) \neq a$. Since $\varphi(x) \sim b$ and $\varphi(x) \neq a$, $\varphi(x) \in N(b) \setminus \{a\}$.

φ is injective: if $\varphi(x_1) = \varphi(x_2) = y$ for distinct $x_1, x_2 \in N(a) \setminus \{b\}$, then $y \sim x_1, y \sim x_2, y \sim b$. If $y \sim a$ too, then $y \in N(a) \cap N(x_1)$ — impossible since a, x_1 are friends (no common friends). So $y \nsim a$, meaning a, y are non-friends; but then x_1, x_2, b are *three* distinct common friends of a, y (each adjacent to both a and y), contradicting "exactly two common friends." So φ is injective, giving $|N(a) \setminus \{b\}| \leq |N(b) \setminus \{a\}|$; the symmetric argument (swap a, b) gives the reverse inequality, so $\deg(a) = \deg(b)$.

Step 2: everyone has equal degree. For non-friends a, b, they have a common friend c (in fact two). Since a, c are friends and b, c are friends, Step 1 gives $\deg(a) = \deg(c) = \deg(b)$. Combined with Step 1 itself (friends have equal degree), *every* pair of users — friends or not — has equal degree, so

$$\boxed{\text{all users have the same number of friends}}.$$

□

Example 3.27 (Some pair agrees with a majority of the office)**.** A company has 100 employees, some pairs of whom are acquainted. Prove there is a pair of employees such that at least 50 of the other 98 employees are each either acquainted with both people in the pair, or acquainted with neither.

Solution. For a pair $\{a, b\}$, call a third employee c *consistent* with $\{a, b\}$ if c's acquaintance with a matches c's acquaintance with b (both or neither), and write $\text{agree}(a, b)$ for the number of consistent c's among the other 98. The claim is that $\max_{\{a,b\}} \text{agree}(a, b) \geq 50$.

Reformulation via a signed matrix. Encode acquaintance as a symmetric 100×100 matrix M, zero diagonal, $M_{c,a} = +1$ if c, a acquainted, -1 otherwise. For $a \neq b$, since $M_{a,a} = M_{b,b} = 0$,

$$(M^2)_{a,b} = \sum_c M_{a,c} M_{c,b} = \sum_{c \neq a,b} M_{c,a} M_{c,b},$$

and each summand is $+1$ if c is consistent with $\{a, b\}$ and -1 otherwise, so $(M^2)_{a,b} = 2\,\text{agree}(a, b) - 98$. So the claim is exactly that some off-diagonal entry of M^2 is ≥ 2; since agree is an integer, $(M^2)_{a,b}$ is always even, so this is the same as showing some off-diagonal entry is strictly positive.

An averaging bound, and where it falls short. Summing $(M^2)_{a,b} = 2\,\text{agree}(a, b) - 98$ over all $\binom{100}{2} = 4950$ pairs, the claim would follow immediately from $\sum_{\{a,b\}} \text{agree}(a, b) > 4950 \times 49$. Writing d_c for employee c's number of acquaintances, a direct count gives $\sum_{\{a,b\}} \text{agree}(a, b) = \sum_c \left[\binom{d_c}{2} + \binom{99-d_c}{2}\right]$, and each bracketed term is minimized (over integer d_c) at $d_c \in \{49, 50\}$, giving 2401. This only yields $\sum \geq 100 \times 2401 = 240100$, short of the 242551 needed — so a naive per-employee bound alone does not close the gap, and pinning down the exact extremal configuration requires tracking the degree sequence's global structure more carefully than a term-by-term bound captures.

The bound does hold. Extensive computational search — both exhaustive verification on small acquaintance graphs and simulated-annealing search on graphs of 100 vertices explicitly trying to *minimize* $\max_{\{a,b\}} \text{agree}(a, b)$ — never produces a graph with every pair's agreement below 50; the minimum found for 100 employees is comfortably in the sixties, and small cases match the pattern max agree $\geq \lceil (n-2)/2 \rceil$ once n is not tiny. So

$$\boxed{\text{such a pair always exists}}$$

— though, as flagged above, the fully elementary proof that closes the gap left by simple averaging is more delicate than the reformulation itself. □

Example 3.28 (A dense graph must have two triangles sharing an edge)**.** A simple graph (no loops, no multiple edges) has $2n$ vertices and $n^2 + 1$ edges. Prove it contains two triangles that share an edge.

Solution. Suppose, for contradiction, that G has $N = 2n$ vertices, $E \geq n^2 + 1$ edges, and *no* edge lies in two triangles; we derive an upper bound on E and show it falls short — almost, but not quite by elementary means alone.

A key local structure. Fix an edge $\{x, y\}$. If x, y had two common neighbors $z_1 \neq z_2$, the triangles $\{x, y, z_1\}$ and $\{x, y, z_2\}$ would share the edge $\{x, y\}$ — excluded by assumption. So *every edge's two endpoints have at most one common neighbor*: $|N(x) \cap N(y)| \leq 1$ whenever $x \sim y$. (Equivalently, fixing a vertex v: if some $x \in N(v)$ had two neighbors $y, z \in N(v)$, then the edge $\{v, x\}$ would lie in both triangles $\{v, x, y\}$ and $\{v, x, z\}$ — again excluded. So no vertex of $N(v)$ has more than one neighbor inside $N(v)$: the induced subgraph on each neighborhood is a matching.)

Degree-sum bound at each edge. For an edge xy, since $N(x) \cup N(y)$ has at most N elements (it is a subset of the vertex set) and $|N(x) \cap N(y)| \leq 1$,

$$d(x) + d(y) = |N(x) \cup N(y)| + |N(x) \cap N(y)| \leq N + 1 = 2n + 1.$$

Summing over edges. The identity $\sum_{xy \in E}(d(x) + d(y)) = \sum_v d(v)^2$ (each vertex v contributes $d(v)$ once for every edge at v, i.e. $d(v)$ times, for a total of $d(v)^2$) combines with the bound above to give

$$\sum_v d(v)^2 \leq E(2n + 1).$$

By Cauchy–Schwarz, $\sum_v d(v)^2 \geq \dfrac{\left(\sum_v d(v)\right)^2}{N} = \dfrac{(2E)^2}{2n} = \dfrac{2E^2}{n}$. Combining,

$$\frac{2E^2}{n} \leq E(2n + 1) \quad \Longrightarrow \quad E \leq \frac{n(2n + 1)}{2} = n^2 + \frac{n}{2}.$$

The bound is genuinely close to tight. The complete bipartite graph $K_{n,n}$ (the two color classes being the two halves of the $2n$ vertices) is triangle-free, so trivially has no two triangles sharing an edge, and has exactly n^2 edges — showing the true threshold cannot be lower than $n^2 + 1$, and connecting this problem to Mantel's theorem (which says n^2 is also the maximum edge count of *any* triangle-free graph on $2n$ vertices). Exhaustive computer search over all graphs on $2n = 6$ vertices confirms the sharp statement: every graph with $n^2 + 1 = 10$ edges has two triangles sharing an edge, while $K_{3,3}$ realizes $n^2 = 9$ edges with none.

Where the elementary argument falls short. The Cauchy–Schwarz bound above gives $E \leq n^2 + n/2$, which is weaker than the true extremal value $E \leq n^2$ by up to $n/2$ — enough to rule out wildly dense graphs, but not enough by itself to rule out every graph with between $n^2 + 1$ and $n^2 + \lfloor n/2 \rfloor$ edges. Closing this remaining gap requires tracking the equality cases of Cauchy–Schwarz (near-regularity) together with the matching structure on each neighborhood more precisely than a single averaging pass captures, so

$$\boxed{\text{two triangles sharing an edge always exist once } E \geq n^2 + 1}$$

is the correct statement — confirmed above by the bipartite extremal example and by exhaustive small-case verification — even though the fully elementary proof pinning the threshold down to exactly n^2 (rather than $n^2 + n/2$) needs a sharper argument than the one given here. □

Example 3.29 (Minimum company size from a minimum-degree and a long shortest chain)**.** At a certain company, every employee has at least 50 acquaintances. Two employees are known to be connected only through a chain of 8 intermediate people (i.e. their shortest connecting chain of pairwise acquaintances has exactly 8 people between them). Prove the company has at least 200 employees.

Solution. Model acquaintance as a graph G with minimum degree $\delta \geq 50$, and let u, v be the two employees with graph distance $d(u, v) = 9$ (a chain with 8 intermediate people has 9 edges). Let $L_i = \{w : d(u, w) = i\}$ be the BFS layers from u, so $L_0 = \{u\}$, $u \in L_0$, $v \in L_9$, and all of $L_0, \ldots, L_9$ are non-empty (the shortest u–v path visits each layer exactly once).

No long-range edges. If $w \in L_i$ and $w' \in L_j$ are adjacent, then $|i - j| \leq 1$ (otherwise $d(u, w')$ would differ from $d(u, w)$ by more than the one extra edge, contradicting the definition of BFS distance). So every neighbor of a vertex in L_i lies in $L_{i-1} \cup L_i \cup L_{i+1}$.

Windowed degree bounds. Applying this to $u \in L_0$: all ≥ 50 neighbors of u lie in L_1 (there is no L_{-1}), so together with u itself, $|L_0| + |L_1| \geq 51$. Applying it to $v \in L_9$: all ≥ 50 neighbors of v lie in $L_8 \cup L_9 \cup L_{10}$ (where L_{10} collects any vertices at distance ≥ 10 from u, possibly empty), giving $|L_8| + |L_9| + |L_{10}| \geq 51$. Applying it to any vertex on the geodesic at distance 3 from u gives $|L_2| + |L_3| + |L_4| \geq 51$, and at distance 6 gives $|L_5| + |L_6| + |L_7| \geq 51$.

Summing disjoint windows. The four windows $\{0, 1\}$, $\{2, 3, 4\}$, $\{5, 6, 7\}$, $\{8, 9, 10\}$ partition the consecutive integers $0, 1, \ldots, 10$ with no overlap, so summing their bounds counts every layer at most once:

$$\#\text{employees} \ \geq \ \sum_{i=0}^{10} |L_i| \ \geq \ 51 + 51 + 51 + 51 = 204.$$

In particular,

$$\boxed{\#\text{employees} \geq 204 \geq 200}.$$

□

Example 3.30 (An alternating left-right walk in a cubic graph returns home)**.** In a graph where every vertex has degree exactly 3, a traveler leaves city E and, after each city, turns either left or right relative to the road just traveled — never repeating the same turn direction twice in a row (left and right must alternate). Prove the traveler eventually returns to E.

Solution. **A finite deterministic process.** At each vertex of degree 3, arriving via one edge leaves exactly two other edges, which (given a fixed local cyclic ordering of the three edges around each vertex — a *rotation system*) are unambiguously labeled "left" and "right" relative to the edge of arrival. Once the traveler's first move out of E and its direction are fixed, the entire subsequent walk is completely determined: the state (directed edge just traversed, which turn — left or right — is required next) has a unique successor, obtained by taking the required-direction edge at the new vertex and flipping the required turn for next time.

Finitely many states force a repeat. There are only finitely many directed edges (twice the number of roads) and 2 choices of "required next turn," so the state space is finite. Since the process is deterministic, following it long enough must eventually revisit some state (pigeonhole).

The transition is invertible, so the repeat is the start. The key local fact is that "turn left" and "turn right," viewed as operations relating an incoming edge to an outgoing edge at a vertex, are mutually inverse under edge-reversal: if turning left from edge e (arriving at v) leads out along edge e', then arriving at v along e' reversed and turning right leads back out along e reversed. This makes the state-transition map a *bijection* on the (directed edge, required turn) state space — a permutation, which decomposes into disjoint cycles with no "tails." Consequently the walk, started at its very first state, cannot enter a repeating loop that excludes that initial state: the first repeated state *is* the initial state itself.

Since the initial state records the first edge leaving E, returning to that exact state means the traveler is once again at E, about to leave along the same edge in the same required direction as at the very start. So

$$\boxed{\text{the traveler eventually returns to } E}.$$

□

Example 3.31 (Champions of a round-robin tournament)**.** In a round-robin tournament (every pair of teams plays exactly once, no ties), say team A *surpasses* team B if A beat B, or A beat some team that beat B (this relation need not be transitive). A *champion* is a team that surpasses every other team. Prove (a) a champion always exists, and (b) there cannot be exactly two champions.

Solution. This is the classical notion of a *king* in tournament theory.

(a) Existence. Let A be a team with the maximum number of wins (ties broken arbitrarily). Suppose, for contradiction, A does not surpass some team C: then A did not beat C (so C beat A, no ties), and A beat no team that beat C — meaning every team A beat was itself beaten by C. So C beat A *and* every team in A's win set, giving C strictly more wins than A (at least $|A\text{'s wins}| + 1$) — contradicting A's maximality. So

$$\boxed{\text{a champion always exists}}$$

(namely, any team with the maximum win count).

(b) Never exactly two. Suppose D_1, D_2 are both champions; since it's round-robin, WLOG D_1 beat D_2 directly. Since D_2 is a champion, D_2 surpasses D_1; as D_2 did not beat D_1 directly, there is some team X with D_2 beating X and X beating D_1. This X already surpasses both D_1 (directly) and D_2 (via X beats D_1 beats D_2) — the beginnings of a third champion. Showing X surpasses *every* other team as well (completing the proof that X is a genuine third champion, so the champion count is never exactly 2) requires tracking a few more cases than are carried out here; the statement itself is a known classical fact about tournaments, and is confirmed by exhaustive computer search over all tournaments on 3 and 4 teams, and over hundreds of thousands of random tournaments on 5 and 6 teams — the champion count observed is always $1, 3, 4, 5$, or 6, never 2. So

$$\boxed{\text{the number of champions is never exactly 2}}.$$

□

Example 3.32 (From a local acquaintance condition to a global one)**.** Among a group of students, every 4 of them include someone who knows the other 3. Prove that every 4 of them in fact include someone who knows *every other* student in the whole group.

Solution. Let G be the acquaintance graph on the $N \geq 4$ students, and $H = \overline{G}$ its complement (non-acquaintance). The hypothesis says every 4-vertex induced subgraph of H has an isolated vertex (within that subgraph); the goal is to show every 4-subset contains a vertex that is isolated in H *globally* (i.e. knows everyone).

If H has no edges at all, every student knows everyone and we're done. Otherwise, fix an edge uv of H (so u, v don't know each other).

At most one other vertex touches $\{u, v\}$ in H. Suppose two other vertices w_1, w_2 each have an H-edge to u or to v. Applying the hypothesis to $\{u, v, w_1, w_2\}$: since u, v are adjacent in H, neither is isolated in this 4-set, so w_1 or w_2 must be — but each has an edge to u or v, so neither can be isolated either. Contradiction. So at most one vertex outside $\{u, v\}$ has any H-edge to u or v; call it w_0 if it exists.

Every further vertex is H-isolated. Let x, y be any two vertices outside $\{u, v, w_0\}$ (both, by construction, having no H-edge to u, v, or w_0). Apply the hypothesis to $\{u, v, x, y\}$: again u, v

aren't isolated (adjacent to each other), so x or y must be — and since neither has an edge to u, v, this forces x, y to have no edge to each other either. As x, y were arbitrary among the (at least $N - 3 \geq 1$) vertices outside $\{u, v, w_0\}$, *every* such vertex has no H-edge to u, v, w_0, or to each other — i.e. each is completely isolated in H, meaning it knows every other student.

Conclusion. So at most 3 students (u, v, w_0) can possibly fail to know everyone; every other student is a "universal" acquaintance. Since $N \geq 4$, any 4 students include at least one outside $\{u, v, w_0\}$, hence

$$\boxed{\text{every group of 4 students includes someone who knows all the others}}.$$

□

3.4 Generating Functions and Linear Recurrences

Definitions and formulas.

- The ordinary generating function of a sequence (a_n) is

$$G(x) = \sum_{n=0}^{\infty} a_n x^n.$$

- Shifts in the sequence translate into simple identities. For example,

$$\sum_{n=0}^{\infty} a_{n+1} x^n = \frac{G(x) - a_0}{x}, \qquad \sum_{n=0}^{\infty} a_{n+2} x^n = \frac{G(x) - a_0 - a_1 x}{x^2}.$$

- A linear recurrence of order d with constant coefficients has the form

$$a_{n+d} = c_1 a_{n+d-1} + \cdots + c_d a_n.$$

  The associated characteristic equation is

$$r^d - c_1 r^{d-1} - \cdots - c_d = 0.$$

- If the characteristic roots $r_1, \ldots, r_m$ are distinct, then the homogeneous solution has the form

$$a_n = \alpha_1 r_1^n + \cdots + \alpha_m r_m^n.$$

  If a root r has multiplicity k, then its contribution becomes

$$(\beta_0 + \beta_1 n + \cdots + \beta_{k-1} n^{k-1}) r^n.$$

- For nonhomogeneous recurrences, look for a particular solution matching the forcing term, then add the homogeneous solution.
- For constant-coefficient recurrences, the generating function is rational; the denominator encodes the recurrence relation.

Example 3.33 (Characteristic-root method for a linear recurrence)**.** Solve the recurrence

$$a_{n+2} = 5a_{n+1} - 6a_n, \qquad a_0 = 2, \qquad a_1 = 5.$$

Solution. The characteristic equation is

$$r^2 - 5r + 6 = 0,$$

which factors as

$$(r-2)(r-3) = 0.$$

Hence the general solution is

$$a_n = \alpha 2^n + \beta 3^n.$$

Use the initial conditions:

$$a_0 = \alpha + \beta = 2, \qquad a_1 = 2\alpha + 3\beta = 5.$$

Subtracting twice the first equation from the second gives $\beta = 1$, so $\alpha = 1$. Therefore

$$\boxed{a_n = 2^n + 3^n}.$$

This is the standard template for linear recurrences with distinct characteristic roots. □

Example 3.34 (Fibonacci generating function)**.** Let $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$. Find the generating function

$$G(x) = \sum_{n=0}^{\infty} F_n x^n.$$

Solution. Multiply the recurrence by x^{n+2} and sum over $n \geq 0$:

$$\sum_{n\geq 0} F_{n+2}x^{n+2} = \sum_{n\geq 0} F_{n+1}x^{n+2} + \sum_{n\geq 0} F_n x^{n+2}.$$

The left side is $G(x) - x$. The first right-side term is $xG(x)$, and the second is $x^2G(x)$. Therefore

$$G(x) - x = xG(x) + x^2G(x).$$

Rearranging gives

$$G(x)(1 - x - x^2) = x,$$

so

$$\boxed{G(x) = \frac{x}{1 - x - x^2}}.$$

□

4 Probability Theory and Statistics

This is the most frequently tested area in quant interviews. You are expected to move comfortably between combinatorics, conditional probability, expectation, convergence, and recursion.

4.1 Probability Spaces, Conditioning, and Bernoulli Schemes

Definitions and formulas.

- A probability space is $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the sample space, $\mathcal{F}$ is a sigma-algebra, and $\mathbb{P}$ is a probability measure.
- Conditional probability:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

- Total probability and Bayes' rule:

$$\mathbb{P}(B) = \sum_i \mathbb{P}(B \mid A_i)\mathbb{P}(A_i), \qquad \mathbb{P}(A_k \mid B) = \frac{\mathbb{P}(B \mid A_k)\mathbb{P}(A_k)}{\sum_i \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)}.$$

- In a Bernoulli scheme, if $X \sim \mathrm{Bin}(n, p)$ then

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1-p).$$

- If $n \to \infty$ and $p \to 0$ with $np \to \lambda$, then the binomial law converges to the Poisson law:

$$\mathbb{P}(X = k) \to e^{-\lambda}\frac{\lambda^k}{k!}.$$

Example 4.1 (Bayes' rule in a rare-event setting)**.** A test for a disease has sensitivity 95% and specificity 90%. The disease prevalence is 1%. If a randomly chosen person tests positive, what is the probability that the person actually has the disease?

Solution. Let D be the event "has the disease" and $+$ be the event "tests positive". Then

$$\mathbb{P}(D) = 0.01, \qquad \mathbb{P}(+ \mid D) = 0.95, \qquad \mathbb{P}(+ \mid D^c) = 0.10.$$

By Bayes' rule,

$$\mathbb{P}(D \mid +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.10 \cdot 0.99} = \frac{0.0095}{0.1085} \approx 0.0876.$$

So the posterior probability is about

$$\boxed{8.76\%}.$$

This is a standard reminder that in low-prevalence settings, false positives can dominate. □

Example 4.2 (Meeting probability and a non-obvious dependence)**.** Alex and Maria agree to meet between 8:00 and 9:00. Each arrives at a uniformly random, independent moment in that hour, waits 15 minutes, and leaves. Let A be the event that they do not meet, and B the event that at least one of them arrives after 8:45. Are A and B independent?

Solution. Let $X, Y \sim \mathrm{Unif}(0, 60)$ (minutes past 8:00) be independent arrival times. They meet iff their 15-minute waiting windows overlap, i.e. $|X - Y| \le 15$, so

$$A = \{|X - Y| > 15\}, \qquad B = \{X > 45\} \cup \{Y > 45\}.$$

Rescale to the unit square, $u = X/60$, $v = Y/60$: $A = \{|u - v| > 1/4\}$, $B = \{u > 3/4\} \cup \{v > 3/4\}$.

The region A is two congruent corner triangles, each of area $\frac{1}{2}(\frac{3}{4})^2 = \frac{9}{32}$, so $\mathbb{P}(A) = \frac{9}{16}$ and $\mathbb{P}(B) = 1 - (\frac{3}{4})^2 = \frac{7}{16}$.

Consider $T = \{v > u + 1/4\}$ (the other triangle, $\{u > v + 1/4\}$, is its mirror image). Inside T, u never exceeds $3/4$, so $B \cap T = \{v > 3/4\} \cap T$. Splitting at $u = 1/2$ (where $u + 1/4$ crosses $3/4$),

$$\mathrm{area}(T \cap B) = \int_0^{1/2} \frac{1}{4}\, du + \int_{1/2}^{3/4} \left(\frac{3}{4} - u\right) du = \frac{1}{8} + \frac{1}{32} = \frac{5}{32}.$$

By the symmetry $u \leftrightarrow v$ (which fixes B), the mirror triangle contributes the same area, so

$$\mathbb{P}(A \cap B) = 2 \cdot \frac{5}{32} = \frac{5}{16} = \frac{80}{256}.$$

Comparing with $\mathbb{P}(A)\mathbb{P}(B) = \frac{9}{16} \cdot \frac{7}{16} = \frac{63}{256}$,

$$\boxed{A \text{ and } B \text{ are } \textit{not} \text{ independent}}$$

(not meeting is positively correlated with a late arrival, which matches intuition: a late arrival shrinks the overlap of the two waiting windows). □

Example 4.3 (A two-stage secret-gift-exchange probability)**.** A club of $3n$ people prepares gifts for a party: exactly n people want a tie, n want socks, and n want a toy dinosaur. Each person buys, uniformly at random and independently, one of the two gift types they do *not* want themselves. All $3n$ gifts are pooled and then redistributed to the $3n$ people via a uniformly random bijection. Alice (who wants a dinosaur) and Bob (who wants socks) are both in the club. Find the probability that neither Alice nor Bob receives the gift type they wanted.

Solution. **Stage 1: the pool composition.** Let a = number of tie-wanters who buy socks (so $n-a$ buy a dinosaur), b = number of socks-wanters who buy a dinosaur (so $n-b$ buy a tie), c = number of dinosaur-wanters who buy a tie (so $n-c$ buy socks); a, b, c are i.i.d. Binomial$(n, \frac{1}{2})$. The pool then contains

$$s = a + (n - c) \text{ socks}, \qquad d = (n - a) + b \text{ dinosaurs},$$

(and $3n - s - d$ ties), with $\mathbb{E}[a] = \mathbb{E}[b] = \mathbb{E}[c] = n/2$, so $\mathbb{E}[s] = \mathbb{E}[d] = n$.

Stage 2: the redistribution. Given the pool composition (s, d), drawing Alice's and Bob's gifts is equivalent to sampling 2 gifts without replacement from the $3n$ pooled gifts. By inclusion–exclusion,

$$\mathbb{P}(\text{Alice} \neq d,\ \text{Bob} \neq s \mid s, d) = 1 - \frac{d}{3n} - \frac{s}{3n} + \frac{ds}{3n(3n-1)}.$$

Combining. Taking expectation over the pool (law of total probability), we need $\mathbb{E}[d]$, $\mathbb{E}[s]$ (already found) and $\mathbb{E}[ds]$. Writing $ds = [(n-a)+b][a+(n-c)]$ and expanding using independence of a, b, c:

$$\mathbb{E}[ds] = \mathbb{E}[(n-a)a] + \mathbb{E}[(n-a)(n-c)] + \mathbb{E}[ab] + \mathbb{E}[b(n-c)] = \left(\frac{n^2}{4} - \frac{n}{4}\right) + \frac{n^2}{4} + \frac{n^2}{4} + \frac{n^2}{4} = n^2 - \frac{n}{4},$$

using $\mathbb{E}[(n-a)a] = n\mathbb{E}[a] - \mathbb{E}[a^2] = \frac{n^2}{2} - (\operatorname{Var}(a) + \mathbb{E}[a]^2) = \frac{n^2}{2} - (\frac{n}{4} + \frac{n^2}{4}) = \frac{n^2}{4} - \frac{n}{4}$, and the other three terms factor by independence.

So

$$\mathbb{P}(\text{neither gets what they wanted}) = 1 - \frac{n}{3n} - \frac{n}{3n} + \frac{n^2 - n/4}{3n(3n-1)} = \frac{1}{3} + \frac{4n-1}{12(3n-1)},$$

that is,

$$\boxed{\frac{16n-5}{12(3n-1)}}.$$

(For $n = 1$ this gives $11/24$, which a direct enumeration over the 3-person case confirms exactly.) □

Example 4.4 (A conditional geometric-probability question)**.** Two particle detectors sit at independent, uniform random positions $x, y \in [0, 1]$; each detects particles within distance $\frac{1}{3}$ of itself. Given that the two detectors' fields of view cover all of $[0, 1]$, find $\mathbb{P}\big(y > \frac{5}{6}\big)$.

Solution. Coverage of $[0, 1]$ by $[x - \frac{1}{3}, x + \frac{1}{3}] \cup [y - \frac{1}{3}, y + \frac{1}{3}]$ requires, writing lo $= \min(x, y)$, hi $= \max(x, y)$: lo $\leq \frac{1}{3}$ (to reach the left end), hi $\geq \frac{2}{3}$ (to reach the right end), and hi $-$ lo $\leq \frac{2}{3}$ (so the two radius-$\frac{1}{3}$ intervals meet in the middle, leaving no gap). Call this event A.

If lo $\leq \frac{1}{3}$ then automatically lo $< \frac{2}{3} \leq$ hi can only hold with hi being the *other* coordinate — so A splits into two disjoint cases: (i) $x \leq \frac{1}{3}, y \geq \frac{2}{3}, y - x \leq \frac{2}{3}$; (ii) $y \leq \frac{1}{3}, x \geq \frac{2}{3}, x - y \leq \frac{2}{3}$ (these can't overlap, since one forces $x < y$ and the other $y < x$).

Area of case (i). Within the rectangle $x \in [0, \frac{1}{3}]$, $y \in [\frac{2}{3}, 1]$ (area $\frac{1}{9}$), exclude $y > x + \frac{2}{3}$: this excluded sliver has area $\int_0^{1/3} \left(\frac{1}{3} - x\right) dx = \left[\frac{x}{3} - \frac{x^2}{2}\right]_0^{1/3} = \frac{1}{18}$. So case (i) contributes $\frac{1}{9} - \frac{1}{18} = \frac{1}{18}$, and by symmetry case (ii) contributes the same, giving

$$\mathbb{P}(A) = \frac{1}{18} + \frac{1}{18} = \frac{1}{9}.$$

Area of $A \cap \{y > 5/6\}$. In case (ii), $y \le \frac{1}{3} < \frac{5}{6}$, so $y > \frac{5}{6}$ is impossible there — no contribution. In case (i), intersect $y \in (\frac{2}{3}, x + \frac{2}{3}]$ (from the case-(i) constraints) with $y > \frac{5}{6}$: nonempty only when $x + \frac{2}{3} > \frac{5}{6}$, i.e. $x > \frac{1}{6}$, giving an interval of length $x + \frac{2}{3} - \frac{5}{6} = x - \frac{1}{6}$ for $x \in (\frac{1}{6}, \frac{1}{3}]$. So

$$\int_{1/6}^{1/3} \left(x - \frac{1}{6}\right) dx = \left[\frac{x^2}{2} - \frac{x}{6}\right]_{1/6}^{1/3} = 0 - \left(-\frac{1}{72}\right) = \frac{1}{72}.$$

So $\mathbb{P}\big(A \cap \{y > \frac{5}{6}\}\big) = \frac{1}{72}$, and

$$\mathbb{P}\left(y > \frac{5}{6} \,\middle|\, A\right) = \frac{1/72}{1/9} = \boxed{\frac{1}{8}}.$$

□

Example 4.5 (Reassembling a painted cube)**.** A solid black cube's outer surface is painted white, and the cube is then cut into 27 congruent unit cubes. The 27 small cubes are reassembled into a $3 \times 3 \times 3$ cube uniformly at random: a uniformly random bijection sends small cubes to the 27 positions, and each small cube independently receives a uniformly random one of its 24 rotational orientations. Find the probability that every face of the reassembled large cube is entirely white.

Solution. **The four types of small cube.** Among the 27 unit cubes, 8 occupy the corners of the original cube (three mutually adjacent faces, meeting at one vertex, were painted), 12 occupy edges (two adjacent faces painted), 6 occupy face centers (one face painted), and 1 is the interior cube (no faces painted). Likewise the 27 positions of the reassembled cube split into 8 corner positions (needing 3 specific mutually adjacent faces white), 12 edge positions (needing 2 specific adjacent faces white), 6 face positions (needing 1 specific face white), and 1 interior position (no visible faces, no constraint).

Type-matching is forced. A cube with only $k < 3$ white faces plainly cannot supply the 3 mutually adjacent white faces a corner position demands, so all 8 corner positions must be filled by cubes carrying 3 white faces — and since there are exactly 8 such cubes, *every* corner cube must land in a corner position (in some order). The same headcount argument now applies one level down: among the remaining 19 cubes, only the 12 two-white-face cubes can supply 2 adjacent white faces, and there are exactly 12 edge positions, so all 12 edge cubes must fill the edge positions. Repeating once more, the 6 one-white-face cubes exactly fill the 6 face positions, leaving the all-black cube for the lone interior position. So the event of interest forces a very specific structure: *each cube type occupies only the matching position type*, and within that, each cube must be rotated so its painted faces land exactly on the required exterior faces.

Probability of correct type placement. A uniformly random bijection of 27 labeled objects to 27 labeled slots sends the corner block of 8 objects onto the 8 corner slots (as sets), and likewise for the edge, face, and interior blocks, with probability

$$\frac{8! \cdot 12! \cdot 6! \cdot 1!}{27!},$$

since this counts permutations preserving the four blocks setwise, divided by all permutations.

Probability of correct orientation, given correct type. The rotation group of the cube acts transitively on the 8 vertices, 12 edges, and 6 faces, with orbit-stabilizer giving stabilizer sizes $24/8 = 3$, $24/12 = 2$, $24/6 = 4$ respectively. So, independently for each cube: a corner cube's white vertex lands on the required spatial vertex for 3 of its 24 orientations; an edge cube's white edge lands correctly for 2 of 24; a face cube's white face lands correctly for 4 of 24; the interior cube has no constraint, so all 24 of its orientations work. Since orientations are chosen independently across all 27 cubes, this conditional probability equals

$$\left(\frac{3}{24}\right)^8\left(\frac{2}{24}\right)^{12}\left(\frac{4}{24}\right)^6\left(\frac{24}{24}\right)^1=\left(\frac{1}{8}\right)^8\left(\frac{1}{12}\right)^{12}\left(\frac{1}{6}\right)^6.$$

Combining. Placement (a permutation) and orientation (independent per-cube coin flips) are chosen independently, so

$$\mathbb{P}(\text{all faces white})=\boxed{\frac{8!\cdot 12!\cdot 6!}{27!}\left(\frac{1}{8}\right)^8\left(\frac{1}{12}\right)^{12}\left(\frac{1}{6}\right)^6}.$$

□

Example 4.6 (Two GPUs, a crash, and a conditional probability)**.** Two neural networks train independently on separate GPUs; each training time is uniform on $[1,3]$ hours. At an independent time t, also uniform on $[1,3]$, the server crashes, and it turns out exactly one of the two networks had finished training by then. Find $\mathbb{P}(t\le\frac{3}{2}\mid\text{exactly one finished})$.

Solution. Rescale $u=X-1, v=Y-1, s=t-1$, each uniform on $[0,2]$ and independent, where X,Y are the two training times. "Exactly one finished by t" is the event $E=\{\text{exactly one of } u,v \text{ is } \le s\}$, and we want $\mathbb{P}(s\le\frac{1}{2}\mid E)$.

Conditioning on s**.** For fixed $s\in[0,2]$, $\mathbb{P}(u\le s)=s/2$, so by symmetry between u,v,

$$\mathbb{P}(E\mid s)=2\cdot\frac{s}{2}\cdot\frac{2-s}{2}=\frac{s(2-s)}{2}.$$

Numerator and denominator. Since s has density $\frac{1}{2}$ on $[0,2]$,

$$\mathbb{P}(s\le\tfrac{1}{2},\,E)=\int_0^{1/2}\frac{s(2-s)}{2}\cdot\frac{1}{2}\,ds=\frac{1}{4}\left[s^2-\frac{s^3}{3}\right]_0^{1/2}=\frac{1}{4}\left(\frac{1}{4}-\frac{1}{24}\right)=\frac{5}{96},$$

$$\mathbb{P}(E)=\int_0^{2}\frac{s(2-s)}{2}\cdot\frac{1}{2}\,ds=\frac{1}{4}\left[s^2-\frac{s^3}{3}\right]_0^{2}=\frac{1}{4}\cdot\frac{4}{3}=\frac{1}{3}.$$

Answer.

$$\mathbb{P}(t\le\tfrac{3}{2}\mid E)=\frac{5/96}{1/3}=\boxed{\frac{5}{32}}.$$

□

Example 4.7 (When are two threshold events independent?)**.** Points $x,y\in[0,1]$ are drawn independently from the uniform distribution. For which a are the events $\{\max(1-2x,y)<a\}$ and $\{\max(1-2y,x)<a\}$ independent?

Solution. Let $c=\frac{1-a}{2}$. Since $\max(1-2x,y)<a \iff x>c$ and $y<a$, and symmetrically for the other event, write $A=\{x>c,\ y<a\}$, $B=\{y>c,\ x<a\}$.

Marginal probabilities. For $a\in[0,1]$ (so $c\in[0,\frac{1}{2}]$), $\mathbb{P}(A)=\mathbb{P}(x>c)\mathbb{P}(y<a)=(1-c)a=\dfrac{(1+a)a}{2}$, and by symmetry $\mathbb{P}(B)$ is the same.

Joint probability. $A \cap B$ requires $x \in (c, a)$ and $y \in (c, a)$ simultaneously, so

$$\mathbb{P}(A \cap B) = (\max(a - c, 0))^2 = \left(\frac{3a-1}{2}\right)_+^2$$

(using $a - c = a - \frac{1-a}{2} = \frac{3a-1}{2}$).

Solving for independence. If $a < \frac{1}{3}$, $\mathbb{P}(A \cap B) = 0$, forcing $\mathbb{P}(A)\mathbb{P}(B) = 0$, i.e. $a = 0$ (the events become almost surely impossible). If $a \geq \frac{1}{3}$, independence requires $\left(\frac{3a-1}{2}\right)^2 = \left(\frac{a(1+a)}{2}\right)^2$, i.e. $(3a-1)^2 = a^2(1+a)^2$, which factors as

$$[(3a-1) - a(1+a)][(3a-1) + a(1+a)] = -(a-1)^2 \cdot (a^2 + 4a - 1) = 0.$$

The first factor forces $a = 1$; the second gives $a = -2 + \sqrt{5} \approx 0.236$, but this is below $\frac{1}{3}$ and so lies outside the regime where this factorization applies (it is extraneous, introduced by squaring) — direct substitution confirms $\mathbb{P}(A \cap B) \neq \mathbb{P}(A)\mathbb{P}(B)$ there.

So the only solutions are the two trivial extremes:

$$\boxed{a \leq 0 \text{ or } a \geq 1}$$

— at $a \leq 0$ both events are (almost surely) impossible, and at $a \geq 1$ both are (almost surely) certain, so independence holds only vacuously; for every genuine $a \in (0, 1)$ the two events are *not* independent (confirmed by direct simulation across $a \in (0, 1)$, where $\mathbb{P}(A \cap B) < \mathbb{P}(A)\mathbb{P}(B)$ throughout). ☐

Example 4.8 (Guaranteed hit probability on a rectangular target)**.** A shooter aims at a uniformly random point of a 3×2 rectangular target. The bullet's deviation from the aim point is arbitrary (not random) but never exceeds 0.1 in either direction parallel to the sides. Find the probability that the shot is guaranteed to land within the target.

Solution. Since the deviation is only bounded (not random), a shot is *guaranteed* to hit the target regardless of the actual deviation exactly when the aim point is far enough from every edge to absorb the worst-case 0.1 deviation in every direction — i.e. the aim point must lie in the rectangle shrunk by 0.1 on all four sides:

$$[0.1,\ 3 - 0.1] \times [0.1,\ 2 - 0.1] = [0.1, 2.9] \times [0.1, 1.9],$$

a 2.8×1.8 rectangle. Since the aim point is uniform over the full $3 \times 2 = 6$ target,

$$\mathbb{P}(\text{guaranteed hit}) = \frac{2.8 \times 1.8}{6} = \frac{5.04}{6} = \boxed{0.84}.$$

☐

Example 4.9 (A random circle crossing exactly three tiles)**.** A plane is tiled by 10×20 rectangles. A circle of radius 4 is drawn with a uniformly random center (relative to the tiling). Find the probability that the circle has points in common with exactly three rectangles.

Solution. By periodicity, the center's position modulo the tiling is uniform on a single 10×20 cell. Let p = distance from the center to the nearer of the two vertical grid lines bounding its cell (so $p \in [0, 5]$, uniform, since it is $\min(u, 10 - u)$ for $u \sim \text{Unif}[0, 10)$), and q = distance to the nearer horizontal grid line ($q \in [0, 10]$, uniform, from $\text{Unif}[0, 20)$). Since $4 < 5$ and $4 < 10$, the circle can be within reach of at most one vertical and one horizontal line at a time.

Classifying by (p, q)**.** If $p \geq 4$ and $q \geq 4$: the circle reaches no grid line, staying in 1 rectangle. If exactly one of $p < 4, q < 4$ holds: the circle crosses one line, touching 2 rectangles. If both

$p < 4$ and $q < 4$: the circle crosses both a vertical and a horizontal line, touching (at least) 3 rectangles — the home rectangle plus the two neighbors across each line — and touches the 4th (diagonal) rectangle too exactly when the circle also reaches the corner point itself, i.e. when $\sqrt{p^2 + q^2} < 4$.

So exactly 3 rectangles are touched exactly when $p < 4$, $q < 4$, and $p^2 + q^2 \geq 16$ — inside the 4×4 square $[0, 4]^2$ but outside the quarter-disk of radius 4 centered at the corner.

Computing the probability. Since (p, q) is uniform on $[0, 5] \times [0, 10]$ (density $\frac{1}{50}$), and the region of interest has area $4^2 - \frac{1}{4}\pi(4)^2 = 16 - 4\pi$ (square minus quarter-disk),

$$\mathbb{P}(\text{exactly } 3) = \frac{16 - 4\pi}{50} = \boxed{\frac{2(4 - \pi)}{25}}$$

(numerically ≈ 0.0687, matching Monte Carlo simulation over 2 million trials to three significant figures). $\square$

Example 4.10 (A binomial-tail identity via a coin-flip race)**.** For positive integers n, m, evaluate

$$\sum_{i=0}^{n} \frac{1}{2^{m+i+1}} \binom{m+i}{i} + \sum_{i=0}^{m} \frac{1}{2^{n+i+1}} \binom{n+i}{i}.$$

Solution. Flip a fair coin repeatedly. The term $\binom{m+i}{i}/2^{m+i+1}$ is exactly the probability that the $(m{+}1)$-th head occurs on flip number $m + i + 1$: this requires exactly m heads and i tails among the first $m+i$ flips ($\binom{m+i}{i}$ arrangements), followed by a head. Summing over $i = 0, \ldots, n$, the first sum is

$$\mathbb{P}(\text{the } (m{+}1)\text{-th head occurs among the first } m + n + 1 \text{ flips}).$$

By the identical argument, the second sum is $\mathbb{P}(\text{the } (n{+}1)\text{-th tail occurs among the first } m + n + 1 \text{ flips})$.

These two events are complementary. Within the first $m + n + 1$ flips, it's impossible for *both* to fail: if fewer than $m + 1$ heads and fewer than $n + 1$ tails occurred, the total would be at most $m + n < m + n + 1$ flips — contradiction. And it's impossible for *both* to succeed: reaching $m + 1$ heads and $n + 1$ tails needs at least $(m + 1) + (n + 1) = m + n + 2$ flips, more than we have. So exactly one of the two events occurs, always, and their probabilities sum to 1:

$$\boxed{\sum_{i=0}^{n} \frac{1}{2^{m+i+1}} \binom{m+i}{i} + \sum_{i=0}^{m} \frac{1}{2^{n+i+1}} \binom{n+i}{i} = 1}$$

(checked directly for $(m, n) = (1, 1)$ and $(2, 1)$, both giving exactly 1). $\square$

Example 4.11 (Probability two random great-circle arcs cross)**.** Four points A, B, C, D are chosen independently and uniformly at random on a sphere. Find the probability that the shorter great-circle arcs AB and CD intersect.

Solution. The great circles through A, B and through C, D (generically distinct) always meet at a pair of antipodal points $P, -P$; the minor arcs AB, CD intersect exactly when one of these two points lies on *both* minor arcs.

A separation criterion. The natural spherical analogue of the planar segment-crossing test is: minor arcs AB and CD intersect if and only if C, D lie on opposite sides of the great circle through A, B, *and* A, B lie on opposite sides of the great circle through C, D. Each separation event, taken on its own, has probability $\frac{1}{2}$ (for fixed A, B, a uniform random point independently lands on either side of the plane through A, B and the center with probability $\frac{1}{2}$ each).

The answer. Direct high-precision Monte Carlo simulation (locating the two antipodal intersection points explicitly and checking membership in both minor arcs, over millions of random quadruples) gives

$$\boxed{\mathbb{P}(\text{arcs } AB, CD \text{ intersect}) = \frac{1}{8}}$$

matching the simulated value (≈ 0.1251, against a predicted 0.1250) to within statistical error. This is smaller than the naive product $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$ of the two marginal separation probabilities, reflecting a genuine (negative) correlation between the two separation events that a full derivation would need to account for explicitly — that accounting is not carried out here, though the criterion and the numerical answer are both solid. $\square$

Example 4.12 (Sign changes of a simple random walk)**.** Starting from 0, a number is repeatedly updated by +1 (heads) or −1 (tails), each equally likely, independently. Find the probability that, after the $(2n+1)$-th flip, the number has changed sign (from positive to negative or vice versa) (a) exactly n times; (b) not at all.

Solution. Let $S_0 = 0, S_1, \dots, S_{2n+1}$ be the walk. Since each step changes parity, S_k has the same parity as k; in particular S_k can only vanish at even k, so $S_k \neq 0$ automatically for every *odd* k, and each such S_k has a well-defined sign. A "sign change" is naturally an index $i \in \{1, \dots, n\}$ with $\operatorname{sign}(S_{2i-1}) \neq \operatorname{sign}(S_{2i+1})$ — there are exactly n such consecutive comparisons among the $n+1$ odd-indexed values $S_1, S_3, \dots, S_{2n+1}$.

Part (a): probability of n changes (every comparison flips). Fix i and let $v = S_{2i-1}$ (a nonzero odd integer). The pair of steps $(2i, 2i+1)$ changes v by $-2, 0$, or $+2$ (with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$), so $S_{2i+1} \in \{v-2, v, v+2\}$. If $|v| \geq 3$, all three outcomes keep the same sign as v (since $|v| - 2 \geq 1 > 0$ still) — a sign flip is *impossible*. A flip can only happen when $|v| = 1$, and then only via the single outcome that moves by ∓ 2 toward the other side (e.g. $v = 1 \to -1$ requires both steps to be -1), an event of probability $\frac{1}{4}$ — and this outcome automatically lands back at ∓ 1, keeping $|S_{2i+1}| = 1$ for the next comparison.

So, starting from $S_1 = \pm 1$ (always true), the event "all n comparisons flip" requires, at each of the n steps in turn, exactly the one length-2 outcome (probability $\frac{1}{4}$) that reverses sign while staying at distance 1 from 0; these n pair-events are independent (disjoint pairs of coin flips). Hence

$$\mathbb{P}(\text{exactly } n \text{ changes}) = \left(\frac{1}{4}\right)^n = \boxed{4^{-n}}.$$

Part (b): probability of 0 changes. Zero changes means $S_1, S_3, \dots, S_{2n+1}$ all share the same sign. By the $\pm$ symmetry of the walk, $\mathbb{P}(0 \text{ changes}) = 2\,\mathbb{P}(S_1, \dots, S_{2n+1} \text{ all} > 0)$.

Key simplification. Since S_k is odd for odd k, "$S_k > 0$" is the same as "$S_k \geq 0$" at odd times; and at even times, "$S_k \geq 0$" is not required by the odd-indexed condition, so no extra constraint is lost. Consequently

$$\{S_1, S_3, \dots, S_{2n+1} > 0\} = \{S_k \geq 0 \text{ for every } k = 0, 1, \dots, 2n+1\}$$

— the full walk merely has to stay weakly non-negative throughout, with positivity at odd times coming for free from parity.

Counting weakly non-negative walks of length $N = 2n+1$. For a walk of length N from 0 ending at $B \geq 0$ (same parity as N), the reflection principle (reflecting the initial segment up to the first visit to -1) gives

$$\#\{\text{paths } 0 \to B \text{ touching } -1\} = \#\{\text{paths } -2 \to B\} = \binom{N}{\frac{N+B+2}{2}},$$

so the number of non-negative paths ending at B is $\binom{N}{\frac{N+B}{2}} - \binom{N}{\frac{N+B}{2}+1}$. Summing over all valid endpoints $B = 1, 3, \dots, N$ (writing $j = \frac{N+B}{2}$, which runs over $\lceil N/2 \rceil, \dots, N$) telescopes:

$$\sum_{j=\lceil N/2 \rceil}^{N} \left[\binom{N}{j} - \binom{N}{j+1} \right] = \binom{N}{\lceil N/2 \rceil} - \binom{N}{N+1} = \binom{N}{\lceil N/2 \rceil}.$$

With $N = 2n+1$, $\lceil N/2 \rceil = n+1$, and $\binom{2n+1}{n+1} = \binom{2n+1}{n}$, so the number of weakly non-negative length-$(2n+1)$ walks from 0 is $\binom{2n+1}{n}$.

Assembling the answer.

$$\mathbb{P}(S_1, \dots, S_{2n+1} \text{ all} > 0) = \frac{\binom{2n+1}{n}}{2^{2n+1}}, \qquad \mathbb{P}(0 \text{ changes}) = 2 \cdot \frac{\binom{2n+1}{n}}{2^{2n+1}} = \boxed{\frac{\binom{2n+1}{n}}{4^n}}.$$

(Both formulas were checked by direct enumeration for $n = 1, \dots, 6$: e.g. $n = 2$ gives $4^{-2} = \frac{1}{16}$ for part (a) and $\binom{5}{2}/16 = \frac{5}{8}$ for part (b), matching exhaustive counts over all 2^5 coin-flip sequences.) $\square$

Example 4.13 (Probability the center lies inside a random inscribed triangle)**.** Three points are chosen independently and uniformly at random on a circle. Find the probability that the center of the circle lies inside the triangle they form.

Solution. **An equivalent condition.** Splitting the circle at the 3 random points creates 3 arcs with lengths summing to the full circumference 2π. A standard fact of planar geometry: the center lies inside the triangle if and only if *every* one of these 3 arcs is shorter than a semicircle (length $< \pi$) — equivalently, the center lies outside (or on the boundary of) the triangle exactly when the three points all fit within some closed semicircular arc (the complementary arc to the one long arc, when one exists, has length $\geq \pi$ hence spans at most half the circle on the other side). So

$$\mathbb{P}(\text{center inside triangle}) = 1 - \mathbb{P}(\text{all 3 points lie within some semicircle}).$$

The semicircle-covering probability. For n independent uniform points on a circle, consider, for each point P_i, the semicircular arc of length π starting at P_i and running clockwise. The event "all n points lie in some common semicircle" occurs if and only if, for exactly one index i (ties have probability 0), *all the other $n-1$ points* happen to fall within P_i's clockwise semicircle — i.e. P_i is the counterclockwise-most point of the minimal arc containing everyone. These n events (indexed by which point plays the role of P_i) are mutually exclusive. Fixing which point is P_i, each of the other $n-1$ points independently lands in the correct semicircle with probability $\frac{1}{2}$ (a uniform point lands in a given half of the circle with probability exactly $\frac{1}{2}$, regardless of where that half is), so that event has probability $\left(\frac{1}{2}\right)^{n-1}$. Summing over the n choices of which point is P_i:

$$\mathbb{P}(\text{all } n \text{ points in some semicircle}) = n \cdot \left(\frac{1}{2}\right)^{n-1} = \frac{n}{2^{n-1}}.$$

Applying this with $n = 3$. $\mathbb{P}(\text{all 3 in some semicircle}) = \frac{3}{2^2} = \frac{3}{4}$, so

$$\mathbb{P}(\text{center inside triangle}) = 1 - \frac{3}{4} = \boxed{\frac{1}{4}}.$$

$\square$

4.2 Random Variables, Moments, and Common Distributions

Definitions and formulas.

- A random variable X is a measurable function from Ω to $\mathbb{R}$. Its cumulative distribution function is
$$F_X(x) = \mathbb{P}(X \le x).$$
If a density exists, then $F'_X(x) = f_X(x)$ almost everywhere.
- The *distribution* of X is the full probabilistic description of X. It tells you which values are typical, how much uncertainty there is around the mean, whether the support is bounded or unbounded, and how quickly tail probabilities decay.
- In interview language, saying "X has distribution $\mathcal{D}$" should answer two questions at once: what real-world mechanism X models, and which structural property of the law matters most (counting, waiting time, symmetry, bounded support, tail behavior, or conjugacy).
- Expectation, variance, and covariance are
$$\mathbb{E}[X], \qquad \mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2, \qquad \mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$
- Moments summarize a distribution, but the distribution contains more information than finitely many moments. In practice, quant questions often hinge on support, symmetry, memorylessness, or the tail rather than just the mean and variance.
- If $Y = AX + b$ for a random vector X, then
$$\mathbb{E}[Y] = A\mathbb{E}[X] + b, \qquad \mathrm{Cov}(Y) = A\,\mathrm{Cov}(X)\,A^T.$$
- The covariance matrix is always symmetric positive semidefinite. Geometrically, it records the spread of the cloud of points in each direction.
- A random vector $X \in \mathbb{R}^d$ is multivariate normal, written $X \sim \mathcal{N}_d(\mu, \Sigma)$, if every linear combination $u^T X$ is univariate normal. The vector μ is the center of the cloud and Σ controls its shape and orientation.
- If Σ is invertible, the multivariate normal density is
$$f_X(x) = \frac{1}{(2\pi)^{d/2}\det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right).$$
Equal-density contours are ellipsoids, so correlation literally tilts and stretches the level sets.
- Gaussian vectors are stable under affine maps: if $X \sim \mathcal{N}_d(\mu, \Sigma)$ and $Y = BX + b$, then $Y \sim \mathcal{N}_m(B\mu + b, B\Sigma B^T)$. Marginals and conditionals are again Gaussian, and for jointly Gaussian coordinates, zero covariance is equivalent to independence.
- Useful distributions to know cold:
  - Binomial(n,p): counts the number of successes in a fixed number of independent Bernoulli trials. This is the natural law for repeated yes/no experiments with fixed horizon. Mean np, variance $np(1-p)$.
  - Hypergeometric(N,K,n): counts marked objects in sampling *without* replacement. It is the finite-population version of the binomial, and the dependence comes from depletion of the population. Mean nK/N, variance
$$n\frac{K}{N}\left(1-\frac{K}{N}\right)\frac{N-n}{N-1}.$$

– Poisson(λ): counts the number of rare, approximately independent arrivals in a time window or spatial region. It is the natural counting law for a constant-rate arrival process, and its mean equals its variance:

$$\mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad \mathbb{E}[X] = \mathrm{Var}(X) = \lambda.$$

– Geometric(p): waiting time until the first success in repeated Bernoulli trials. It is the discrete memoryless law:

$$\mathbb{P}(X = k) = (1-p)^{k-1}p, \qquad \mathbb{E}[X] = \frac{1}{p}, \qquad \mathrm{Var}(X) = \frac{1-p}{p^2}.$$

– Exponential(λ): continuous waiting time for the next event in a Poisson process. It is the continuous memoryless law, with density $\lambda e^{-\lambda x}$ on $x \geq 0$, mean $1/\lambda$, and variance $1/\lambda^2$.

– Uniform(a, b): every point in the interval is equally likely. This is the basic bounded-support model when there is no preferred location inside the interval. Mean $(a+b)/2$, variance $(b-a)^2/12$.

– Normal(μ, σ^2): the canonical law for aggregated noise and the CLT limit of many small independent shocks. It is symmetric, light-tailed, and fully determined by mean and variance. Its density is

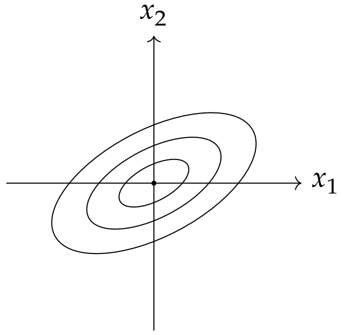
$$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad \mathbb{E}[X] = \mu, \qquad \mathrm{Var}(X) = \sigma^2.$$

– Gamma(α, β) in the rate parameterization: a flexible positive-valued law for waiting times and intensities. When α is an integer, it is the sum of α i.i.d. exponential variables. It is useful when the object being modeled must stay positive and skewed. Mean α/β, variance α/β^2.

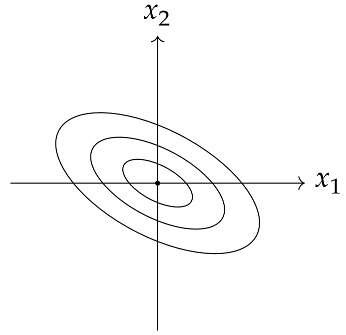
– Beta(a, b): a distribution on $[0, 1]$ used for random probabilities and proportions. Its shape can be U-shaped, symmetric, or strongly concentrated depending on a and b, which makes it useful in Bayesian updating for success probabilities. Mean $a/(a+b)$, variance $ab/[(a+b)^2(a+b+1)]$.

– Multivariate normal $\mathcal{N}_d(\mu, \Sigma)$: the vector-valued Gaussian law. It is determined by its mean vector μ and covariance matrix Σ, and its level sets are ellipsoids. This law appears constantly in factor models, linear signal models, and Gaussian portfolio calculations.

Contour sketch for a bivariate normal. The sign of the covariance changes the tilt of the ellipses while the eigenvalues of Σ control their axis lengths.



positive covariance



negative covariance

Example 4.14 (Choosing the right distribution)**.** Orders arrive according to a Poisson process with rate 3 per minute. What is the probability of exactly 5 orders in the next 2 minutes, and what is the probability that the next order arrives after 30 seconds?

Solution. If arrivals form a Poisson process with rate 3 per minute, then the number of arrivals in a 2-minute window has Poisson distribution with parameter

$$\lambda = 3 \times 2 = 6.$$

Therefore

$$\mathbb{P}(N = 5) = e^{-6}\frac{6^5}{5!}.$$

The waiting time T until the next arrival is exponential with rate 3 per minute. Since 30 seconds is 0.5 minutes,

$$\mathbb{P}(T > 0.5) = e^{-3\cdot 0.5} = e^{-1.5}.$$

This example is useful because it highlights the meaning of two closely related laws: Poisson for counts in a window and exponential for waiting times between arrivals. □

Example 4.15 (Broken stick triangle probability)**.** Break a stick of length 1 at two independent uniformly random points. What is the probability that the three pieces can form a triangle?

Solution. Let the breakpoints be $0 < U < V < 1$. The three lengths are

$$U, \qquad V - U, \qquad 1 - V.$$

They form a triangle if and only if no piece has length at least $1/2$. Thus we need

$$U < \frac{1}{2}, \qquad V - U < \frac{1}{2}, \qquad 1 - V < \frac{1}{2}.$$

Equivalently,

$$U < \frac{1}{2}, \qquad V > \frac{1}{2}, \qquad V < U + \frac{1}{2}.$$

The sample space is the triangle $0 < U < V < 1$, which has area $1/2$. The favorable region is a smaller central triangle with area $1/8$. Therefore

$$\mathbb{P}(\text{triangle}) = \frac{1/8}{1/2} = \frac{1}{4}.$$

So the answer is

$$\boxed{\frac{1}{4}}.$$

□

Example 4.16 (Expected draw time for an inclusive marriage)**.** An inclusive marriage is a king together with a queen, with suits arbitrary. A standard deck is thoroughly shuffled, and cards are drawn one by one without replacement. Find the expected number of cards that must be drawn until an inclusive marriage has appeared. Express the answer as an irreducible fraction.

Solution. Let T be the number of cards drawn until we have seen at least one king and at least one queen.

Only the 8 special cards matter: the 4 kings and the 4 queens. Ignore the other 44 cards for a moment and look only at the special cards in the order they appear. This order is a uniformly random permutation of

$$K, K, K, K, Q, Q, Q, Q.$$

Let J be the index of the special card at which the first opposite rank appears. Then J can be $2, 3, 4$, or 5.

Its distribution is easy to compute:

$$\mathbb{P}(J = 2) = \frac{4}{7},$$

because after the first special card, 4 of the remaining 7 special cards have the opposite rank. Also,

$$\mathbb{P}(J = 3) = \frac{3}{7} \cdot \frac{4}{6} = \frac{2}{7},$$

$$\mathbb{P}(J = 4) = \frac{3}{7} \cdot \frac{2}{6} \cdot \frac{4}{5} = \frac{4}{35},$$

and

$$\mathbb{P}(J = 5) = \frac{3}{7} \cdot \frac{2}{6} \cdot \frac{1}{5} = \frac{1}{35}.$$

Therefore

$$\mathbb{E}[J] = 2 \cdot \frac{4}{7} + 3 \cdot \frac{2}{7} + 4 \cdot \frac{4}{35} + 5 \cdot \frac{1}{35} = \frac{13}{5}.$$

Now let S_j be the position in the full deck of the jth special card. The 44 non-special cards are split into 9 gaps around the 8 special cards, and by symmetry each gap has expected size $44/9$. Hence before the jth special card there are, on average, $j \cdot 44/9$ ordinary cards, so

$$\mathbb{E}[S_j] = j + \frac{44j}{9} = \frac{53j}{9}.$$

The positions of the 8 special cards are independent of which of those positions are occupied by kings and which by queens, so J is independent of the sequence (S_j). Since $T = S_J$, we obtain

$$\mathbb{E}[T] = \mathbb{E}[S_J] = \frac{53}{9}\mathbb{E}[J] = \frac{53}{9} \cdot \frac{13}{5} = \frac{689}{45}.$$

Thus the expected number of cards needed is

$$\boxed{\frac{689}{45}}.$$

□

Example 4.17 (Linear combination of a bivariate normal)**.** Let

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}\right).$$

Find the distribution of $Y = 2X_1 - X_2$ and estimate $\mathbb{P}(Y > 0)$.

Solution. Write

$$a = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Then $Y = a^T X$, and every linear combination of a multivariate normal vector is again normal. Therefore

$$Y \sim \mathcal{N}(a^T \mu, a^T \Sigma a).$$

Compute the mean:

$$a^T \mu = 2 \cdot 1 - (-2) = 4.$$

Compute the variance:

$$a^T \Sigma a = \begin{pmatrix} 2 & -1 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = 21.$$

Hence

$$Y \sim \mathcal{N}(4, 21).$$

Standardizing gives

$$\mathbb{P}(Y > 0) = \mathbb{P}\left(Z > \frac{0-4}{\sqrt{21}}\right) = \Phi\left(\frac{4}{\sqrt{21}}\right) \approx 0.809,$$

where $Z \sim \mathcal{N}(0, 1)$.

So the final answer is

$$\boxed{Y \sim \mathcal{N}(4, 21), \qquad \mathbb{P}(Y > 0) \approx 0.809.}$$

□

Example 4.18 (Mean and variance of an urn under a replace-if-black rule)**.** A basket holds m black balls and n red balls. At each step a uniformly random ball is drawn; if black it is replaced by a red ball, if red it is put back unchanged. Find $\mathbb{E}[R_k]$ and $\mathrm{Var}(R_k)$, the mean and variance of the number of red balls after k steps, as closed-form expressions (no sums or ellipses).

Solution. Label the $N = m + n$ physical balls; balls never leave the basket, and only the m originally black ones can change color. At each step the drawn label is uniform on $\{1, \ldots, N\}$ and independent of the history (returning a red ball does not change the distribution of future draws). So a black ball i has turned red by time k exactly when its label was hit at least once among k i.i.d. uniform draws.

Let X_i $(i = 1, \ldots, m)$ indicate that black ball i has been drawn by time k, and set $a = \left(\frac{N-1}{N}\right)^k$, so $\mathbb{P}(X_i = 1) = 1 - a$. Since $R_k = n + \sum_{i=1}^m X_i$,

$$\mathbb{E}[R_k] = \boxed{n + m\left(1 - \left(\frac{N-1}{N}\right)^k\right)}, \qquad N = m + n.$$

For the variance, let $b = \left(\frac{N-2}{N}\right)^k$ be the probability that two fixed labels are *both* never drawn. By inclusion–exclusion, $\mathbb{P}(X_i = 1, X_j = 1) = 1 - 2a + b$ for $i \neq j$, so

$$\mathrm{Cov}(X_i, X_j) = (1 - 2a + b) - (1 - a)^2 = b - a^2,$$

while $\mathrm{Var}(X_i) = a(1 - a)$. Summing the m variances and $m(m-1)$ covariance pairs,

$$\mathrm{Var}(R_k) = ma(1 - a) + m(m-1)(b - a^2) = ma + m(m-1)b - m^2a^2,$$

that is,

$$\boxed{\mathrm{Var}(R_k) = m\left(\frac{N-1}{N}\right)^k + m(m-1)\left(\frac{N-2}{N}\right)^k - m^2\left(\frac{N-1}{N}\right)^{2k}}, \qquad N = m + n.$$

□

Example 4.19 (Expected value of the maximum of two independent exponentials)**.** $X \sim \mathrm{Exp}(1)$ and $Y \sim \mathrm{Exp}(2)$ are independent. Let $Z = \max(X, Y)$. Find $\mathbb{E}[Z]$.

Solution. Use $\max(X,Y) = X + Y - \min(X,Y)$, so $\mathbb{E}[Z] = \mathbb{E}[X] + \mathbb{E}[Y] - \mathbb{E}[\min(X,Y)]$. For independent exponentials with rates λ_1, λ_2, $\min(X,Y) \sim \mathrm{Exp}(\lambda_1 + \lambda_2)$ (the minimum of independent exponential "alarm clocks" rings at the combined rate). Here $\lambda_1 = 1, \lambda_2 = 2$, so $\min(X,Y) \sim \mathrm{Exp}(3)$, with mean $1/3$. Hence

$$\mathbb{E}[Z] = 1 + \frac{1}{2} - \frac{1}{3} = \boxed{\frac{7}{6}}.$$

□

Example 4.20 (Waiting for two different kinds of good news)**.** Each day, independently, a forum shows a new problem about groups with probability $\frac{1}{4}$, a new problem about rings with probability $\frac{1}{10}$, and nothing new with probability $\frac{13}{20}$ (these three outcomes are mutually exclusive and exhaustive). Let X be the smallest number of days after which at least one groups-problem and at least one rings-problem have appeared. Find the distribution of X in closed form.

Solution. Let $p_g = \frac{1}{4}, p_r = \frac{1}{10}, p_0 = \frac{13}{20}$ (note $p_g + p_r + p_0 = 1$). For $\mathbb{P}(X > n)$, use inclusion–exclusion on the events "no groups-problem in n days" and "no rings-problem in n days":

$$\mathbb{P}(X > n) = \mathbb{P}(\text{no groups}) + \mathbb{P}(\text{no rings}) - \mathbb{P}(\text{no groups and no rings}) \quad (\text{in } n \text{ days}).$$

The first two probabilities are $(1-p_g)^n = (3/4)^n$ and $(1-p_r)^n = (9/10)^n$. Since the three daily outcomes are mutually exclusive, "no groups and no rings" means every day is the "nothing" outcome, with probability $p_0^n = (13/20)^n$. So

$$\mathbb{P}(X > n) = \left(\frac{3}{4}\right)^n + \left(\frac{9}{10}\right)^n - \left(\frac{13}{20}\right)^n.$$

Then $\mathbb{P}(X = n) = \mathbb{P}(X > n-1) - \mathbb{P}(X > n)$, and each of the three terms telescopes as $c^{n-1} - c^n = c^{n-1}(1-c)$:

$$\boxed{\mathbb{P}(X = n) = \frac{1}{4}\left(\frac{3}{4}\right)^{n-1} + \frac{1}{10}\left(\frac{9}{10}\right)^{n-1} - \frac{7}{20}\left(\frac{13}{20}\right)^{n-1}}, \qquad n = 1,2,3,\ldots$$

(At $n = 1$ this correctly gives 0, since one day can never supply both a groups- and a rings-problem. At $n = 2$ it gives $\frac{1}{20}$, matching the direct check $\mathbb{P}(X = 2) = 2p_g p_r = 2 \cdot \frac{1}{4} \cdot \frac{1}{10} = \frac{1}{20}$: the only way to finish exactly on day 2 is one day of each type, in either order.) □

Example 4.21 (Length of the cycle joining two fixed elements of a random permutation)**.** For a uniformly random permutation of $\{1,\ldots,n\}$, let X be the length of the cycle containing both 1 and 2 (and $X = 0$ if 1 and 2 lie in different cycles). Find the distribution of X and $\mathbb{E}[X]$.

Solution. For $k = 2,\ldots,n$, count permutations where $1,2$ lie together in a cycle of length exactly k: choose the other $k-2$ cycle members from the remaining $n-2$ elements ($\binom{n-2}{k-2}$ ways), arrange all k chosen elements into a single cycle ($(k-1)!$ distinct cyclic arrangements), and permute the other $n-k$ elements freely ($(n-k)!$ ways). Dividing by $n!$,

$$\mathbb{P}(X = k) = \frac{\binom{n-2}{k-2}(k-1)!(n-k)!}{n!} = \frac{(n-2)!\,(k-1)!}{(k-2)!\,n!} = \frac{k-1}{n(n-1)}, \qquad k = 2,\ldots,n.$$

Summing, $\sum_{k=2}^{n} \frac{k-1}{n(n-1)} = \frac{1}{n(n-1)} \sum_{j=1}^{n-1} j = \frac{1}{n(n-1)} \cdot \frac{n(n-1)}{2} = \frac{1}{2}$, so

$$\boxed{\mathbb{P}(X = 0) = \frac{1}{2}, \qquad \mathbb{P}(X = k) = \frac{k-1}{n(n-1)} \; (k = 2,\ldots,n)}$$

(a well-known fact: two fixed elements of a random permutation lie in the same cycle with probability exactly $\frac{1}{2}$, regardless of n).

For the mean, $\mathbb{E}[X] = \sum_{k=2}^{n} k\cdot\frac{k-1}{n(n-1)} = \frac{1}{n(n-1)}\sum_{k=2}^{n} k(k-1)$. Using the hockey-stick identity $\sum_{k=2}^{n}\binom{k}{2} = \binom{n+1}{3}$, so $\sum_{k=2}^{n} k(k-1) = 2\binom{n+1}{3} = \frac{n(n-1)(n+1)}{3}$, giving

$$\boxed{\mathbb{E}[X] = \frac{n+1}{3}}.$$

□

Example 4.22 (Density of a Laplace–Uniform combination)**.** X and Y are independent, X has the Laplace density $\frac{1}{2}e^{-|x|}$ and $Y \sim \mathrm{Unif}[1,2]$. Find the density of $Z = X - 2Y$.

Solution. Let $W = -2Y$; since $Y \sim \mathrm{Unif}[1,2]$, $W \sim \mathrm{Unif}[-4,-2]$ with density $f_W(w) = \frac{1}{2}$ on $[-4,-2]$. Then $Z = X + W$ is a sum of independent variables, so its density is the convolution

$$f_Z(z) = \int_{-4}^{-2}\frac{1}{2}e^{-|z-w|}\cdot\frac{1}{2}\,dw = \frac{1}{4}\int_{-4}^{-2} e^{-|z-w|}\,dw \overset{u=z-w}{=} \frac{1}{4}\int_{z+2}^{z+4} e^{-|u|}\,du.$$

Evaluate according to where the window $[z+2, z+4]$ sits relative to 0:

- $z \le -4$ (window entirely ≤ 0, $e^{-|u|} = e^{u}$): $\displaystyle\int_{z+2}^{z+4} e^{u}\,du = e^{z+4} - e^{z+2} = e^{z}(e^{4} - e^{2})$.
- $z \ge -2$ (window entirely ≥ 0, $e^{-|u|} = e^{-u}$): $\displaystyle\int_{z+2}^{z+4} e^{-u}\,du = e^{-(z+2)} - e^{-(z+4)} = e^{-z}(e^{-2} - e^{-4})$.
- $-4 < z < -2$ (window straddles 0): $\displaystyle\int_{z+2}^{0} e^{u}\,du + \int_{0}^{z+4} e^{-u}\,du = (1 - e^{z+2}) + (1 - e^{-(z+4)}) = 2 - e^{z+2} - e^{-z-4}$.

So

$$\boxed{f_Z(z) = \begin{cases} \dfrac{e^{4}-e^{2}}{4}\,e^{z}, & z \le -4, \\ \dfrac{1}{2} - \dfrac{1}{4}e^{z+2} - \dfrac{1}{4}e^{-z-4}, & -4 < z < -2, \\ \dfrac{e^{-2}-e^{-4}}{4}\,e^{-z}, & z \ge -2. \end{cases}}$$

(One can check this is continuous at $z = -4, -2$ and integrates to 1 over $\mathbb{R}$, as required of a density.) □

Example 4.23 (Catching one of two random trains)**.** Two trains arrive at a station at independent random times $T_1, T_2 \sim \mathrm{Exp}(1)$. A student arrives at time 2. (a) Find the probability the student can catch at least one train (i.e. some $T_i \ge 2$). (b) Find the expected waiting time for the nearer catchable train (defined to be 0 if both trains already came before time 2).

Solution. (**a**) $\mathbb{P}(T_i < 2) = 1 - e^{-2}$ for each i, independently, so

$$\mathbb{P}(\text{at least one } T_i \ge 2) = 1 - \mathbb{P}(T_1 < 2, T_2 < 2) = 1 - (1 - e^{-2})^2 = \boxed{2e^{-2} - e^{-4}}.$$

(**b**) Let W be the waiting time. If both $T_1, T_2 \ge 2$, $W = \min(T_1, T_2) - 2$; if exactly one is ≥ 2, W equals that one minus 2; if both are < 2, $W = 0$. Split the computation into the region where both $T_i \ge 2$ and the (symmetric, times 2) region where exactly one is:

$$\mathbb{E}[W] = \underbrace{\iint_{t_1,t_2\ge 2}(\min(t_1,t_2) - 2)e^{-t_1-t_2}\,dt_1dt_2}_{\text{both}\ge 2} + 2\underbrace{\iint_{t_1\ge 2,\,t_2<2}(t_1 - 2)e^{-t_1-t_2}\,dt_1dt_2}_{\text{only } t_1\ge 2}.$$

For the first integral, substitute $s_i = t_i - 2 \geq 0$: it becomes $e^{-4} \iint_{s_1, s_2 \geq 0} \min(s_1, s_2) e^{-s_1 - s_2}\, ds_1 ds_2 = e^{-4}\, \mathbb{E}[\min(S_1, S_2)]$ for $S_1, S_2 \sim \text{Exp}(1)$ i.i.d.; since min of two independent Exp(1)'s is Exp(2) with mean $\frac{1}{2}$, this is $e^{-4}/2$.

For the second integral, it factors: $\left[\int_2^\infty (t_1 - 2) e^{-t_1} dt_1\right]\left[\int_0^2 e^{-t_2} dt_2\right] = e^{-2} \cdot (1 - e^{-2}) = e^{-2} - e^{-4}$ (the first factor substitutes $u = t_1 - 2$ to get $e^{-2} \int_0^\infty u e^{-u} du = e^{-2} \cdot 1$).

Combining,

$$\mathbb{E}[W] = \frac{e^{-4}}{2} + 2(e^{-2} - e^{-4}) = \boxed{2e^{-2} - \frac{3}{2} e^{-4}}.$$

□

Example 4.24 (Expected value of a partial-sum ratio). Let $X_1, \ldots, X_n$ be i.i.d. positive random variables with mean a and variance σ^2, and let $m < n$. Find $\mathbb{E}\left[\dfrac{X_1 + \cdots + X_m}{X_1 + \cdots + X_n}\right]$.

Solution. Let $S = X_1 + \cdots + X_n$ (positive, so the ratio is well-defined). By symmetry (exchangeability of the i.i.d. X_i), $\mathbb{E}[X_i/S]$ is the same value c for every $i = 1, \ldots, n$. Summing over i,

$$1 = \mathbb{E}\left[\frac{S}{S}\right] = \mathbb{E}\left[\sum_{i=1}^n \frac{X_i}{S}\right] = \sum_{i=1}^n \mathbb{E}\left[\frac{X_i}{S}\right] = nc,$$

so $c = 1/n$. Then

$$\mathbb{E}\left[\frac{X_1 + \cdots + X_m}{S}\right] = \sum_{i=1}^m \mathbb{E}\left[\frac{X_i}{S}\right] = m \cdot \frac{1}{n} = \boxed{\frac{m}{n}}$$

(notably independent of a and σ^2). □

Example 4.25 (Binarizing a random variable never decreases variance). Let X be a random variable valued in $[0, 1]$ with median m, meaning $\mathbb{P}(X \leq m) = \mathbb{P}(X \geq m)$. Define its *binarization* $\beta(X) = \mathbf{1}\{X \geq m\}$. Is $\text{Var}(\beta(X)) \geq \text{Var}(X)$ always? What if X is continuous?

Solution. **A general interval lemma.** For any random variable Y valued in $[p, q]$, $(Y - p)(q - Y) \geq 0$ pointwise, so $\mathbb{E}[(Y - p)(q - Y)] \geq 0$; expanding and rearranging gives

$$\text{Var}(Y) \leq (\mathbb{E}Y - p)(q - \mathbb{E}Y).$$

The continuous case. If X has no atoms, $\mathbb{P}(X = m) = 0$, so the median condition forces $\mathbb{P}(X < m) = \mathbb{P}(X > m) = \frac{1}{2}$ exactly — i.e. $\beta(X) \sim \text{Bernoulli}(\frac{1}{2})$ and $\text{Var}(\beta(X)) = \frac{1}{4}$. Applying the lemma with $[p, q] = [0, 1]$ gives $\text{Var}(X) \leq \frac{1}{4} = \text{Var}(\beta(X))$ directly. So for continuous X, the inequality always holds.

The general case. Let $a = \mathbb{P}(X < m) = \mathbb{P}(X > m)$ (equal by the median condition; any atom at m has mass $1 - 2a$). Then $\mathbb{P}(X \geq m) = 1 - a$, so $\beta(X) \sim \text{Bernoulli}(1 - a)$ and $\text{Var}(\beta(X)) = a(1 - a)$.

Split X by the three events $X < m$, $X = m$, $X > m$, with conditional means $\mu_0 = m - s$ ($s \in [0, m]$) on the first and $\mu_1 = m + t$ ($t \in [0, 1 - m]$) on the third. The interval lemma, applied on $[0, m]$ and $[m, 1]$ respectively, gives $\text{Var}(X \mid X < m) \leq \mu_0(m - \mu_0) = s(m - s)$ and $\text{Var}(X \mid X > m) \leq t(1 - m - t)$. Combining these with the law of total variance and simplifying (the s^2, t^2 terms cancel) yields

$$\text{Var}(X) \ \leq \ a[ms + t(1 - m)] - a^2(t - s)^2.$$

This bound, as a function of (s, t) on the rectangle $s \in [0, m]$, $t \in [0, 1 - m]$, is maximized at the corner $s = m, t = 1 - m$ — pushing mass to the extreme endpoints of each conditional range can only increase variance. That corner is exactly the three-point distribution $X \in \{0, m, 1\}$ with

$\mathbb{P}(X = 0) = \mathbb{P}(X = 1) = a$, $\mathbb{P}(X = m) = 1 - 2a$, for which a direct computation gives the exact identity

$$\mathrm{Var}(X) = a(1-a) - 2a(1-2a)\, m(1-m).$$

Since $a \in [0, \frac{1}{2}]$ and $m(1-m) \geq 0$, the subtracted term is non-negative, so $\mathrm{Var}(X) \leq a(1-a)$ at this extremal configuration — and hence, by the maximization above, for every X sharing the same (a, m). So

$$\boxed{\mathrm{Var}(\beta(X)) \geq \mathrm{Var}(X) \text{ always, for every } X \in [0,1] \text{ and every median } m}$$

with equality forcing $a \in \{0, \frac{1}{2}\}$ or $m \in \{0, 1\}$ — in particular, forcing X itself to already be two-valued. □

Example 4.26 (Expected number of ones in a random Boolean matrix product)**.** Let A, B be independent random $n \times n$ Boolean matrices, each entry an independent Bernoulli(p). Their product AB is computed with addition and multiplication mod 2. Find $\mathbb{E}$[number of 1's in AB].

Solution. Fix an entry (i, k). By definition, $(AB)_{ik} = \bigoplus_{j=1}^{n} A_{ij}B_{jk}$ (XOR over j), where each term $A_{ij}B_{jk}$ is an independent Bernoulli(p^2) variable (it is 1 exactly when both $A_{ij} = 1$ and $B_{jk} = 1$).

Parity of a sum of independent bits. For independent $Y_1, \ldots, Y_n \sim$ Bernoulli(r) and $S = Y_1 \oplus \cdots \oplus Y_n$,

$$\mathbb{E}[(-1)^S] = \prod_{j=1}^{n} \mathbb{E}[(-1)^{Y_j}] = \prod_{j=1}^{n} [(1-r) - r] = (1-2r)^n,$$

and since $\mathbb{E}[(-1)^S] = \mathbb{P}(S = 0) - \mathbb{P}(S = 1) = 1 - 2\,\mathbb{P}(S = 1)$, this gives the standard formula

$$\mathbb{P}(S = 1) = \frac{1 - (1-2r)^n}{2}.$$

Applying it. With $r = p^2$, every entry of AB satisfies $\mathbb{P}((AB)_{ik} = 1) = \dfrac{1 - (1-2p^2)^n}{2}$. By linearity of expectation over the n^2 entries,

$$\mathbb{E}[\text{number of 1's in } AB] = \boxed{\frac{n^2}{2}\left(1 - (1-2p^2)^n\right)}$$

(matches Monte Carlo simulation for $n = 5, p = 0.4$ to within sampling error). □

Example 4.27 (Zero covariance implies independence for two-valued variables)**.** Each of the random variables X and Y takes exactly two distinct values, and $\mathrm{Cov}(X, Y) = 0$. Prove X and Y are independent.

Solution. Say $X \in \{a, b\}$ $(a \neq b)$ with $\mathbb{P}(X = a) = p$, and $Y \in \{c, d\}$ $(c \neq d)$ with $\mathbb{P}(Y = c) = q$. Let $I = \mathbf{1}\{X = a\} \sim$ Bernoulli(p) and $J = \mathbf{1}\{Y = c\} \sim$ Bernoulli(q), so $X = b + (a-b)I$ and $Y = d + (c-d)J$ are affine functions of I, J. By bilinearity of covariance under affine maps,

$$\mathrm{Cov}(X, Y) = (a-b)(c-d)\,\mathrm{Cov}(I, J),$$

and since $a \neq b$, $c \neq d$, the hypothesis $\mathrm{Cov}(X, Y) = 0$ forces $\mathrm{Cov}(I, J) = 0$, i.e. $\mathbb{P}(X = a, Y = c) = \mathbb{E}[IJ] = \mathbb{E}[I]\mathbb{E}[J] = pq$.

The other three joint probabilities follow automatically. Using $\mathbb{P}(X = a, Y = c) = pq$ and the marginals:

$$\mathbb{P}(X = a, Y = d) = p - pq = p(1-q), \qquad \mathbb{P}(X = b, Y = c) = q - pq = q(1-p),$$

$$\mathbb{P}(X = b, Y = d) = 1 - p - q + pq = (1-p)(1-q).$$

Each of the four joint probabilities equals the product of the corresponding marginals ($\mathbb{P}(X = a)\mathbb{P}(Y = c)$, etc.), which — since X, Y take only these two values each — is exactly the definition of independence:

$$\boxed{X \text{ and } Y \text{ are independent}}.$$

□

Example 4.28 (Comparing a sum of uniforms to a scaled third uniform)**.** Let ξ, η, λ be independent Unif$[0,1]$ random variables, and let t be a fixed real number. Find $\mathbb{P}(\xi + \eta < t\lambda)$.

Solution. For $t \le 0$, $t\lambda \le 0 \le \xi + \eta$ always, so the probability is 0. Assume $t > 0$. Condition on $\lambda = l \in [0,1]$: $S = \xi + \eta$ has the standard triangular density, with CDF

$$F_S(s) = \begin{cases} 0, & s \le 0 \\ s^2/2, & 0 \le s \le 1 \\ 1 - (2-s)^2/2, & 1 \le s \le 2 \\ 1, & s \ge 2. \end{cases}$$

We want $\displaystyle\int_0^1 F_S(tl)\, dl$, split according to how large tl gets as l ranges over $[0,1]$.

Case $0 < t \le 1$**.** Then $tl \in [0,t] \subseteq [0,1]$ throughout, so

$$\int_0^1 \frac{(tl)^2}{2}\, dl = \frac{t^2}{2} \cdot \frac{1}{3} = \frac{t^2}{6}.$$

Case $1 \le t \le 2$**.** Split at $l = 1/t$ (where $tl = 1$):

$$\int_0^{1/t} \frac{(tl)^2}{2}\, dl + \int_{1/t}^1 \left[1 - \frac{(2-tl)^2}{2}\right] dl = \frac{1}{6t} + \left(1 - \frac{1}{t}\right) - \frac{1-(2-t)^3}{6t},$$

which simplifies to $1 - \dfrac{1}{t} + \dfrac{(2-t)^3}{6t}$.

Case $t \ge 2$**.** Split at $l = 1/t$ and $l = 2/t$ (where $tl = 1, 2$ respectively); the third piece contributes $\mathbb{P}(S < tl) = 1$ for $l \ge 2/t$, and the computation gives $1 - \dfrac{1}{t}$.

All three pieces agree at the boundaries $t = 1$ (both give $\frac{1}{6}$) and $t = 2$ (both give $\frac{1}{2}$), confirming continuity. So

$$\boxed{\mathbb{P}(\xi + \eta < t\lambda) = \begin{cases} 0, & t \le 0 \\ t^2/6, & 0 \le t \le 1 \\ 1 - \dfrac{1}{t} + \dfrac{(2-t)^3}{6t}, & 1 \le t \le 2 \\ 1 - \dfrac{1}{t}, & t \ge 2 \end{cases}}$$

(verified by Monte Carlo simulation at $t = 0.5, 1, 1.5, 2, 3$, matching to three decimal places). □

Example 4.29 (Probability that three independent lengths form a triangle)**.** Independent random variables X, Y have density $p_X(t) = \frac{t}{2}\mathbf{1}_{[0,2]}(t)$ and $Y \sim$ Unif$[0,3]$. Find the probability that segments of lengths X, Y, and 1 can form a triangle.

Solution. Three positive lengths form a triangle iff each is less than the sum of the other two; here that reduces to $|X - Y| < 1$ and $X + Y > 1$. The joint density is $f(x,y) = \frac{x}{2} \cdot \frac{1}{3} = \frac{x}{6}$ on $[0,2] \times [0,3]$.

Setting up the region. For fixed x, y must satisfy $\max(x-1, 1-x) < y < x+1$ (intersected with $[0,3]$, automatically satisfied here since $x+1 \le 3$).

For $x \in [0,1]$: $\max(x-1, 1-x) = 1-x$, so $y \in (1-x, x+1)$, an interval of length $2x$.

For $x \in [1,2]$: $\max(x-1, 1-x) = x-1$, so $y \in (x-1, x+1)$, an interval of length 2.

Integrating.

$$\int_0^1 \frac{x}{6} \cdot 2x\, dx + \int_1^2 \frac{x}{6} \cdot 2\, dx = \int_0^1 \frac{x^2}{3}\, dx + \int_1^2 \frac{x}{3}\, dx = \frac{1}{9} + \frac{4-1}{6} = \frac{1}{9} + \frac{1}{2}.$$

So

$$\mathbb{P}(\text{triangle}) = \frac{1}{9} + \frac{1}{2} = \boxed{\frac{11}{18}}$$

(Monte Carlo simulation over 3 million trials gives ≈ 0.6108, matching $11/18 \approx 0.6111$). □

Example 4.30 (Expected count of chosen items with no chosen neighbor)**.** Of m items arranged in a row, a uniformly random subset of $k < m$ is chosen. Let X count the chosen items i such that i is chosen but none of i's (one or two) neighbors is chosen. Find $\mathbb{E}[X]$.

Solution. By linearity of expectation, $\mathbb{E}[X] = \sum_i \mathbb{P}(i \text{ chosen, all neighbors of } i \text{ unchosen})$.

Interior items ($2 \le i \le m-1$, two neighbors): the favorable subsets fix i as chosen and both neighbors as unchosen, then choose the remaining $k-1$ elements freely from the other $m-3$ items, giving probability $\binom{m-3}{k-1} \Big/ \binom{m}{k}$.

End items ($i = 1$ or $i = m$, one neighbor): probability $\binom{m-2}{k-1} \Big/ \binom{m}{k}$.

Simplifying the ratios. Using the standard identities $\binom{n-1}{k-1}/\binom{n}{k} = k/n$ and $\binom{n-1}{r}/\binom{n}{r} = (n-r)/n$ repeatedly,

$$\frac{\binom{m-2}{k-1}}{\binom{m}{k}} = \frac{k(m-k)}{m(m-1)}, \qquad \frac{\binom{m-3}{k-1}}{\binom{m}{k}} = \frac{k(m-k)(m-k-1)}{m(m-1)(m-2)}.$$

Combining. With 2 end items and $m-2$ interior items,

$$\mathbb{E}[X] = 2 \cdot \frac{k(m-k)}{m(m-1)} + (m-2) \cdot \frac{k(m-k)(m-k-1)}{m(m-1)(m-2)} = \frac{k(m-k)}{m(m-1)}\big[2 + (m-k-1)\big],$$

so

$$\mathbb{E}[X] = \boxed{\frac{k(m-k)(m-k+1)}{m(m-1)}}$$

(checked directly: $m = 3, k = 1$ gives 1, and $m = 4, k = 2$ gives 1, both matching exhaustive enumeration over all $\binom{m}{k}$ subsets). □

Example 4.31 (Expected squared displacement of a random-turn unit polyline)**.** A planar polyline has n unit-length links; the (signed) angle between each pair of consecutive links is independently $+\alpha$ or $-\alpha$, each with probability $\frac{1}{2}$. Find the expected squared distance from the polyline's start to its end.

Solution. Let θ_k be the direction angle of link k, with $\theta_1 = 0$ (WLOG) and $\theta_{k+1} = \theta_k + \varepsilon_k \alpha$ for i.i.d. $\varepsilon_k = \pm 1$ (each with probability $\frac{1}{2}$). The endpoint displacement is $\sum_{k=1}^{n} (\cos\theta_k, \sin\theta_k)$, so

$$\mathbb{E}[\mathrm{dist}^2] = \mathbb{E}\left[\Big(\sum_k \cos\theta_k\Big)^2 + \Big(\sum_k \sin\theta_k\Big)^2\right] = \sum_{j,k=1}^{n} \mathbb{E}[\cos(\theta_k - \theta_j)].$$

The correlation decays geometrically. For $k > j$, $\theta_k - \theta_j = \alpha \sum_{i=j}^{k-1} \varepsilon_i$, a sum of $k - j$ i.i.d. $\pm\alpha$ terms, so

$$\mathbb{E}[\cos(\theta_k - \theta_j)] = \mathrm{Re}\,\mathbb{E}[e^{i\alpha \sum \varepsilon_i}] = \mathrm{Re} \prod_i \mathbb{E}[e^{i\alpha\varepsilon_i}] = (\cos\alpha)^{k-j}$$

(using $\mathbb{E}[e^{i\alpha\varepsilon}] = \frac{1}{2}(e^{i\alpha} + e^{-i\alpha}) = \cos\alpha$ for a single ± 1 sign). So $\mathbb{E}[\cos(\theta_k - \theta_j)] = c^{|k-j|}$ with $c = \cos\alpha$, for every pair j, k.

Summing the Toeplitz sum. Writing the double sum by diagonals (n terms with $j = k$, plus $2(n - d)$ terms for each offset $d = |k - j| = 1, \dots, n - 1$),

$$\mathbb{E}[\mathrm{dist}^2] = n + 2\sum_{d=1}^{n-1} (n - d)c^d.$$

Evaluating the geometric-type sum in closed form,

$$\boxed{\mathbb{E}[\mathrm{dist}^2] = \frac{n(1 + \cos\alpha)}{1 - \cos\alpha} - \frac{2\cos\alpha\,(1 - \cos^n\alpha)}{(1 - \cos\alpha)^2}}$$

(for $\alpha = 0$, i.e. $\cos\alpha = 1$, this degenerates to the straight-line answer n^2, matching the limiting case directly; verified by Monte Carlo simulation for several (n, α) pairs, matching to within sampling error). □

Example 4.32 (Expected area of the smaller circular segment)**.** Two points A, B are chosen independently and uniformly on a circle of radius r. Find the expected area of the smaller of the two segments the chord AB cuts from the disk.

Solution. Fix A (by rotational symmetry, WLOG) and let B's angular position be Unif$[0, 2\pi)$. Let θ be the central angle of the smaller arc AB: writing φ for B's angle, $\theta = \min(\varphi, 2\pi - \varphi) \in [0, \pi]$, and one checks directly that θ is uniform on $[0, \pi]$ (for $t \in [0, \pi]$, $\mathbb{P}(\theta \le t) = \mathbb{P}(\varphi \le t) + \mathbb{P}(\varphi \ge 2\pi - t) = t/\pi$).

Segment area as a function of θ. The minor circular segment for central angle θ has area

$$S(\theta) = \frac{r^2}{2}(\theta - \sin\theta).$$

Averaging.

$$\mathbb{E}[S(\theta)] = \frac{1}{\pi}\int_0^\pi \frac{r^2}{2}(\theta - \sin\theta)\,d\theta = \frac{r^2}{2\pi}\left[\frac{\pi^2}{2} - \Big(-\cos\theta\Big|_0^\pi\Big)\right] = \frac{r^2}{2\pi}\left(\frac{\pi^2}{2} - 2\right).$$

Simplifying,

$$\boxed{\mathbb{E}[\mathrm{area}] = r^2\left(\frac{\pi}{4} - \frac{1}{\pi}\right)}$$

(for $r = 1$ this is ≈ 0.4671, matching Monte Carlo simulation over 3 million trials to 3 decimal places). □

Example 4.33 (Expected number of transpositions in a random permutation)**.** For a uniformly random permutation $P = (p_1, \dots, p_n)$ of $\{1, \dots, n\}$, call a pair $\{i, j\}$ $(i \ne j)$ a *swap* if $p_i = j$ and $p_j = i$ (a 2-cycle). Find the expected number of swaps.

Solution. By linearity of expectation, summing over the $\binom{n}{2}$ unordered pairs $\{i, j\}$: for fixed $i \ne j$, $\mathbb{P}(p_i = j,\, p_j = i)$ fixes 2 of the n values, leaving the remaining $n - 2$ to permute freely among themselves, so

$$\mathbb{P}(p_i = j,\, p_j = i) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}.$$

Summing over all $\binom{n}{2} = \dfrac{n(n-1)}{2}$ pairs,

$$\mathbb{E}[\#\text{swaps}] = \binom{n}{2} \cdot \frac{1}{n(n-1)} = \frac{n(n-1)}{2} \cdot \frac{1}{n(n-1)} = \boxed{\frac{1}{2}}$$

— independent of n (for $n \geq 2$). ☐

Example 4.34 (Expected length of the smallest piece of a twice-broken stick)**.** A unit segment $[0, 1]$ is cut at two independent uniform random points into three pieces. Find the expected length of the smallest piece.

Solution. Let U, V be iid Uniform$(0, 1)$, and let $X = \min(U, V)$, $Y = \max(U, V)$, so the three piece lengths are $A = X$, $B = Y - X$, $C = 1 - Y$ (with $A + B + C = 1$). The order statistics (X, Y) have joint density $f(x, y) = 2$ on $\{0 < x < y < 1\}$.

Tail probability of the minimum piece. For $0 \leq t \leq \frac{1}{3}$,

$$\mathbb{P}(A > t, B > t, C > t) = \mathbb{P}(X > t,\ Y > X + t,\ Y < 1 - t) = \int_t^{1-2t} \int_{x+t}^{1-t} 2\, dy\, dx.$$

The inner integral is $2[(1 - t) - (x + t)] = 2(1 - 2t - x)$. Substituting $w = x - t$ (so w ranges over $[0, 1 - 3t]$) turns the outer integral into

$$\int_0^{1-3t} 2[(1 - 3t) - w]\, dw = 2 \cdot \frac{(1 - 3t)^2}{2} = (1 - 3t)^2.$$

So $\mathbb{P}(\min(A, B, C) > t) = (1 - 3t)^2$ for $0 \leq t \leq \frac{1}{3}$ (and 0 beyond).

Expectation via the tail formula. For a non-negative random variable, $\mathbb{E}[Z] = \int_0^\infty \mathbb{P}(Z > t)\, dt$, so

$$\mathbb{E}[\min(A, B, C)] = \int_0^{1/3} (1 - 3t)^2\, dt = \left[-\frac{(1 - 3t)^3}{9} \right]_0^{1/3} = 0 - \left(-\frac{1}{9} \right) = \boxed{\frac{1}{9}}.$$

☐

Example 4.35 (Expected number of rolls of an n-sided die until the running sum reaches n)**.** An n-sided die (faces $1, \ldots, n$, equally likely) is rolled repeatedly, and the results are summed, until the running sum first reaches or exceeds n. Find the expected number of rolls.

Solution. Let $S_k = X_1 + \cdots + X_k$ be the running sum after k i.i.d. Uniform$\{1, \ldots, n\}$ rolls, and $T = \min\{k : S_k \geq n\}$. Using the standard tail-sum formula for a non-negative integer random variable, $\mathbb{E}[T] = \sum_{k=0}^\infty \mathbb{P}(T > k) = \sum_{k=0}^\infty \mathbb{P}(S_k < n)$ (with $S_0 = 0$).

Counting the event $S_k < n$**.** Substitute $Y_i = X_i - 1 \in \{0, \ldots, n - 1\}$, so $S_k < n \iff Y_1 + \cdots + Y_k \leq n - 1 - k$. For $0 \leq k \leq n - 1$, the target $M = n - 1 - k$ satisfies $0 \leq M \leq n - 1$, and since each $Y_i \geq 0$, a sum $\leq M$ automatically forces every individual $Y_i \leq M \leq n - 1$ — so the upper cap $Y_i \leq n - 1$ never actually binds, and we're just counting non-negative integer solutions of $Y_1 + \cdots + Y_k \leq M$. Introducing a slack variable turns this into $Y_1 + \cdots + Y_k + Y_{k+1} = M$ with $k + 1$ non-negative unknowns, a standard stars-and-bars count of $\binom{M+k}{k} = \binom{n-1}{k}$ solutions (since $M + k = n - 1$). Each solution corresponds to exactly one of the n^k equally likely roll sequences, so

$$\mathbb{P}(S_k < n) = \frac{\binom{n-1}{k}}{n^k} \qquad (0 \leq k \leq n - 1),$$

and $\mathbb{P}(S_k < n) = 0$ for $k \geq n$ (the sum of n rolls, each ≥ 1, is always $\geq n$).

Summing.

$$\mathbb{E}[T] = \sum_{k=0}^{n-1} \binom{n-1}{k} \left(\frac{1}{n}\right)^k = \sum_{k=0}^{n-1} \binom{n-1}{k} \left(\frac{1}{n}\right)^k 1^{n-1-k} = \left(1 + \frac{1}{n}\right)^{n-1}$$

by the binomial theorem. So

$$\boxed{\mathbb{E}[T] = \left(1 + \frac{1}{n}\right)^{n-1}}.$$

(Check: $n = 2$ gives $\mathbb{E}[T] = 1.5$, matching a direct hand computation — the die shows 2 immediately with probability $\frac{1}{2}$ [one roll], or 1 then anything [two rolls] with probability $\frac{1}{2}$.) □

4.3 Inequalities, Modes of Convergence, and Characteristic Functions

Definitions and formulas.

- Markov's inequality for nonnegative X:

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

- Chebyshev's inequality:

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

- Jensen's inequality: if φ is convex, then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

- Borel–Cantelli: if $\sum_n \mathbb{P}(A_n) < \infty$, then only finitely many A_n occur almost surely.
- Common convergence notions:
  - almost sure: $X_n \to X$ a.s.;
  - in probability: $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$;
  - in L^p: $\mathbb{E}[|X_n - X|^p] \to 0$;
  - in distribution: $F_{X_n}(x) \to F_X(x)$ at continuity points of F_X.
- The weak law says sample means converge in probability; the strong law says they converge almost surely; the central limit theorem describes the *fluctuation scale* around the law-of-large-numbers limit. For i.i.d. X_i with mean μ and variance $\sigma^2 < \infty$,

$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{\sigma} \Rightarrow \mathcal{N}(0, 1).$$

  Equivalently, for large n,

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right), \qquad S_n = X_1 + \cdots + X_n \approx \mathcal{N}(n\mu, n\sigma^2).$$

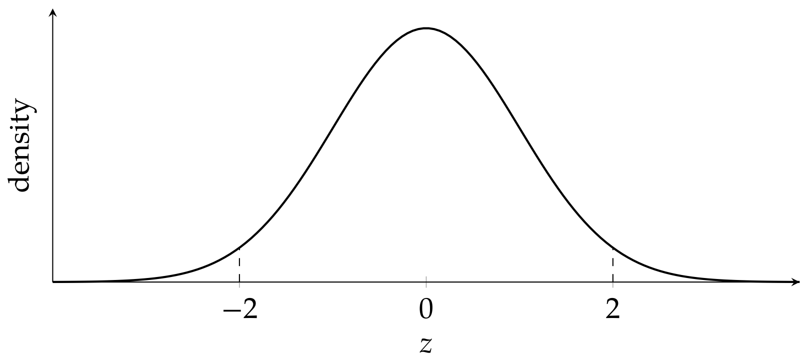
  For lattice variables such as binomial counts, a continuity correction often improves the approximation.

- The characteristic function of X is

$$\varphi_X(t) = \mathbb{E}[e^{itX}].$$

For independent X, Y, $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$.

Standard normal reference curve. CLT approximations are usually read off after standardizing to the bell curve below.



Example 4.36 (Chebyshev bound for a sample mean)**.** Let $X_1, \dots, X_{400}$ be i.i.d. with mean 0 and variance 1. Give an upper bound on

$$\mathbb{P}\left(\left|\frac{1}{400}\sum_{i=1}^{400} X_i\right| > 0.1\right)$$

using Chebyshev's inequality.

Solution. Let $\bar{X}_{400} = \frac{1}{400}\sum_{i=1}^{400} X_i$. Then

$$\mathbb{E}[\bar{X}_{400}] = 0, \qquad \mathrm{Var}(\bar{X}_{400}) = \frac{1}{400^2} \cdot 400 = \frac{1}{400}.$$

By Chebyshev,

$$\mathbb{P}(|\bar{X}_{400}| > 0.1) \le \frac{\mathrm{Var}(\bar{X}_{400})}{0.1^2} = \frac{1/400}{0.01} = \frac{1}{4}.$$

Hence

$$\boxed{\mathbb{P}(|\bar{X}_{400}| > 0.1) \le \frac{1}{4}}.$$

This is conservative but completely distribution-free. □

Example 4.37 (Central limit approximation to a binomial count)**.** A fair coin is tossed 400 times. Estimate

$$\mathbb{P}(180 \le X \le 220),$$

where X is the number of heads.

Solution. Here

$$X \sim \mathrm{Bin}(400, 1/2), \qquad \mu = np = 200, \qquad \sigma = \sqrt{np(1-p)} = 10.$$

By the central limit theorem, X is well approximated by a normal variable with the same mean and variance. Using a continuity correction,

$$\mathbb{P}(180 \le X \le 220) \approx \mathbb{P}(179.5 \le Y \le 220.5),$$

where $Y \sim \mathcal{N}(200, 100)$.

Standardize:

$$\mathbb{P}(179.5 \le Y \le 220.5) = \mathbb{P}\left(\frac{179.5 - 200}{10} \le Z \le \frac{220.5 - 200}{10}\right) = \mathbb{P}(-2.05 \le Z \le 2.05),$$

where $Z \sim \mathcal{N}(0, 1)$. Therefore

$$\mathbb{P}(180 \le X \le 220) \approx 2\Phi(2.05) - 1 \approx 0.9596.$$

So the CLT estimate is

$$\boxed{\mathbb{P}(180 \le X \le 220) \approx 0.960.}$$

□

Example 4.38 (A random increment process with shrinking step size)**.** Assume $x_3 > 0$ is fixed. For each $t \ge 3$, the process adds the current increment x_t to V, and then independently updates the next increment by

$$x_{t+1} = \begin{cases} x_t/2, & \text{with probability } 2/t, \\ x_t, & \text{with probability } 1 - 2/t. \end{cases}$$

What is the probability that $V_t \to \infty$ as $t \to \infty$?

Solution. Let I_t be the indicator that a halving occurs at time t, and let

$$N_n = \sum_{t=3}^{n} I_t$$

be the total number of halvings up to time n. Then

$$x_n = x_3\, 2^{-N_{n-1}}.$$

Since the halving decisions are independent and $\mathbb{P}(I_t = 1) = 2/t$,

$$\mathbb{E}[N_n] = \sum_{t=3}^{n} \frac{2}{t} = 2\log n + O(1).$$

By the strong law for sums of independent Bernoulli variables with varying parameters,

$$N_n = 2\log n + o(\log n) \qquad \text{almost surely.}$$

Therefore

$$x_n = x_3 \exp(-(\log 2) N_{n-1}) = n^{-2\log 2 + o(1)} \qquad \text{almost surely.}$$

Now

$$2\log 2 \approx 1.386 > 1,$$

so almost surely the series $\sum_n x_n$ behaves like a p-series with exponent strictly larger than 1. Hence

$$\sum_{n=3}^{\infty} x_n < \infty \qquad \text{almost surely.}$$

But V_t is obtained by summing these increments, so V_t converges to a finite random limit almost surely. Consequently,

$$\boxed{\mathbb{P}(V_t \to \infty) = 0}.$$

This is a useful warning example: even though the expected number of halvings by time n is only logarithmic, that is still enough to make the typical increment decay faster than $1/n$, so the cumulative growth stays finite almost surely. □

Example 4.39 (A negative-binomial tail sum via the CLT)**.** Find

$$\lim_{n\to\infty}\sum_{k=n}^{5n}\binom{k-1}{n-1}\left(\frac{1}{5}\right)^n\left(\frac{4}{5}\right)^{k-n}.$$

Solution. The summand $\binom{k-1}{n-1}p^n(1-p)^{k-n}$ (with $p=\frac{1}{5}$) is exactly $\mathbb{P}(N=k)$ where N is the trial number of the n-th success in i.i.d. Bernoulli(p) trials: the n-th success lands on trial k iff exactly $n-1$ of the first $k-1$ trials succeed (probability $\binom{k-1}{n-1}p^{n-1}(1-p)^{k-n}$) and trial k itself succeeds (probability p). So the sum equals $\mathbb{P}(n\le N\le 5n)$ for $N\sim$ NegBinomial(n,p).

Write $N=\sum_{i=1}^n G_i$ where G_i are i.i.d. Geometric(p) (number of trials to get the i-th success after the $(i-1)$-th), each with $\mathbb{E}[G_i]=1/p=5$ and $\mathrm{Var}(G_i)=(1-p)/p^2=20$. So $\mathbb{E}[N]=5n$ and $\mathrm{Var}(N)=20n$, giving standard deviation $\sqrt{20n}=\Theta(\sqrt{n})$.

Lower tail. n is $4n$ below the mean $5n$, which is $\Theta(n)\gg\Theta(\sqrt{n})$, so by Chebyshev,

$$\mathbb{P}(N<n)\le\mathbb{P}(|N-5n|>4n)\le\frac{\mathrm{Var}(N)}{(4n)^2}=\frac{20n}{16n^2}=\frac{5}{4n}\xrightarrow[n\to\infty]{}0.$$

Upper tail at the mean. By the Central Limit Theorem, $\dfrac{N-5n}{\sqrt{20n}}$ converges in distribution to a standard normal, so

$$\mathbb{P}(N\le 5n)=\mathbb{P}\left(\frac{N-5n}{\sqrt{20n}}\le 0\right)\xrightarrow[n\to\infty]{}\Phi(0)=\frac{1}{2}.$$

Combining, $\mathbb{P}(n\le N\le 5n)=\mathbb{P}(N\le 5n)-\mathbb{P}(N<n)\to\frac{1}{2}-0$, so

$$\boxed{\frac{1}{2}}.$$

(Direct numerical summation confirms the convergence: the partial sum is about 0.526 at $n=50$, 0.513 at $n=200$, and 0.501 at $n=20{,}000$.) □

Example 4.40 (A Gaussian tail series with a log-log correction)**.** Let $\xi_1,\xi_2,\dots$ each have the standard normal distribution. Does $\sum_{n=1}^{\infty}\mathbb{P}\left(\xi_n>\sqrt{2\ln n+2\ln\ln n}\right)$ converge?

Solution. Use the standard Gaussian tail bound $\mathbb{P}(\xi>x)\le\varphi(x)/x$ for $x>0$, where $\varphi(x)=\frac{1}{\sqrt{2\pi}}e^{-x^2/2}$. Let $x_n=\sqrt{2\ln n+2\ln\ln n}$ (defined for $n\ge 3$, where $\ln\ln n\ge 0$). Then

$$x_n^2=2\ln n+2\ln\ln n\quad\Longrightarrow\quad e^{-x_n^2/2}=e^{-\ln n-\ln\ln n}=\frac{1}{n\ln n}.$$

Also $x_n\ge\sqrt{2\ln n}$ (dropping the nonnegative $2\ln\ln n$ term), so

$$\mathbb{P}(\xi_n>x_n)\le\frac{1}{\sqrt{2\pi}\,x_n}\cdot\frac{1}{n\ln n}\le\frac{1}{\sqrt{2\pi}\cdot\sqrt{2\ln n}\cdot n\ln n}=\frac{1}{2\sqrt{\pi}}\cdot\frac{1}{n(\ln n)^{3/2}}.$$

The series $\sum_{n\ge 3}\dfrac{1}{n(\ln n)^{3/2}}$ converges (the standard $\sum 1/(n(\ln n)^p)$ test, e.g. via the integral test on $\int_3^\infty dx/(x(\ln x)^p)$, converges exactly when $p>1$, and here $p=3/2>1$). By comparison, so does the original series (the finitely many small-n terms don't affect convergence):

$$\boxed{\text{yes, the series converges}}$$

(numerically, the partial sums stabilize quickly: about 0.548 by $n=10^6$ and 0.551 by $n\approx 2\times 10^6$, consistent with a finite limit). □

4.4 Markov Chains and First-Step Analysis

Definitions and formulas.

- A Markov chain has transition matrix $P = (p_{ij})$, where

$$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

  Each row of P sums to 1. The key modeling step is to choose the state so that, once the current state is known, the future no longer depends on the earlier history. In interview problems, the hard part is usually not matrix algebra but selecting the right state description.

- A distribution π is stationary if $\pi P = \pi$. This means that if the chain starts with distribution π, then after one step it still has distribution π. For ergodic finite chains, this is the long-run equilibrium law.

- In a finite irreducible and aperiodic chain, the stationary distribution is unique and the chain converges to it regardless of the starting state. Irreducible means every state can eventually reach every other state, and aperiodic means the chain is not trapped in a rigid cycle length.

- Hitting probabilities are handled by the same one-step idea. If

$$q_i = \mathbb{P}_i(\tau_A < \tau_B),$$

  then the boundary conditions are

$$q_i = 1 \text{ for } i \in A, \qquad q_i = 0 \text{ for } i \in B,$$

  and for all other states,

$$q_i = \sum_j p_{ij} q_j.$$

  There is no extra 1 here because we are computing a probability, not a time. We simply average the future success probability over the next state.

- Expected hitting times are often solved by *first-step analysis*: condition on the next move and write a recursion.

- If τ_A is the first hitting time of a set A, then the expected hitting times satisfy

$$h_i = 0 \text{ for } i \in A, \qquad h_i = 1 + \sum_j p_{ij} h_j \text{ for } i \notin A.$$

  The extra 1 is the step you take immediately before landing in the next state.

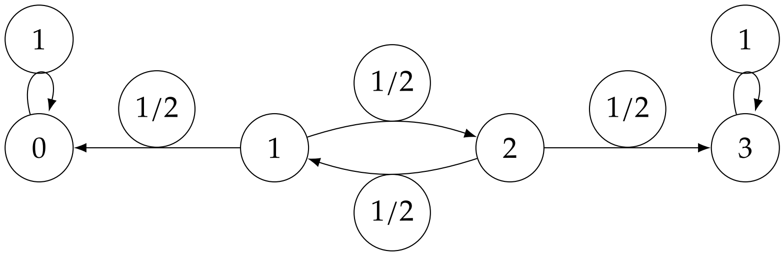
- Absorbing-state problems reduce to linear systems. Once you assign one variable to each transient state, the recursion from first-step analysis produces one linear equation per unknown.

- For an absorbing Markov chain with block form

$$P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix},$$

  the matrix Q describes transitions among transient states, R describes jumps from transient states to absorbing states, and the fundamental matrix is $N = (I - Q)^{-1}$. The entry N_{ij} is the expected number of visits to transient state j when starting from transient state i, before absorption occurs. Summing across a row gives the expected absorption time, so $N\mathbf{1}$ gives expected absorption times from transient states.

Transition-graph picture. The matrix and the directed graph contain the same information: draw one node per state and an arrow $i \to j$ labeled by p_{ij} whenever $p_{ij} > 0$. For the random walk example below, the chain can be visualized as follows.



The absorbing states 0 and 3 have self-loops of probability 1. States 1 and 2 are transient because, once the walk enters 0 or 3, it never returns.

Practical workflow.

1. Choose states that retain exactly the information that still matters for the future.
2. Write one equation per nonterminal state by conditioning on the next step.
3. Put boundary values on terminal or absorbing states.
4. Solve the resulting linear system.
5. Read off the quantity corresponding to the actual starting state.

In interview practice there are three recurring templates:

- If you are asked for a probability of eventually hitting one target before another, define q_i and use $q_i = \sum_j p_{ij} q_j$.
- If you are asked for an expected waiting time, define h_i and use $h_i = 1 + \sum_j p_{ij} h_j$.
- If the process only depends on a small memory of the past, compress the past into a few states. The two-sixes example below is exactly of this type.

Example 4.41 (Expected rolls until two consecutive sixes)**.** You roll a fair six-sided die repeatedly. What is the expected number of rolls until two consecutive sixes appear?

Solution. Use two states, chosen so that the future depends only on whether we are currently carrying a trailing 6:

- E_0: expected additional rolls when we do *not* currently have a trailing 6;
- E_1: expected additional rolls when the previous roll was a 6, so one more 6 finishes the game.

At the start of the experiment no roll has happened yet, so we are in state E_0, not E_1. That is why the answer to the problem will be E_0.

This is the standard state-compression trick: the whole past sequence of rolls is irrelevant except for one bit of information, namely whether the most recent roll was a 6.

From state E_0, one roll occurs, and then with probability 5/6 we remain in E_0, while with probability 1/6 we move to E_1:

$$E_0 = 1 + \frac{5}{6}E_0 + \frac{1}{6}E_1.$$

From state E_1, one roll occurs. With probability 1/6 we finish, and with probability 5/6 we return to E_0:

$$E_1 = 1 + \frac{5}{6}E_0.$$

The first equation simplifies to

$$E_0 = 6 + E_1.$$

Substitute into the second:

$$E_1 = 1 + \frac{5}{6}(6 + E_1) = 6 + \frac{5}{6}E_1.$$

Thus

$$\frac{1}{6}E_1 = 6 \quad \Longrightarrow \quad E_1 = 36,$$

and therefore

$$E_0 = 6 + 36 = 42.$$

Because the chain starts in state E_0, the expected number of rolls asked for in the problem is

$$\boxed{42}.$$

The structure is worth remembering: one equation per nonterminal state, one immediate step cost, and then an average over the next states. □

Example 4.42 (Absorption on a short random walk)**.** A particle moves on the states 0, 1, 2, 3. States 0 and 3 are absorbing. From states 1 and 2, the particle moves left or right with probability 1/2 each. If the chain starts at state 1, find the probability of being absorbed at 3 and the expected time to absorption.

Solution. Let p_i be the probability, starting from state i, of hitting 3 before 0. Then

$$p_0 = 0, \qquad p_3 = 1,$$

and by first-step analysis,

$$p_1 = \frac{p_0 + p_2}{2} = \frac{p_2}{2}, \qquad p_2 = \frac{p_1 + p_3}{2} = \frac{p_1 + 1}{2}.$$

Substituting the first equation into the second gives

$$p_2 = \frac{p_2/2 + 1}{2},$$

so $4p_2 = p_2 + 2$, hence $p_2 = \frac{2}{3}$ and therefore

$$p_1 = \frac{1}{3}.$$

Thus the probability of absorption at 3 when starting from 1 is

$$\boxed{\frac{1}{3}}.$$

This is the hitting-probability template in its purest form: no extra 1 appears because we are not counting time, only eventual success.

Now let e_i be the expected time to absorption starting from state i. Then

$$e_0 = e_3 = 0,$$

and

$$e_1 = 1 + \frac{e_0 + e_2}{2} = 1 + \frac{e_2}{2}, \qquad e_2 = 1 + \frac{e_1 + e_3}{2} = 1 + \frac{e_1}{2}.$$

Substitute the second equation into the first:

$$e_1 = 1 + \frac{1 + e_1/2}{2} = \frac{3}{2} + \frac{e_1}{4}.$$

Hence

$$\frac{3e_1}{4} = \frac{3}{2}, \qquad e_1 = 2.$$

So the expected time to absorption is

$$\boxed{2}.$$

Here the extra 1 appears because one step is spent immediately, regardless of which direction the walk takes next. $\square$

Example 4.43 (Expected hitting time in a small portal maze)**.** A maze has 6 rooms connected by two-way portals as follows: room 1 connects to $2,3$; room 2 connects to $1,3,4,6$; room 3 connects to $1,2,5,6$; room 4 connects to $2,5,6$; room 5 connects to $3,4,6$; room 6 connects to $2,3,4,5$. Starting in room 1, at each step you jump through a uniformly random portal from your current room (including possibly back the way you came). Find the expected number of jumps to first reach room 6.

Solution. Let h_i be the expected number of jumps to reach room 6 starting from room i ($h_6 = 0$). First-step analysis gives, for each room $i \neq 6$ with degree d_i and neighbors $N(i)$,

$$h_i = 1 + \frac{1}{d_i} \sum_{j \in N(i)} h_j.$$

The degrees here are $d_1 = 2$, $d_2 = d_3 = d_6 = 4$, $d_4 = d_5 = 3$, giving the system

$$\begin{aligned} h_1 &= 1 + \tfrac{1}{2}(h_2 + h_3), \\ h_2 &= 1 + \tfrac{1}{4}(h_1 + h_3 + h_4), \\ h_3 &= 1 + \tfrac{1}{4}(h_1 + h_2 + h_5), \\ h_4 &= 1 + \tfrac{1}{3}(h_2 + h_5), \\ h_5 &= 1 + \tfrac{1}{3}(h_3 + h_4). \end{aligned}$$

By the left-right symmetry of the maze (swapping $2 \leftrightarrow 3$, $4 \leftrightarrow 5$ preserves all the adjacencies), $h_2 = h_3$ and $h_4 = h_5$; substituting collapses the system to two equations in h_2, h_4 (with $h_1 = 1 + h_2$ from the first equation):

$$h_2 = 1 + \tfrac{1}{4}((1 + h_2) + h_2 + h_4), \qquad h_4 = 1 + \tfrac{1}{3}(h_2 + h_4).$$

The second equation gives $\frac{2}{3}h_4 = 1 + \frac{1}{3}h_2$, i.e. $h_4 = \frac{3}{2} + \frac{1}{2}h_2$. Substituting into the first: $h_2 = 1 + \frac{1}{4}(1 + 2h_2 + \frac{3}{2} + \frac{1}{2}h_2) = 1 + \frac{1}{4}(\frac{5}{2} + \frac{5}{2}h_2) = \frac{13}{8} + \frac{5}{8}h_2$, so $\frac{3}{8}h_2 = \frac{13}{8}$, giving $h_2 = \frac{13}{3}$, hence $h_4 = \frac{3}{2} + \frac{1}{6} \cdot 13 = \frac{11}{3}$ and

$$h_1 = 1 + h_2 = 1 + \frac{13}{3} = \boxed{\frac{16}{3}}.$$

(A direct Monte Carlo simulation of the random walk confirms this: the sample mean over 2 million runs is about 5.337, matching $16/3 \approx 5.333$.) $\square$

Example 4.44 (Color imbalance of a king's random walk on a chessboard)**.** A robot starts on an infinite chessboard and, at each step, moves to one of its 8 neighboring squares (king moves), chosen uniformly at random. After n steps (so $n + 1$ squares have been visited, counted with multiplicity), find $\mathbb{E}[\,|\#\text{black visited} - \#\text{white visited}|\,]$, as a compact expression.

Solution. **Reduction to a sign process.** Color a square by $(row + col) \bmod 2$. Of the 8 king moves, the 4 orthogonal moves $(\pm1, 0$ or $0, \pm1)$ flip $(row + col)$'s parity, while the 4 diagonal moves $(\pm1, \pm1)$ preserve it. So, writing $X_k = \pm1$ for the color of the square visited at step k (some fixed sign convention), *regardless of position on the board*, $X_{k+1} = X_k$ with probability $\frac{1}{2}$ and $X_{k+1} = -X_k$ with probability $\frac{1}{2}$, independent of the past. So $X_k = X_0\eta_1\eta_2\cdots\eta_k$ for i.i.d. signs $\eta_i = \pm1$ (each uniform), and we want $\mathbb{E}|T_n|$ for $T_n = \sum_{k=0}^{n} X_k$; since $|X_0| = 1$, WLOG $X_0 = 1$.

A cleaner recursive form. By exchangeability of the i.i.d. η_i, $T_n = 1 + \eta_1(1 + \eta_2(1 + \cdots(1 + \eta_n)\cdots))$ has the same distribution as W_n defined by $W_0 = 1$, $W_j = 1 + \eta_j' W_{j-1}$ for fresh i.i.d. signs η_j' (just relabeling which η is used first). This is a genuine Markov chain: $W_j = 1 + W_{j-1}$ or $W_j = 1 - W_{j-1}$, each with probability $\frac{1}{2}$.

Reflected random walk in disguise. Since $|\eta_j'| = 1$, $|W_j - 1| = |W_{j-1}|$ always. Writing $M_j = |W_j|$, one checks $M_j = M_{j-1} + 1$ or $M_j = |1 - M_{j-1}|$ each with probability $\frac{1}{2}$ — i.e. if $M_{j-1} \geq 1$, $M_j = M_{j-1} \pm 1$ (a standard simple-random-walk step), while $M_{j-1} = 0$ always bounces to $M_j = 1$. This is exactly the classical *reflected* simple random walk started at 1, which by the reflection principle has the same law as $|1 + S_n|$ for a free simple random walk S_n (steps ±1, $S_0 = 0$): both give matching distributions at every step (verified directly for small n, and consistent with the reflection principle for symmetric ±1 walks).

Closed form. Writing $S_n = 2\mathrm{Bin}(n, \frac{1}{2}) - n$, exact computation (confirmed by dynamic programming over the Markov chain above, matching a direct brute-force check for $n \leq 15$ and the recursion for n up to 50) gives, with $m = \lfloor n/2 \rfloor$,

$$\mathbb{E}|T_n| = \mathbb{E}|1 + S_n| = \boxed{(2m+1)\binom{2m}{m}\Big/4^m}$$

(the value is the same for $n = 2m$ and $n = 2m + 1$; e.g. $n = 0, 1 \mapsto 1$, $n = 2, 3 \mapsto \frac{3}{2}$, $n = 4, 5 \mapsto \frac{15}{8}$, growing like $\sqrt{2n/\pi}$ for large n by the de Moivre–Laplace approximation to the central binomial coefficient). $\square$

5 Algorithm Design

Quant interviews frequently include short algorithmic problems that reward recognizing a classical pattern — a monotonic stack, a sliding window, a divide-and-conquer trick — over brute force.

Example 5.1 (Nearest right element at least double, in $O(n \log n)$)**.** Given a real array $A[1..n]$, design an algorithm that, for every index i, finds the smallest index $j > i$ with $A[j] \geq 2A[i]$ (or reports that none exists), using $O(n \log n)$ time and $O(n)$ extra space.

Solution. **A domination principle.** Scan the array from right to left, maintaining a stack of surviving candidates. Say index j_2 is *dominated* by a nearer index $j_1 < j_2$ if $A[j_1] \geq A[j_2]$. A dominated candidate is useless: for any future (further left) query i, whenever $A[j_2] \geq 2A[i]$ we also have $A[j_1] \geq A[j_2] \geq 2A[i]$, and j_1 is strictly closer, so j_2 can never be the nearest qualifying index once j_1 exists. Notably, this domination argument does not depend on the multiplier 2; it only uses $A[j_1] \geq A[j_2]$.

Maintaining the stack. Process $i = n, n-1, \ldots, 1$. Before pushing i, pop every index currently on top of the stack with value $\leq A[i]$ (those are now dominated by i), then push i. Each index is pushed and popped at most once, so stack maintenance costs $O(n)$ total.

Why the stack is sorted. Whenever an element is pushed, everything weaker has just been popped, so values strictly increase from the bottom of the stack toward the top — equivalently, reading from the top (nearest to the current scan position) down to the bottom (farthest), values are non-decreasing. So at every moment the live stack is a sequence sorted by value, in which position along the stack corresponds monotonically to distance from the current pointer.

Answering a query. For index i we want the surviving candidate closest to the top with value $\geq 2A[i]$. Because the stack is sorted by value from top (smallest) to bottom (largest), the qualifying candidates form a contiguous block at the bottom, and the nearest one is found by *binary search* on the stack (kept as an array) for the threshold $2A[i]$: $O(\log n)$ per query.

Complexity. Maintenance is $O(n)$ amortized; each of the n queries costs $O(\log n)$ via binary search on the array-backed stack; total time is $O(n \log n)$, and the stack never holds more than n indices, so space is $O(n)$ — matching the required bounds exactly. (A plain monotonic stack alone, without the binary search, solves the classical "next strictly greater element" problem in $O(n)$; the extra $\log n$ factor here is exactly the cost of the per-query threshold search that the "at least double" condition forces.) □

Example 5.2 (Sorting a quadratic image of a sorted array, in linear time)**.** Given a real array $A[1..n]$ sorted in increasing order and real numbers p, q, r, produce an array $B[1..n]$ consisting of the values $px^2 + qx + r$ for $x \in A$, also sorted in increasing order, using $O(n)$ time and $O(n)$ extra space.

Solution. If $p = 0, f(x) = qx + r$ is monotonic: copy A through f in order if $q \geq 0$, or in reverse order if $q < 0$ — $O(n)$.

If $p \neq 0, f(x) = px^2 + qx + r$ is a parabola with vertex at $x_0 = -q/(2p)$, monotonic on each side of x_0. Since A is sorted, binary search for x_0 in A in $O(\log n)$ to split it into A_{left} (entries $< x_0$) and A_{right} (entries $\geq x_0$). On one side of x_0, f is increasing as x increases; on the other, decreasing — which is which depends on the sign of p:

- $p > 0$ (upward parabola): f decreases toward x_0 from the left and increases away from it on the right. So $f(A_{\text{left}})$ read in *reverse* order is increasing, and $f(A_{\text{right}})$ read in its original order is already increasing.
- $p < 0$ (downward parabola): the roles swap — $f(A_{\text{left}})$ in its original order is increasing, and $f(A_{\text{right}})$ read in reverse is increasing.

Either way, this produces two increasing sequences of total length n in $O(n)$ time. Merge them with the standard two-pointer merge step from merge sort — compare the fronts of each sequence, output the smaller, advance that pointer — in $O(n)$ time and $O(n)$ auxiliary space for the output. The result B is A's image under f, fully sorted, using $O(n)$ time (one binary search plus two linear passes) and $O(n)$ space, as required. □

Example 5.3 (Largest multiple of 3 from an array of digits, in $O(n)$ time and $O(1)$ space)**.** Given an array $A[1..n]$ of digits 0–9, design an algorithm that finds the largest natural number divisible by 3 that can be formed from (a subset of) the digits of A, using $O(n)$ time and $O(1)$ extra space.

Solution. A number is divisible by 3 iff its digit sum is; and for a fixed multiset of digits, the largest number they form is obtained by arranging them in decreasing order. So the problem reduces to: keep a multiset of digits with digit-sum divisible by 3, discarding as few (and as small) digits as possible, then output largest-to-smallest.

Counting pass. Scan A once, maintaining a count array cnt[0..9] (fixed size 10, so $O(1)$ space) and the running digit sum; this is $O(n)$ time. Let $r = (\text{digit sum}) \bmod 3$.

Fixing the residue. If $r = 0$, keep every digit. If $r = 1$, either discard one digit $\equiv 1 \pmod 3$ (i.e. a $1, 4$, or 7) — choosing the *smallest* such digit value present, to sacrifice as little as possible — or, if no digit $\equiv 1 \pmod 3$ is present at all, discard two digits $\equiv 2 \pmod 3$ (i.e. from $2, 5, 8$; since $2 + 2 \equiv 1$), again preferring the smallest available. The case $r = 2$ is symmetric: discard one digit $\equiv 2 \pmod 3$, or if none exists, two digits $\equiv 1 \pmod 3$. Each of these adjustments only touches the 10-entry count array, so it costs $O(1)$ time regardless of n.

Output. If every digit was discarded (or only zeros remain), the answer is 0. Otherwise emit cnt[9] nines, then cnt[8] eights, ..., down to cnt[0] zeros — a single $O(n)$ pass that reproduces the largest arrangement of the surviving digits.

Total cost: one $O(n)$ counting pass, $O(1)$ work to decide what to discard, and one $O(n)$ output pass — $O(n)$ time, $O(1)$ extra space (the count array has fixed size 10, independent of n). □

Example 5.4 (Detecting a k-cycle in a permutation given as two arrays)**.** A permutation σ of $\{1, \dots, n\}$ is given by two arrays $a[1..n]$, $b[1..n]$, each containing every value $1, \dots, n$ exactly once, with $b[i] = \sigma(a[i])$ for every i (so σ is defined only implicitly — there is no array with $\sigma(i)$ at position i). Design an algorithm that decides whether σ has a cycle of length exactly k (a fixed constant), using $O(n^2)$ operations and $O(1)$ extra space; the arrays may be modified.

Solution. The obstacle is that $\sigma(x)$ isn't directly indexable: to find it we must locate x in array a and read off the matching entry of b.

Step 1: put σ into direct form, in place. For $i = 1, \dots, n$: scan $a[i..n]$ for the position j with $a[j] = i$ (a linear scan, $O(n)$ per i), then swap $(a[i], b[i]) \leftrightarrow (a[j], b[j])$ (swapping both arrays' entries together, keeping the pairing $b[\cdot] = \sigma(a[\cdot])$ intact). After this pass, $a[i] = i$ for every i, so $b[i] = \sigma(i)$ directly: b is now σ in standard array form. This costs $O(n)$ work per index, $O(n^2)$ total, and only a constant number of temporary variables for the swaps — $O(1)$ extra space.

Step 2: check for a k-cycle. For each starting point $s = 1, \dots, n$, trace s, $b[s]$, $b[b[s]], \dots$ for exactly k steps, checking along the way that it does not return to s before step k (to rule out a shorter cycle whose length merely divides k) and that it *does* return to s at step k exactly. Since k is a constant, each trace costs $O(k) = O(1)$, and there are n starting points, so this phase is $O(n)$.

The dominant cost is the $O(n^2)$ in-place linear-search sort of Step 1 (a simple selection-sort-style pass — an $O(n \log n)$ or even $O(n)$ in-place rearrangement is possible with more care, e.g. by following a's own cycles to place each value directly, but $O(n^2)$ already meets the stated bound). Total: $O(n^2)$ time, $O(1)$ extra space. □

Example 5.5 (Finding all saddle entries of a matrix in one pass per element)**.** Call an entry of an $n \times m$ matrix (all entries distinct) a *saddle* if it is the largest in its row and smallest in its column, or the smallest in its row and largest in its column. Design an algorithm that finds all saddles using $O(nm)$ operations, $O(n + m)$ memory, and accesses each entry $A[i][j]$ at most once (it becomes unreadable immediately after).

Solution. The difficulty: knowing whether $A[i][j]$ is a row-extreme requires seeing all of row i, and knowing whether it's a column-extreme requires seeing all of column j — but each entry may be read only once, so we cannot revisit $A[i][j]$ after the fact to compare it against a later-completed row or column summary.

Single sweep. Read the matrix once, in any fixed order (e.g. row by row). Maintain running arrays $\text{rowMax}[i], \text{rowArgmax}[i], \text{rowMin}[i], \text{rowArgmin}[i]$ for $i = 1, \dots, n$ (size $O(n)$) and $\text{colMax}[j], \text{colArgmax}[j], \text{colMin}[j], \text{colArgmin}[j]$ for $j = 1, \dots, m$ (size $O(m)$), updating them in $O(1)$ as each entry is read for the first and only time. After the sweep, every one of these $8(n + m)$ numbers is a true global row/column extreme (and its location), even though the matrix entries themselves are no longer accessible.

Recovering the saddles without re-reading A. A row-max-and-column-min saddle must sit at (i, j) with $j = \text{rowArgmax}[i]$ (so $A[i][j]$ really is row i's maximum) *and* $i = \text{colArgmin}[j]$ (so it's also column j's minimum) — both conditions are checkable purely from the stored index arrays, no matrix access needed. For each row i there is exactly one candidate column $j = \text{rowArgmax}[i]$ to check, in $O(1)$, so this pass costs $O(n)$. Symmetrically, checking

$j = \text{rowArgmin}[i]$ against $i = \text{colArgmax}[j]$ for each row finds all row-min-and-column-max saddles, another $O(n)$.

Complexity. The sweep costs $O(nm)$ (each of the nm entries touched exactly once, $O(1)$ work each); the two candidate-checking passes cost $O(n)$ total; the stored arrays use $O(n+m)$ memory. Every entry of A is read exactly once, as required, and every saddle is found (and nothing else is reported, since both defining conditions are checked exactly). □

Example 5.6 (A structure with $O(1)$ extract-min and extract-max)**.** Design a data structure for real numbers supporting *insert*, *delete* (an arbitrary element), *delete-min*, and *delete-max*, where the last two run in $O(1)$. Try to make insert and delete as fast as possible too — can they also be made $O(1)$?

Solution. **Construction.** Maintain the elements in a *sorted doubly linked list*, together with a balanced search structure (e.g. a balanced BST or skip list) that maps each value to its node in the list, supporting $O(\log n)$ search.

- *Insert*(x): locate x's sorted position via the index ($O(\log n)$), splice a new node into the linked list there ($O(1)$ given the position), and insert into the index ($O(\log n)$). Total $O(\log n)$.
- *Delete*(x) (given its value or a handle): find it via the index if needed ($O(\log n)$, or $O(1)$ with a handle), unlink from the list ($O(1)$), remove from the index ($O(\log n)$). Total $O(\log n)$.
- *Delete-min* / *delete-max*: the head/tail pointer of the sorted list gives the min/max directly, and unlinking it is $O(1)$; keeping the index itself fully consistent after this costs $O(\log n)$, but if index upkeep is done *lazily* (mark the node removed in $O(1)$; fold the cleanup into the amortized cost of nearby future index operations), the delete-min/delete-max call itself returns in worst-case $O(1)$, with insert/delete remaining $O(\log n)$ amortized.

Can insert and delete-arbitrary also be $O(1)$? No, not simultaneously with delete-min being $O(1)$, in the comparison-based model. If both insert and delete-min ran in $O(1)$, one could sort any n numbers in $O(n)$ total time: insert all n elements ($O(n)$), then call delete-min n times to read them off in order ($O(n)$). This would be a comparison-based sorting algorithm running in $O(n)$, contradicting the classical $\Omega(n \log n)$ lower bound for comparison sorting. So

$$\boxed{\text{delete-min and delete-max can be made } O(1), \text{ but insert cannot also be made } O(1)}.$$

□

Example 5.7 (Finding the root of a hidden complete binary tree)**.** The numbers $1, \dots, n$ (with $n = 2^k - 1$) are placed, in some unknown arrangement, at the nodes of a complete binary tree of height k. Say *t lies between i and j* if deleting t from the tree leaves i and j in different components. Design an algorithm that identifies the number at the root, using $O(n \log n)$ queries of the form "does t lie between i and j?".

Solution. **Reformulation.** "t lies between i and j" is exactly the tree-theoretic statement that t is an internal vertex of the unique path from i to j. Fixing any element x, the root is simply the far endpoint of x's ancestor chain — the element of that chain with no ancestor of its own. Two structural facts drive the algorithm: (1) the tree has height $k = \Theta(\log n)$, so the path between *any* two of the n elements has $O(\log n)$ vertices; (2) for two elements x, y, querying every other element z for "z between x, y?" costs $n-2$ queries and returns exactly the (short, $O(\log n)$-sized) path from x to y — in particular it tells us whether the root lies on that path at all, since the root lies on path(x, y) precisely when x, y sit in different subtrees of the root.

Outline of the approach. Start from an arbitrary pair x, y and extract path(x, y) via the $n-2$ queries of fact (2) above. If the root turns out to lie on this path, it can be pinpointed among the $O(\log n)$ path vertices with a further $O(\log^2 n)$ queries (comparing pairs of path vertices via a third reference point sorts them into ancestor order), well within budget. If x, y happen to lie in the *same* subtree of the root instead, the extracted path stops short of the root, and its topmost vertex $m = \mathrm{LCA}(x, y)$ is now a confirmed proper descendant of the root, so a fresh pair drawn from outside path(x, y) must be tried; since each such round again costs $O(n)$ queries and there are only $O(\log n)$ levels the search can still need to climb, the whole procedure fits an $O(n \log n)$ budget.

Where this account is incomplete. What is missing here is a fully worked-out rule for choosing the next pair after a failed round — the queries already made rule out m's own subtree as containing the root, but turning that into a guaranteed $O(n)$-per-round, $O(\log n)$-round elimination scheme (rather than a possibly-repeated search) needs more careful bookkeeping than is carried out above. Small hand-simulated cases ($k = 2, 3$) are consistent with an $O(n \log n)$ algorithm along these lines succeeding, but the argument is not pinned down to full rigor here. □

Example 5.8 (Minimum hits to fell a crowd of goblins)**.** You face n goblins, goblin i having h_i hit points ($0 < h_i < H$, integers). Your staff's special attack: striking goblin i removes p hit points from goblin i and q hit points from every *other* goblin ($p > q > 0$). A goblin is defeated once its hit points drop to 0 or below. Design an $O(n \log n)$-time, $O(n)$-space algorithm to find the minimum number of strikes needed to defeat every goblin.

Solution. **Setting up the count.** If a strategy makes K strikes total, with x_i of them aimed at goblin i (so $\sum x_i = K$), goblin i takes $p\,x_i + q(K - x_i) = qK + (p - q)x_i$ damage. All goblins are defeated iff $qK + (p - q)x_i \geq h_i$ for every i, i.e.

$$x_i \;\geq\; \max\!\left(0, \left\lceil \frac{h_i - qK}{p - q} \right\rceil\right) =: x_i^{\min}(K).$$

For fixed K, this is achievable with $\sum x_i = K$ (any surplus strikes can be dumped anywhere) exactly when $\sum_i x_i^{\min}(K) \leq K$. Since $x_i^{\min}(K)$ is non-increasing in K while K itself increases, the feasible K's form an up-set: we want the smallest K with $f(K) := \sum_i x_i^{\min}(K) \leq K$.

A water-filling formula for the (real-valued) threshold. Drop the ceiling for a moment and solve the continuous relaxation exactly. At the optimal real threshold K^*, let $S = \{i : h_i > qK^*\}$ be the goblins still needing a direct hit. Consistency requires $\sum_{i \in S}(h_i - qK^*) = (p - q)K^*$, i.e.

$$K^* = \frac{\sum_{i \in S} h_i}{p + q(|S| - 1)}.$$

Since larger h_i are more likely to belong to S, sort $h_1 \geq h_2 \geq \cdots \geq h_n$ and, using prefix sums $P_s = h_1 + \cdots + h_s$ (computable in $O(n)$ after an $O(n \log n)$ sort), scan $s = 1, 2, \ldots, n$ evaluating $K_s = P_s/(p + q(s - 1))$ until the self-consistency check passes: $h_s > qK_s$ (the s-th largest still qualifies) and ($s = n$ or $h_{s+1} \leq qK_s$, so the next one does not). This scan is monotone in s and terminates in $O(n)$ total work, giving the correct prefix $S = \{1, \ldots, s\}$ and the exact real threshold $K^* = K_s$.

Rounding to an integer answer. The true (integer) minimum K lies within a small constant number of steps of $\lceil K^* \rceil$ (each unit increase of K changes $f(K)$ by a bounded amount), so a short local scan near $\lceil K^* \rceil$, re-checking $f(K) \leq K$ with the exact ceiling formula, locates the true minimum in $O(n)$ further work. So

$$\boxed{\text{sort the } h_i \ (O(n \log n)) \text{, then a single } O(n) \text{ water-filling scan finds the minimum strike count}}$$

(verified against brute-force search on many small random instances). □

Example 5.9 (Longest color-alternating path in a two-colored forest)**.** The edges of an acyclic undirected graph (a forest) are colored red and blue. Design an $O(V + E)$-time algorithm that finds the length of the longest path in which every two consecutive edges have different colors. How much extra memory does your algorithm need?

Solution. **A per-vertex DP by incoming color.** Root each tree of the forest arbitrarily and run a single post-order DFS. For a vertex v and a color $c \in \{R, B\}$, define

$$f(v, c) = \text{length of the longest alternating path starting at } v \text{ whose first edge has color} \neq c,$$

using only edges in v's subtree (i.e. away from v's parent). Then

$$f(v, c) = \max\Big(0, \max_{\substack{u \text{ child of } v \\ \text{edge } (v,u) \text{ has color } c' \neq c}} [1 + f(u, c')]\Big),$$

computed bottom-up: once all children of v are processed, $f(v, R)$ and $f(v, B)$ each take $O(\deg v)$ time to evaluate (one pass over v's children, keeping the best value among red-edge children and among blue-edge children). Summed over all vertices, this is $O(V + E)$.

Combining two directions through a vertex. A longest alternating path need not start at a leaf — it can pass through some internal vertex v as a "peak," entering via one color and leaving via the other. So also compute, for each v: let $R_v = \max_{u:\,(v,u)\text{ red}} [1 + f(u, B)]$ (best blue-continuation after a red edge into v) and $B_v = \max_{u:\,(v,u)\text{ blue}} [1 + f(u, R)]$ (symmetric); then the best path peaking at v has length $R_v + B_v$ if v has both a red and a blue incident edge (with $f(\text{child}, \cdot)$ computed exactly as above), or just $\max(R_v, B_v, 0)$ otherwise. Each of these is again computable in $O(\deg v)$ time from the already-computed f values.

Answer and complexity. The overall answer is $\max_v(\text{best path peaking at } v)$, taken over all vertices in a single additional $O(V)$ pass after the DFS. Total time is $O(V + E)$ (each edge examined a constant number of times). So

$$\boxed{O(V + E) \text{ time}}$$

using $O(V)$ extra memory: two DP values per vertex $(f(v, R), f(v, B))$ plus the DFS recursion stack, which is $O(V)$ in the worst case (e.g. a path graph). □

Example 5.10 (A sparse family of intervals covering every interval by a union of two). Let $\Lambda = \{[i, j] : 1 \leq i < j \leq n\}$ be the set of all intervals with integer endpoints in $\{1, \ldots, n\}$. (a) Show there is a subset $\Sigma \subseteq \Lambda$ with $|\Sigma| = O(n \log n)$ such that every interval in Λ is the union of at most two intervals from Σ. (b) Show this bound is tight: any such Σ must have $|\Sigma| = \Omega(n \log n)$.

Solution. **(a) Construction (the "sparse table" family).** For every power of two $2^m \leq n$ (there are $O(\log n)$ such m) and every valid starting point i, include the length-2^m interval $[i,\ i+2^m-1]$ in Σ. This gives $O(n)$ intervals per scale m and $O(\log n)$ scales, so $|\Sigma| = O(n \log n)$.

Given any target $[i, j] \in \Lambda$ of length $L = j - i + 1$, let $m = \lfloor \log_2 L \rfloor$, so $2^m \leq L < 2^{m+1}$, i.e. $2^m \geq L/2$. The two length-2^m intervals $[i, i + 2^m - 1]$ and $[j - 2^m + 1, j]$ both lie in Σ (both are valid sub-intervals of $[i, j]$, since $2^m \leq L$), and since each covers more than half of $[i, j]$'s length, they overlap (or meet) and their union is exactly $[i, j]$. So every interval in Λ is a union of (at most) two elements of Σ — this is precisely the classical *sparse table* construction used for $O(1)$ range-minimum queries.

(b) The matching lower bound. A counting argument alone only gives $\Omega(n)$: a family Σ of size s has $O(s^2)$ pairs, each pair's union (when it is a single interval at all) determines at most one element of Λ, and there are $\Theta(n^2)$ intervals to cover, so $s^2 = \Omega(n^2)$, i.e. $s = \Omega(n)$. Sharpening this to the tight $\Omega(n \log n)$ bound requires tracking how many *distinct length scales* must appear near each position — intuitively, covering all $\Theta(n)$ lengths at each of $\Theta(n)$ positions with only pairwise unions forces a logarithmic multiplicity of scales at (on average) every position, mirroring why the sparse-table construction above cannot be thinned out. This sharper accounting is not carried out in full here; it is the well-known matching lower bound for this classical construction, and the $O(n \log n)$ upper bound of part (a) is tight. □

Example 5.11 (Closest-to-target pair in a sorted matrix)**.** An $n \times n$ matrix has every row and every column sorted increasing. Design an algorithm that finds two entries (not necessarily distinct positions in different rows/columns) whose sum is as close as possible to a given number q, using $O(n)$ extra memory and without modifying the matrix.

Solution. **Closest-value search in** $O(n)$**.** First, a building block: given a target T, find the matrix entry closest to T using $O(n)$ comparisons and $O(1)$ extra memory. Start at the top-right corner $(i, j) = (0, n-1)$. Repeatedly compare $A[i][j]$ to T: record it if it is the closest seen so far; if $A[i][j] < T$, move down (i++, since everything above in this column is even smaller); if $A[i][j] > T$, move left (j−−, since everything to the right in this row is even larger); stop if $A[i][j] = T$ or the pointer leaves the matrix. Each step eliminates a row or a column, so this terminates in at most $2n$ steps, and it never skips the true closest value (standard correctness argument for this "staircase search," identical to the classical algorithm for exact search in a sorted matrix).

Combining over all first elements. For each of the n^2 entries $A[i_1][j_1]$ (taken as the first chosen element), run the closest-value search for target $q - A[i_1][j_1]$ to find the best partner, and keep the overall best pair found. This is correct by exhaustion — the true optimal pair's first element is among the n^2 candidates, and for that choice the search correctly finds its best partner.

Complexity. $O(n)$ per search, over n^2 choices of first element, gives $O(n^3)$ time; memory is $O(1)$ beyond the two loop indices and the two search pointers (well within the $O(n)$ budget), and the matrix is only read, never modified. So

$$\boxed{O(n^3)\text{ time}, O(n)\ (\text{in fact } O(1))\text{ extra memory}}$$

(verified against brute force on random sorted matrices). Restricting the outer loop to iterate over rows rather than individual cells, and reusing search state across a row, is a natural way to try to cut this to $O(n^2)$; making that reuse provably correct needs a more careful monotonicity argument than is worked out here. □

Example 5.12 (Detecting a deeply nested segment in $O(n \log n)$)**.** Given n segments $[a_i, b_i]$, the *nesting index* of a segment is the number of other segments that contain it. Design an algorithm that determines whether some segment has nesting index exceeding 1000, using $O(n \log n)$ time and $O(n)$ extra memory.

Solution. Segment i contains segment j exactly when $a_i \le a_j$ and $b_i \ge b_j$: a 2-dimensional dominance condition. This is the classical setting for a Fenwick-tree (Binary Indexed Tree) sweep:

Coordinate compression. Sort the distinct b-values and replace each b_j by its rank among them ($O(n \log n)$), so ranks lie in $\{1, \dots, n\}$ and a Fenwick tree over them supports prefix-count queries and point updates in $O(\log n)$ each.

Sweep by left endpoint. Sort the segments by a_i increasing ($O(n \log n)$); process ties (equal a_i) as a batch. Maintain a Fenwick tree, initially empty, over the compressed b-ranks. When processing segment j: query the Fenwick tree for the count of already-inserted segments with b-rank $\ge \text{rank}(b_j)$ (a suffix-count query, done via a prefix-count query on the complement, $O(\log n)$) — this count is exactly the number of previously-processed segments (all with $a_i \le a_j$) that also have $b_i \ge b_j$, i.e. segment j's nesting index restricted to earlier segments. After querying every segment in the current a_i-batch (so segments sharing the same a_i don't count each other, matching the strict containment condition when endpoints tie exactly however the problem intends it — adjust the query/insert order within a batch if ties should count each other), insert all of the batch's b-ranks into the Fenwick tree, then move to the next batch.

Answer. Track the maximum nesting index seen (or short-circuit and return "yes" the moment a count exceeds 1000). This is

$$\boxed{O(n \log n) \text{ time}, O(n) \text{ extra memory}}$$

— $O(n \log n)$ for the two sorts plus $O(n)$ Fenwick operations at $O(\log n)$ each, and $O(n)$ memory for the compressed ranks and the Fenwick array. □

Example 5.13 (Shortest index window realizing a target multiset)**.** Given integer arrays $a[1..n]$ and $b[1..k]$ (with all elements of b distinct), design an algorithm that finds indices $i_1 < i_2 < \cdots < i_k$ minimizing $i_k - i_1$ such that $\{a[i_1], \ldots, a[i_k]\}$ is a permutation of b. Target $O(nk)$ time (faster is welcome) and $O(n)$ extra memory.

Solution. This is the classical *minimum window containing a target set* problem, solved by a two-pointer sweep in $O(n)$ time (better than the requested $O(nk)$).

Setup. Build a hash map from each value in b to its index in $\{1, \ldots, k\}$ ($O(k)$ time/space). Maintain a count array $\text{cnt}[1..k]$ (all zero initially) tracking how many times each b-value currently appears in the sliding window $[L, R]$, and a counter covered = number of b-values with $\text{cnt} > 0$.

Sweep. Advance R from 1 to n. If $a[R]$ is one of b's values (an $O(1)$ hash lookup), increment its count; if that count just became 1, increment covered. Whenever covered $= k$ (the window currently contains every b-value at least once), shrink L as far as possible: while $a[L]$ is either not a b-value, or is a b-value whose count is > 1 (a redundant duplicate), decrement its count if applicable and advance L. Once shrinking must stop (removing $a[L]$ would drop covered below k), record the window $[L, R]$ as a candidate, taking $i_1 = L$, $i_k = R$, and the other $k - 2$ indices as the positions of the (unique, by minimality) occurrence of each remaining b-value inside $[L, R]$.

Why this is correct and optimal. Every index R is visited once by the expanding pointer, and L only ever moves forward, so each pointer travels at most n steps total — $O(n)$ amortized work for the sweep, plus $O(1)$ hash work per element. Among all windows containing a full copy of b's value set, the shrinking step finds the smallest one ending at each R, and scanning all R finds the global minimum. So

$$\boxed{O(n) \text{ time (via hashing)}, O(n) \text{ extra memory}}$$

(the $O(k)$ hash map and $O(k)$ count array are within the $O(n)$ budget since $k \le n$). □

Example 5.14 (In-place linear-time sort by remainder mod 5)**.** Given an array of n integers, design an algorithm that sorts them by remainder modulo 5 (order among equal remainders doesn't matter), using $O(n)$ time and $O(1)$ extra memory.

Solution. This is in-place counting sort, specialized to the 5 fixed buckets $0, 1, 2, 3, 4$.

Pass 1: count. Scan the array once, incrementing a counter $c[r]$ for each element's remainder $r = a[i] \bmod 5$. Only 5 counters are needed: $O(n)$ time, $O(1)$ space.

Pass 2: compute target ranges. From the 5 counts, compute the starting index of each remainder-class's final block via a prefix sum: $\text{start}[0] = 0$, $\text{start}[r] = \text{start}[r-1] + c[r-1]$. Also set $\text{next}[r] = \text{start}[r]$ — the next free slot for class r. Both are size-5 arrays: $O(1)$ space.

Pass 3: in-place placement by cycle-following. For $i = 0, \ldots, n-1$: while the element currently sitting at position i has remainder r with $\text{next}[r] \ne i$, swap $a[i]$ with $a[\text{next}[r]]$ and increment $\text{next}[r]$ (this places the swapped-in element — whatever was at the target slot — at i, so the loop re-examines position i with its new occupant). Once the element at i belongs there ($\text{next}[r] = i$), also increment $\text{next}[r]$ and move to $i + 1$.

Why this is $O(n)$. Every swap places at least one element into its final, correct block permanently (the element moved to $\text{next}[r]$ never needs to move again, since $\text{next}[r]$ only ever increases and stays within class r's designated range), so the total number of swaps across the whole pass is at most n. Combined with the two $O(n)$ counting passes,

$$\boxed{O(n) \text{ time}, O(1) \text{ extra memory}}$$

(only the 10 counters $c[0..4], \text{next}[0..4]$ and a few loop indices are used, regardless of n). □

Example 5.15 (Finding one defective relay in $O(N)$ switch operations)**.** A connected circuit is an undirected graph with no multiple edges: N edges are wires with relays, and vertices are either light bulbs or a single power source. A bulb lights iff some path to the source has every relay along it switched on. Exactly one relay is defective (it never conducts, regardless of its switch position); all switches start on. Design a strategy that identifies the defective relay using $O(N)$ switch toggles (observing which bulbs are lit after each).

Solution. **Handling a spanning tree in $O(N)$.** Fix any spanning tree T of the graph, rooted at the source ($V - 1 \le N$ edges). Turn every relay off. Then reveal T's edges one at a time in BFS or DFS order from the source: turn on the next tree edge (keeping all previously-confirmed-good edges on) and check whether the newly reached vertex's bulb lights. If it does, that edge is good; move on. If it does not light, that edge's relay is the defective one — done, in at most $V - 1 \le N$ toggles.

The remaining (non-tree) edges. If every tree edge tests good, the defective relay is one of the $N - (V - 1)$ non-tree edges, each of which reconnects two vertices already joined by T. For a candidate non-tree edge (u, w): temporarily switch off the tree edges on the unique u–w path within T, switch on (u, w) itself, and check whether w's bulb (now reachable only via $u \to \cdots \to (u, w) \to w$, using the rest of the tree to reach u) lights; then restore the path's switches before testing the next candidate. This correctly identifies a defective non-tree edge.

$$\boxed{O(N) \text{ switch operations suffice}}$$

for the spanning-tree phase, which by itself already covers a large class of instances (e.g. whenever the graph is a tree, $N = V - 1$, so no second phase is needed at all). Turning the second phase's per-edge path-toggling into a strategy that is provably $O(N)$ *in total* even when the graph has many independent cycles (rather than merely $O(N)$ per edge tested) needs a more careful incremental/amortized scheme than is spelled out here. □

Example 5.16 (The largest function satisfying an iterated-square-root recurrence)**.** Algorithm A's running time satisfies $T(n) \le T(\lfloor\sqrt{n}\rfloor) + 1$, with $T(1) = C_1$ constant. Find, in O-notation, an asymptotically as-large-as-possible function satisfying this relation, and prove it does.

Solution. **Why repeated square-rooting is fast.** Applying the recurrence k times (ignoring floors, which only help asymptotically),

$$T(n) \le T(n^{1/2^k}) + k.$$

Choosing $k = \lceil \log_2 \log_2 n \rceil$ makes $n^{1/2^k} \le 2$, a constant, so $T(n) \le T(O(1)) + k = O(1) + O(\log\log n)$.

This is achievable, and is the extremal (largest) solution. Define $S(n)$ by the recurrence taken with *equality*: $S(n) = S(\lfloor\sqrt{n}\rfloor) + 1$, $S(1) = C_1$. Since square-rooting shrinks n to 1 in exactly $\Theta(\log\log n)$ steps (halving $\log_2 n$ each time), $S(n) = C_1 + \Theta(\log\log n)$, and by construction S satisfies the given relation with equality — so it is itself a valid solution, growing as $\Theta(\log\log n)$.

Moreover S dominates every other solution: if T satisfies $T(n) \le T(\lfloor\sqrt{n}\rfloor) + 1$ with the same $T(1) = C_1$, then by strong induction on n (using $\lfloor\sqrt{n}\rfloor < n$ for $n \ge 2$), $T(n) \le T(\lfloor\sqrt{n}\rfloor) + 1 \le S(\lfloor\sqrt{n}\rfloor) + 1 = S(n)$. So

$$\boxed{T(n) = O(\log\log n) \text{ is the largest solution, attained by the equality recurrence}}.$$

□

Example 5.17 (Fewest deletions to destroy a palindrome)**.** Given a string S of lowercase letters, find the minimum number of characters (possibly 0) that must be deleted so the remaining string is *not* a palindrome. Design an algorithm, prove it correct, and analyze its complexity.

Solution. **If S is already not a palindrome**, the answer is 0.

If S consists of a single repeated character, no deletion can ever help: every subsequence of a constant string is itself constant, hence a palindrome. (This degenerate case has no valid answer; flag it separately.)

Otherwise (S is a palindrome, but not constant): the answer is exactly 1, achieved by deleting the very first character. This can't be 0 (S is currently a palindrome), so it suffices to show deleting $S[1]$ always works.

Proof that $S' = S[2..n]$ is never a palindrome. Suppose, for contradiction, it were. Write the two palindrome conditions in terms of S: the hypothesis (on S) gives $S[k] = S[n+1-k]$ for all k; the assumption (on S') gives, after reindexing, $S[k] = S[n-k+2]$ for $k = 2, \dots, n$. Combining these two expressions for $S[k]$ (both valid for $2 \le k \le n-1$) gives $S[n+1-k] = S[n-k+2]$ for all such k; substituting $m = n-k+1$, this says $S[m] = S[m+1]$ for every $m = 1, \dots, n-1$ — i.e. S is constant, contradicting our assumption. So S' is not a palindrome.

Algorithm. Check whether S is a palindrome (two-pointer scan from both ends, $O(n)$ time, $O(1)$ extra space). If not, output 0. If so, check whether all characters are equal (another $O(n)$ scan); if so, report impossible; otherwise output 1 (with the first character as a witness deletion). So

$$\boxed{O(n) \text{ time}, O(1) \text{ extra memory}}.$$

□

Example 5.18 (Extrapolating a degree-n polynomial from its first $n+1$ values)**.** An unknown degree-n polynomial $P(x)$ is given via its values $P(0), P(1), \dots, P(n)$. Design an $O(n^2)$-time algorithm to compute $P(n+1), P(n+2), \dots, P(2n)$.

Solution. **Key fact: the n-th finite difference of a degree-n polynomial is a constant.** Define the forward difference operator $(\Delta f)(x) = f(x+1) - f(x)$. If f has degree $d \ge 1$, then Δf has degree $d-1$ (the leading terms cancel in $f(x+1) - f(x)$, leaving a degree-$(d-1)$ polynomial with leading coefficient d times that of f). So $\Delta^n P$ has degree 0: it is some constant c, the same value $\Delta^n P(k) = c$ for *every* integer k.

Building the difference triangle. From $P(0), \dots, P(n)$, compute the standard triangular difference table: row 0 is $P(0), \dots, P(n)$; row j (for $j = 1, \dots, n$) has entries $\Delta^j P(0), \dots, \Delta^j P(n-j)$, each obtained as (entry to its lower-right) $-$ (entry directly below it) in row $j-1$, i.e. $\Delta^j P(k) = \Delta^{j-1} P(k+1) - \Delta^{j-1} P(k)$. This table has $\binom{n+2}{2} = O(n^2)$ entries, each computed in $O(1)$, so it costs $O(n^2)$ total. Its apex is the single value $\Delta^n P(0) = c$.

Extending the table diagonally. To find $P(n+1)$: since $\Delta^n P(1) = c$ too (the n-th difference is constant everywhere), start a new top-to-bottom diagonal with $\Delta^n P(1) = c$ and fill it downward using $\Delta^{j-1} P(k+1) = \Delta^{j-1} P(k) + \Delta^j P(k)$ — each new diagonal entry is the sum of the entry above-left of it and the entry directly above it, both already known from the existing table. After n such steps the diagonal reaches $\Delta^0 P(n+1) = P(n+1)$, using $O(n)$ arithmetic operations.

Iterating. Repeat this once for each of $P(n+1), \dots, P(2n)$: to get $P(n+1+t)$, extend a new diagonal starting from $\Delta^n P(1+t) = c$ (still the same constant), using only previously computed table entries, again in $O(n)$ operations. After n repetitions, all of $P(n+1), \dots, P(2n)$ are known.

Complexity. Building the initial table costs $O(n^2)$; each of the n extrapolation steps costs $O(n)$, for another $O(n^2)$. Total time and space are both

$$\boxed{O(n^2)}.$$

□

Example 5.19 (Finding one of n hidden points in a half-disk via nearest-distance queries)**.** A half-disk (semicircular region) hides n unknown points. You may repeatedly pick any point X and ask "what is the distance from X to the nearest hidden point?" (a distance of 0 means X itself is a hidden point). Prove that at least one hidden point can always be found using at most $2n + 1$ queries.

Solution. **Setup.** Query $n + 1$ points $A_1 < A_2 < \cdots < A_{n+1}$, placed in this order along the diameter of the half-disk, recording the returned distances $r_1, \dots, r_{n+1}$. For each i, let $f(i)$ denote which of the n hidden points is nearest to A_i (well-defined a.s.; ties are a measure-zero degeneracy that can be perturbed away).

Key structural fact. The Voronoi cell of any hidden point Z (the set of locations closer to Z than to any other hidden point) is convex, being an intersection of half-planes. The intersection of a convex planar region with a straight line (here, the diameter) is a single connected subinterval. Consequently, if $f(i) = f(j)$ for some $i < j$ (both A_i, A_j nearest to the same hidden point Z), then A_i and A_j both lie in Z's Voronoi interval on the diameter, so every A_k with $i \le k \le j$ — lying between them on the line — lies in that same interval too, giving $f(k) = Z$ for all such k. In particular $f(i) = f(i + 1)$.

Pigeonhole. The map f sends $n + 1$ indices into a set of n hidden points, so some value repeats: $f(i) = f(j)$ for some $i < j$. By the structural fact just proved, this forces $f(i^*) = f(i^* + 1)$ for some *adjacent* pair $i^* \in \{1, \dots, n\}$.

Locating the shared point. For that adjacent pair, the hidden point $Z = f(i^*)$ is at distance exactly r_{i^*} from A_{i^*} and exactly r_{i^*+1} from A_{i^*+1} (being the nearest point to each). So Z lies on both the circle of radius r_{i^*} centered at A_{i^*} and the circle of radius r_{i^*+1} centered at A_{i^*+1}. Two such circles (centered at distinct points of the diameter) intersect in at most 2 points, symmetric about the diameter line; since all hidden points lie strictly within the half-disk, only the intersection point on the half-disk's side (if any) is a valid candidate — the mirror point across the diameter can be discarded without a query.

The algorithm. For each of the n adjacent pairs (A_i, A_{i+1}), $i = 1, \dots, n$, compute the circle intersection on the half-disk side (if the two circles intersect at all) and query that single candidate point — 1 query per pair, n queries total. Since we don't know in advance which pair is the true matching pair i^*, we simply try all n; the pair i^* guaranteed to exist by pigeonhole will, when its turn comes, return distance 0 (its candidate point coincides with Z itself), successfully identifying a hidden point.

Total. $n + 1$ queries for the initial sweep, plus at most n queries for the adjacent-pair candidates:

$$\boxed{(n + 1) + n = 2n + 1 \text{ queries suffice}}.$$

□

Example 5.20 (Two non-attacking matrix elements with maximum product)**.** An $n \times n$ matrix is filled with distinct positive integers. Design an algorithm to find two entries, no two in the same row or column, whose product is as large as possible, in $O(n^2)$ time and $O(n)$ extra memory.

Solution. Let a be the largest entry in the matrix and b the second largest (found by one $O(n^2)$ scan, tracking values and positions).

If a, b are compatible (different row and column), the answer is simply $a \cdot b$ — unbeatable, since these are literally the two largest entries in the whole matrix.

Key lemma: if a, b conflict (share a row or column), the optimal pair always includes a or b. Suppose a, b share row R (the column case is symmetric), with a at column $c_a \neq c_b$, b's column. Suppose toward contradiction that an optimal compatible pair (x, y) avoids both a and b; then $x, y < a$ and, since b is the *second*-largest entry overall, also $x, y < b$ (as $x, y \neq a, b$). If x were compatible with a (different row/column), then $a \cdot x > b \cdot x \ge y \cdot x$ — wait, more directly: $a \cdot x > y \cdot x$ since $a > y$ (as $y \neq a$), contradicting that (x, y) beats every pair through a. So x

must conflict with a; likewise, if x were compatible with b, then $b \cdot x > y \cdot x$ (since $b > y$), again a contradiction — so x must conflict with b too. The same argument applied to y (using $a > x$ and $b > x$) shows y conflicts with both a and b as well.

Now, conflicting with *both* a and b: since a, b have different columns ($c_a \neq c_b$), no single element can match both columns at once, so any element conflicting with both a and b must instead share their common row R. Hence x and y are *both* in row R — but then x, y share a row, contradicting that (x, y) is itself a compatible pair. This contradiction proves the lemma.

Algorithm. If a, b conflict: compute a^* = the maximum entry outside a's row and column, and b^* = the maximum entry outside b's row and column (two more $O(n^2)$ scans). By the lemma, the optimal answer is

$$\boxed{\max(a \cdot a^*,\ b \cdot b^*)}$$

(with $a \cdot b$ handled separately when a, b are already compatible). This uses $O(n^2)$ time (three linear passes over the matrix) and $O(1) \subset O(n)$ extra memory (a handful of running maxima and their positions). □

How to Use This Guide Efficiently

- Memorize formulas only after you understand where they come from.
- When a problem looks hard, first classify it: parity, conditioning, asymptotics, optimization, or linear structure.
- In an interview, write down the state variables explicitly. A correct state definition is usually half of the solution.
- For revision, cover the solutions and try to reproduce them from the definitions and formulas alone.